

NEURAL BELLMAN OPERATORS: PRESERVATION SUPPLEMENT TO REVISION R12

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This supplement preserves the complete preceding R11 paper and its entire preservation supplement without deleting equations, proofs, numerical tables, or unsuccessful calculations. The current manuscript is ECTA_R12. Earlier precision and comparative-performance statements are dated results, not new claims. The R12 revision adds a certified continuous-cost policy library, a finite exact breakpoint test, an adaptive refinement record, and complete new mathematical proofs. All historical source files remain unchanged.

READING MAP

The R12 manuscript contains the original economy, all R11 main results and proofs, all prior studies and accuracy calculations, together with continuous-cost certification and its full proof. A separate point-by-point response identifies the changes against every finding of the latest R9 referee report. The immutable source, result, build, and review identities are recorded in the revision directory.

The following pages reproduce the complete R11 manuscript. They are followed by the complete R11 supplement, which already preserves R9, R8, and earlier development material. In particular, recursive utility, sophisticated temporal selves, games, stochastic traces, action safeguards, unsuccessful earlier certificates, and all retained external experiments remain available. These earlier documents must be read with their original version labels. R12 does not turn an earlier failed tolerance or an exploratory numerical result into a current certificate merely by reproducing it.

PART I: COMPLETE R11 MANUSCRIPT

We study neural policy evaluation, improvement, and independent verification in a continuous economy with costly preference adjustment. An affine-preference market-deflator dual and validated policy evaluation certify regret below .01 for three adjustment-cost coefficients under the original utility, adaptive controls, and first-exit contract. Access to adjustment and the welfare cost of more expensive adjustment are resolved by intervals narrower than their respective effects. A quantitative output-error allowance, interval neural differentiation, and a structured global action oracle connect implemented policies to verified quantities. We add prospectively frozen method-specific tuning and independent holdout evidence without replacing the earlier comparison. Reference-solvable tracking, mean-reverting, and ill-conditioned control problems provide absolute-accuracy frontiers for stored held-action actors. Their exact error decomposition separates deployment error from implemented optimization error. Proofs, retained unsuccessful outcomes, and independent recomputations distinguish policy evaluation, optimality, sampling uncertainty, and the economic inequalities established.

KEYWORDS: Stochastic control, Neural policy iteration, Endogenous preferences, Recursive utility, Verification, Computational economics.

1. INTRODUCTION

numerical decision rule is useful for economic analysis only when its error is small relative to the comparison being made. This requirement is distinct from a small training loss. It is also distinct from outperforming another learning algorithm. When a decision maker can change a reference state, an apparent welfare gain may reflect either the adjustment technology or an accurate evaluation of the policy without that technology. When termination is an economic contract, an inaccurate treatment of first exit can dominate both effects.

This paper studies Neural Bellman Operators (NBO): separated policy evaluation and feasible policy improvement, implemented by neural networks and assessed by independent verification. A motivating economy combines consumption, portfolio choice, stochastic risk aversion, costly preference adjustment. The analysis retains its original running utility, continuous controls, correlated Brownian shocks, and first-exit liquidation covenant. Neither reflecting the state at a training boundary nor altering the running payoff is used to establish the economic results below.

The main numerical result is a certified .01-optimal policy in the original continuous economy at the central initial state, for adjustment-cost coefficients .5, 2, and 8. The respective regret upper bounds are .00987440, .00971727, and .00996896. Each lower endpoint evaluates a feasible policy; each upper endpoint covers all original adaptive controls. The computation does not replace the first-exit contract, restrict the original optimum to time controls, or use a finite action grid to claim a continuous guarantee.

The economic consequences are quantitative. Raising the adjustment-cost coefficient from .5 to eight lowers optimal welfare by an amount in $[-.02318257, .04286880]$. The width of this interval is smaller than its lower endpoint. Access to adjustment at coefficient two raises optimal welfare by an amount in $[-.04096859, .05973218]$ relative to the *optimal* no-adjustment economy, rather than relative to a selected no-adjustment rule. Discounted expenditures of the deployed

not identified by relabeling a learned policy as an optimizer. A computational argument separates policy generation from proof. A learned time-control is polished within the feasible output class and evaluated by Gaussian polynomial moments exact first exit. An affine-preference market-deflator representation bounds the flexible n. Portfolio terms cancel for every adapted original portfolio rule. A supporting source retains flexible adjustment, and a conditional-variance correction pays explicitly for marginal value. Outward weighted Taylor boxes enclose the remaining one-Gaussian s. Explicit continuation potentials cover artificial verification faces without altering the c domain.

constructions improve on the preceding wealth-independent witness, whose $k = 2$ regret as .10634. That witness and its complete proofs remain available; the new certificate is relabeling of its evaluation precision as optimality precision. The restricted economy's market-deflator bound is retained, together with its independently evaluated feasible t gives an initial restricted-value interval narrower than .00905. Combining the two es' bounds establishes the welfare inequalities above.

second contribution addresses the inner optimization needed for verification. In a concave-convex control family, all nonconvex coupling depends on the sum of the actions. dimensional branch-and-bound calculation, together with strongly convex scalar dual lems, certifies the full continuous action box without a tensor action grid. Executed tes reach gaps below 10^{-5} through 128 action dimensions. At the eight- and sixteen-onal learned-critic test states, interval differentiation of the actual network weights also the error in the floating-point critic gradient. This is an implemented bridge between parameters and an action certificate, not an assumption that an optimizer's stationary globally greedy.

same nonconvex family is given an absolute accuracy axis. A pointwise running-cost on and removal of control bounds admit a Cole–Hopf representation that retains the nonquadratic terminal payoff. Orthogonal Gaussian coordinates integrate analytically, a positive scalar integral with an outward enclosure. The resulting number is a lower or the unchanged original minimization problem. Relative performance is evaluated ly in a frozen holdout experiment: three dimensions, two training budgets, twelve seed r panel, equal learning-rate search budgets, warmed initialization, randomized method d a third, independently specified feedback rule. The pinned author implementation of artNet is not modified.

ical verification, monotone approximation, and primal–dual numerical methods supply nt foundations. The stopped Itô argument is not claimed as a new general verification e. Nor is alternating neural evaluation and improvement claimed as a new algorithm. The contribution is the constructive combination of original-contract verification, dent economic upper and lower bounds, computable neural-jet allowances, and a non-ative global action audit, with each error quantity matched to an executed calculation. 10 compares these objects theorem by theorem with neighboring methods. Recursive ophisticated temporal selves, dynamic games, stochastic-trace identities, counterexam- the complete preceding manuscripts are preserved in the supplement rather than being enlarge the empirical claims of the present application.

present revision also tests two previously separate parts of the numerical argument. A new, ively frozen comparison expands method-specific tuning under equal nominal training ities, implements a predeclared boundary-extension rule, and uses entirely new holdout ndependently reference-solvable tracking, mean-reverting, and ill-conditioned control s give absolute-accuracy frontiers for implemented held-action actors. A two-error

the state is $s = (u, x)$, where u denotes relative risk aversion and x denotes wealth. An admissible action is $a = (c, \theta, p)$ in the compact set

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

controls are progressively measurable and have well-defined stopped state equations. The controls and time controls actually evaluated below are bounded and satisfy this requirement. Before termination,

$$du_t = \theta_t dt + .05 dZ_t^u, \quad (1)$$

$$dx_t = [(.02 + .06p_t)x_t - c_t]dt + .2p_t x_t dZ_t^x, \quad d\langle Z^u, Z^x \rangle_t = -.25 dt. \quad (2)$$

horizon is $T = 1$, discounting is $\rho = .04$, and the open state domain is $D = (1.2, 2.8) \times (0, 2)$. Write $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$. For $k > 0$, running utility and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$G(u, x) = -.02(u-2)^2 + .1 \log x, \quad g(t, u, x) = G(u, x) - 8(1-t). \quad (4)$$

policy payoff and optimal value are

$$J_k^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(r-t)} \ell_k(S_r, a_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V_k = \sup_\pi J_k^\pi. \quad (5)$$

no-adjustment value V_R restricts θ to zero and retains *all* admissible consumption and portfolio choices. It is independent of k . The cost experiment holds the controls, transition law, stopping rule, settlement, utility normalization, and consumption units fixed. A state-dependent information of utility would generally define another economy.

The early settlement fee is a liquidation covenant. There is no continuation consumption or wealth injection following exit. At the lower wealth boundary the largest admissible wealth is negative. At the upper boundary, $p = 0$ eliminates the wealth diffusion and permits an upward drift. Thus neither a silently viable rectangle nor a theorem requiring uniform controlled-boundary regularity can be substituted for this contract. The verification below requires only equalities for smooth test functions on the trace at which the process is actually stopped.

For a smooth function v , put

$$\mathcal{L}^a v = \theta v_u + [(.02 + .06p)x - c]v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$R_a v = v_t + \mathcal{L}^a v + \ell_k - \rho v. \quad (7)$$

The mixed derivative is retained. Its sign is not inferred solely from the negative shock correlation.

PROPOSITION 1—Complete selection at a fixed jet: *For the generator in (7), a global minimizing action can be selected componentwise. Consumption is $\text{proj}_{[.05, .8]}(v_x^{-1/u})$ when*

$= .04x^2v_{xx}$. If $h_2 < 0$, the portfolio is $\text{proj}_{[-.5, .8]}(-h_1/h_2)$; if $h_2 \geq 0$, an endpoint in $h_1p + h_2p^2/2$ is selected, with a fixed tie rule.

statement includes convex and zero-curvature portfolio objectives and nonpositive wealth value. It applies to the continuous generator, not to the interpolated, killed-Euler step target. It says nothing about global optimization over shared neural parameters.

NBO: R12 PRESERVATION SUPPLEMENT

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3. LEARNING AND VERIFICATION ARE SEPARATE OPERATIONS

fixed policy, exact evaluation solves $R_\pi v = 0$ with the stopping contract. Exact improvement maximizes the Hamiltonian of that fixed evaluation. Denote these operators by $\mathcal{E}$ and $\mathcal{I}$; policy iteration is $\pi_{j+1} = \mathcal{I}(\mathcal{E}\pi_j)$. A neural implementation approximates both operators, an independent evaluator measures the payoff of the policy that will actually be used.

In raw NBO implementation, critic updates hold the actor fixed. Actor updates hold parameters fixed, but retain differentiation through the controlled successor state. A parameters, when present in a game, are also frozen during unilateral improvement. This prevents an actor loss from lowering itself by changing the object being evaluated. It makes the joint network optimization convex. The ideal policy-iteration compactness and bounding arguments are retained in the supplement; no such argument is invoked to certify the learned Adam trajectory.

Inherited finite safeguard is a distinct algorithm. For $\mathcal{T}_n^a w = r_n^a + \delta P_n^a w$, positive sub- P_n^a , and $\delta = e^{-\rho\Delta}$, define the policy-evaluation residual $e_n = \|v_n - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$ and candidate improvement gap $\eta_n = \|\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$. If the terminal error is at most b , backward comparison yields

$$\|V_{h,n}^* - V_{h,n}^\pi\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

The ρ -dependent envelope and threshold-correction proof are reproduced in the supplement. The audit can reduce the fraction of decisions changed while leaving the cost of a complete audit essentially unchanged. In particular, the retained 125-action tensor audit is not an enumerative certifier of Section 6. Its floating-point envelope is not relabeled as an continuous-time bound.

3.1. A quantitative bridge from a fitted network

$\rho(h) = (1 - e^{-\rho h})/\rho$, with $C_0(h) = h$.

ASSUMPTION 1—Admissible witnesses: *Witnesses are bounded, continuous on the stopped trace and $C^{1,2}$ in the interior of each time slab. Localized Itô identities pass to the stopping time. Uniform integrability. At slab interfaces an upper witness has nonpositive forward jumps and a lower witness has nonnegative forward jumps. The implemented policy has a well-defined state equation.*

LEMMA 1—Stopped one-sided comparison: *Suppose Assumption 1 holds, $U \geq g - b_U$ and b_L on the stopped trace, $\sup_a R_a U \leq \epsilon_U$, and $R_\pi L \geq -\epsilon_L$, where all four allowances*

$$J_k^\pi \geq L - b_L - C_\rho(T - t)\epsilon_L, \quad V_k \leq U + b_U + C_\rho(T - t)\epsilon_U, \quad (9)$$

$$0 \leq V_k - J_k^\pi \leq U - L + b_U + b_L + C_\rho(T - t)(\epsilon_U + \epsilon_L). \quad (10)$$

inequalities do not require an attained optimizer or a smooth optimal value.

THEOREM 2—Cover-and-oracle certificate for neural output: *Let a continuous neural witness and a feasible neural policy π satisfy Assumption 1, with stopped-trace error at most b . Let $\mathcal{C}$ cover the full audited time–state domain and have representatives z_j . Suppose independent evaluation proves, throughout cell j ,*

$$|R_\pi v(z) - R_\pi v(z_j)| \leq \omega_j, \quad |\Gamma(z) - \Gamma(z_j)| \leq \nu_j, \quad \Gamma = \sup_a R_a v - R_\pi v.$$

Then the representative evaluation intervals imply $|R_\pi v(z_j)| \leq e_j$ and a global action oracle satisfies $\Gamma(z_j) \leq q_j$, including jet, inner-solve, and arithmetic allowances. Set $e = \max_j(e_j + \omega_j)$ and $q = \max_j(q_j + \nu_j)$. Then

$$0 \leq V_k - J_k^\pi \leq 2b + C_\rho(T - t)(2e + q). \quad (11)$$

In particular, certified derivative bounds M_j, N_j and cell radii h_j permit $\omega_j = M_j h_j$ and $\nu_j = N_j h_j$. Any sequence of learned outputs for which these verified allowances vanish has vanishing regret, irrespective of whether its parameters converge.

This is a quantitative a posteriori connection to network errors, not a claim that an empirical mean-square loss controls a supremum. Interval differentiation or weight-based derivative bounds can supply the cell moduli. They may be conservative, and covering a high-dimensional domain remains costly. Section 6 executes the jet-to-action part for actual learned critics. The economic calculation instead fits an independent polynomial witness; it does not falsely claim to have certified a whole-state neural critic merely because its action audit passed at selected states.

4. A CONSTRUCTIVE CERTIFICATE FOR THE FLEXIBLE ECONOMY

The principal numerical target is a regret bound of .01 for the original problem (5), not for the approximation. The certificate below attains this target at $(0, 2, 1.25)$ for each of the three adjustment prices. It uses a feasible policy to bound value from below and a separate, preference-market-deflator construction to bound the optimum from above. The latter involves arbitrary adapted consumption, portfolio choice, and preference adjustment. Neither a lattice nor an optimal-value label enters its construction.

4.1. An affine source that preserves the original opportunities

Let $\phi(u) = -.02(u - 2)^2$, $h = 1 - t$, and let Y satisfy (28), with its initial level $y_0 > 0$ now given as a witness parameter. Define

$$S_0(u, y) = \Psi(u, y) + .01y - .004 \log(.5), \quad (12)$$

$$\beta(u, y) = \partial_u \Psi(u, y), \quad (13)$$

$$m(t) = 2 + \int_0^t \bar{\theta}_s ds, \quad 0 \leq \bar{\theta}_s \leq .2, \quad (14)$$

meter of an upper witness, not a restriction on the decision maker's adjustment policy.
 $= \text{proj}_{[.2, 12]} y$ and

$$Q(u) = \phi(m(1)) + \phi'(m(1))(u - m(1)) - \frac{1}{2} \phi''(m(1))(u - m(1))^2$$

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fication rectangle is $u \in [1.5, 2.5]$, $y \in [.2, 12]$. On it, the following supporting inequality

$$S_0(u, y) \leq S_0(m, y) + \beta(m, y)(u - m), \quad 2 \leq m \leq 2.2. \quad (15)$$

e-sided inequality on the whole rectangle, not a fitted tangent at sampled states. Its proof
 s derivative identities and interval inequalities with explicit margins; see Appendix B.1.
 oncavity of S_0 outside this rectangle is not assumed.

$S_0(u, y) = S_0(m(r), \hat{y}) + \beta(m(r), \hat{y})(u - m(r))$. Consider the full-line auxiliary control

$$A_k(t, u, y) = \sup_{|\theta| \leq .2} \mathbb{E}_{t, u, y} \left[\int_t^1 e^{-\rho(r-t)} \{ \hat{S}_r(U_r, Y_r) - k\theta_r^2/2 \} dr \right. \\ \left. + e^{-\rho(1-t)} Q(U_1) \right]. \quad (16)$$

$r_r = \theta_r dr + .05 dZ_r^u$, and the auxiliary control remains adapted. The affine source makes
 inal value of u an explicitly defined conditional expectation, which can be bounded
 solving another neural control problem. Put

$$Z_u(h, u) = 8h \{ e^{-120(u-1.5)+42h} + e^{-120(2.5-u)+42h} \}, \quad (17)$$

$$Z_y(h, y) = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (18)$$

REM 3—Affine-preference dual and a computable initial upper bound: *For $k \geq .5$,
 $2, 2.4]$, and any deterministic $\bar{\theta} \in [0, .2]$, the original flexible optimum satisfies*

$$V_k(0, 2, 1.25) \leq .75y_0 + .1 \log(.5) + A_k(0, 2, y_0) + Z_u(1, 2) + Z_y(1, y_0). \quad (19)$$

ate the right side, put $\beta_t = \beta(m(t), \hat{Y}_t)$ and

$$b(s) = \int_s^1 e^{-\rho(t-s)} \mathbb{E} \beta_t dt + e^{-\rho(1-s)} b_T, \quad (20)$$

$$g_k(q, a) = H_k(q) - qa + ka^2/2, \quad (21)$$

$$C = .00375 \int_0^1 t e^{-\rho t} \mathbb{E} \left[\partial_t \beta(m(t), \hat{Y}_t) \right] dt, \quad (22)$$

the l derivative includes the clipping map and $H_k(q) = \max_{|a| \leq .2} (qa - ka^2/2)$. Then

$$A_k(0, 2, y_0) \leq e^{-\rho} \phi(m(1)) + \int_0^1 e^{-\rho t} \left\{ \mathbb{E} S_0(m(t), \hat{Y}_t) - k\bar{\theta}_t^2/2 \right\} dt + C \\ + \int_0^1 e^{-\rho s} g_k(b(s), \bar{\theta}_s) ds + \frac{.09D^2}{24k}, \quad (23)$$

$$D^2 = \frac{e^{aM + .045a^2}}{16} \{(M + .09a)^2 + .09\}, \quad a = 12/11, \quad M = \log y_0. \quad (24)$$

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 expectations in (22)–(23) are one-Gaussian integrals. They involve no unknown optimal
 e or policy.

the cancellation of the portfolio is the same accounting identity used in the restricted dual:
 $(x) = (.04Yx - Yc)dt +$ a local martingale. The new step is to retain flexible adjustment
 e replacing its source by a supporting affine function. The variance allowance in (23) pays
 icitly for the random conditional marginal value of adjustment. Replacing that marginal
 e by its unconditional mean without this allowance would generally produce an invalid
 er bound. The covariance term C also matters: it retains the original correlation between
 erence and market shocks and is nonpositive here. It is evaluated, not silently discarded or
 used with a change of measure.

the source bound $S_0 \geq -7.1$ and $|\beta| \leq 1.1$ imply $A_k - Q \geq -7.925272h$ on the verification
 angle, by choosing zero adjustment in the auxiliary problem. Hence the factors $8h$ in (18)
 for continuation at artificial faces. On actual wealth exits, the uncorrected affine witness
 y dominates the liquidation settlement. The global fallback $V_k \leq G$ covers continuation
 r an artificial exit. These facts connect (23) to the original stopped economy rather than to a
 ogate full-line economy.

4.2. Policy generation, verified refinement, and the stopping rule

the lower-bound policies start from the retained one-input, twelve-unit neural time-control
 rs. A constrained L-BFGS-B step polishes their sixteen consumption and adjustment outputs
 g the original payoff and the same exact budget projection. This is an explicit enlargement
 n a fixed network’s image to the feasible time-control outputs; it is not described as addi-
 al Adam iterations or as a theorem that the neural optimizer converges. Consumption and
 stment constants are rounded inward relative to the *exact decimal* action bounds, and feasi-
 y is checked using their exact binary rational values. The finite-node training objective is a
 osal mechanism, not an asserted untruncated Gaussian expectation of singular CRRA utility.
 independent polynomial-moment evaluator in Appendix H then encloses each deployed
 inuous-time policy under exact first preference exit. The full network histories, original
 uts, polished outputs, and output-optimization records remain separate.

the dual reference is fitted with Gaussian quadrature, but its fitted objective is never used
 bound. Verification uses $t = v^2$, subdivides each of the sixteen time slabs, and covers the
 dard Gaussian coordinate on $[-6, 6]$. Constant and linear Taylor terms are integrated with
 rval-enclosed weighted moments. Hessian ranges enclose the entire remainder, including
 es crossing the consumption constraint. Gaussian tails, the terminal tangent, the conditional-
 ance allowance, localization, and outward arithmetic are separately included. The small
 dimensional interval tests establishing (15) precede every certificate execution.

the stopping rule is simple. Given a policy lower bound L and an upper bound U from
 rem 3, return the policy with tolerance ε only when $U - L \leq \varepsilon$. Otherwise refine the
 drature or return the explicit unachieved target at the declared work limit. Soundness does
 require optimizer convergence. Nor does soundness imply that an arbitrary tolerance can be
 hed with a fixed affine relaxation. For the three declared .01 instances, the finite executions
 w supply the termination certificate.

k	Feasible policy payoff	Optimal-value upper	Regret upper
0.5	$[-1.27267126, -1.27265474]$	-1.26279686	0.00987440
2	$[-1.29502451, -1.29501124]$	-1.28530723	0.00971727
8	$[-1.32817604, -1.32816919]$	-1.31820708	0.00996896

regret bounds are below .01. Each upper bound covers the full original continuous control set and stopping. Each lower bound evaluates the specified feasible time-control policy. Endpoints are rounded outward; gaps are unrounded endpoints. These are central-state certificates, not uniform claims over every initial condition.

TABLE II

VERIFIED WORK NEEDED TO CROSS THE ABSOLUTE-ERROR TARGET

k	Source boxes	Regret upper	Verification seconds	.01 met
0.5	4,096	0.01027899	5.46	No
0.5	16,384	0.00995838	11.73	Yes
0.5	65,536	0.00987440	27.54	Yes
2	4,096	0.01013162	5.25	No
2	16,384	0.00980424	11.30	Yes
2	65,536	0.00971727	27.06	Yes
8	4,096	0.01037189	5.29	No
8	16,384	0.01005494	11.48	No
8	65,536	0.00996896	26.79	Yes

policy is independently evaluated once at the finest stated precision. Time includes that evaluation, dual verification, and cumulative dual verification through each row, on one CPU thread. It excludes inherited network witness fitting, imports, and setup. This is an absolute-accuracy *verification* frontier, not an all-in training comparison or an external matched-accuracy superiority claim. Raw stage clocks are retained.

TABLE III

SEPARATE UPPER-BOUND ALLOWANCES AT $k = 2$

Boxes	Source remainder	Control gap	Variance allowance	Localization
4,096	0.000339131	0.000063959	0.000081080	0.000000247
16,384	0.000082675	0.000041416	0.000081080	0.000000247
65,536	0.000020398	0.000032100	0.000081080	0.000000247

Source remainder is the Taylor allowance on the covered Gaussian region. Analytic Gaussian tails, interval weight minimal evaluation, and the covariance interval are included in the full records. The variance allowance does not arise merely by subdividing quadrature boxes. No row is obtained by treating a mesh difference as a rigorous error estimate.

Finest upper bounds are approximately -1.26279687 , -1.28530724 , and -1.31820708 . Combined with the feasible-policy lower endpoints, they give regret upper bounds .00987440, .00971727, and .00996896 after outward display rounding. The central target is attained without candidate-action grid, an Euler-error assumption, or an assumption that the policy is within its neural class. The preceding wealth-independent witness and its larger gaps are noted in Appendix D, where their loss of wealth information can be compared with the construction.

the bound applies to the stored deployment constants themselves. A further perturbation quantifies its robustness to control evaluation error, rather than inferring robustness from a full training loss.

PROPOSITION 2—Feasible-output error allowance: *Consider two deterministic time-control policies with $p = 0$, $c, \tilde{c} \in [.7, .8]$, $\theta, \tilde{\theta} \in [0, .2]$, and strictly feasible budgets as in (59). Let*

$$\delta_c = \int_0^1 |c_t - \tilde{c}_t| dt, \quad \delta_\theta = \int_0^1 |\theta_t - \tilde{\theta}_t| dt.$$

For original stopped payoffs satisfy

$$|J_k(c, \theta) - J_k(\tilde{c}, \tilde{\theta})| \leq (3 + .2e^{.02})\delta_c + (25 + .2k + .032)\delta_\theta + 8(14.1 + .02k)e^{-72}. \quad (25)$$

Subsequently, a validated output error bounded by the right side is an additive allowance for the returned policy's regret certificate. Feasibility must still be checked separately; small unconstrained control error alone does not rule out early wealth exit.

For the stored policies the output allowance is zero: their binary64 constants are treated as exact by the evaluator. Proposition 2 instead gives an explicit interface for a budget-preserving implementation with perturbed outputs. Together with the neural-jet bound of Theorem 2, it reflects observable implementation errors to certified quantities without attributing an unproved local convergence rate to Adam, RMSprop, or L-BFGS-B.

5. CERTIFYING THE VALUE OF PREFERENCE ADJUSTMENT

For each feasible policy write $J_k(\pi) = A(\pi) - kB(\pi)$, where

$$B(\pi) = \mathbb{E}^\pi \int_t^\tau e^{-\rho(r-t)} \theta_r^2 / 2 dr, \quad 0 \leq B(\pi) \leq .02C_\rho(T - t).$$

policy set is common to every k .

THEOREM 4—Accuracy-calibrated cost comparisons: *The optimal value is convex and increasing in k . If optimizers exist, their adjustment budgets are nonincreasing across distinct prices. More generally, let π_i be ε_i -optimal at k_i , with $k_2 > k_1$. Then*

$$B(\pi_2) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \quad (26)$$

For $k \in [\underline{V}(k), \bar{V}(k)]$, the budget of an optimizer at k , for $k_- < k < k_+$, lies in the primitive regret range intersected with

$$\left[\frac{\underline{V}(k) - \bar{V}(k_+)}{k_+ - k}, \frac{\bar{V}(k_-) - \underline{V}(k)}{k - k_-} \right]. \quad (27)$$

The theorem concerns discounted adjustment expenditures, not a pointwise ordering of controls. A re-solved V_u/k is not a comparative-static proof. The new full-value enclosures enable I now resolve strict welfare losses across the cost grid. At $k = 2$, (27) also gives the

lower endpoint is informative, while its upper endpoint is still the primitive bound. The enclosed expenditures in Table V describe the specified learned-and-polished policies, identified optimal budget.

5.1. A market-deflator witness for the restricted optimum

Inter comparison becomes possible by exploiting the restricted economy's structure. The auxiliary process

$$dY_t = .02Y_t dt - .3Y_t dZ_t^x, \quad Y_0 = 1.55. \quad (28)$$

is a martingale process constructed from the original shocks. It does not restrict portfolio information, wealth, or add a traded asset. For

$$W_0(u, x, y) = y(x - .5) + .1 \log(.5) - .02(u - 2)^2,$$

terms in the generator of Yx cancel exactly. With $\theta = 0$, define

$$\Psi(u, y) = \max_{.05 \leq c \leq .8} \left\{ \frac{c^{1-u}}{1-u} - yc \right\}, \quad (29)$$

$$S(u, y) = \Psi(u, y) + .01y - .00005 - .004 \log(.5) + .0008(u - 2)^2. \quad (30)$$

For $u \in [1.5, 2.6]$, $y \in [.2, 12]$, one has $-8 < S < 0$. Extend S to a bounded function $\tilde{S} \in [-8, 0]$ by setting its two arguments at these verification faces. Let

$$F_R(h, u, y) = \mathbb{E}_{u,y} \int_0^h e^{-.04r} \tilde{S}(U_r, Y_r) dr, \quad (31)$$

where $U_r = .05 dZ_r^u$ and Y follows (28). Thus $-8h \leq F_R \leq 0$.

LEMMA 5—A restricted-economy upper bound: *Let Z_R be the sum of the following adaptive localization potentials:*

$$Z_{R,u} = 8h \{ e^{-200(u-1.5)+50h} + e^{-240(2.6-u)+72h} \}, \quad (32)$$

$$Z_{R,y} = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (33)$$

$$V_R(0, 2, 1.25) \leq W_0(2, 1.25, 1.55) + F_R(1, 2, 1.55) + Z_R(1, 2, 1.55). \quad (34)$$

This upper bound covers all adaptive original consumption and portfolio controls with $\theta = 0$. Equivalently, for any feasible flexible policy with certified payoff lower bound $\underline{J}_k$ and any upper bound $\bar{V}_R$ from (34),

$$V_k(0, 2, 1.25) - V_R(0, 2, 1.25) \geq \max\{0, \underline{J}_k - \bar{V}_R\}. \quad (35)$$

The expectation in (31) has an explicit two-Gaussian representation. We transform time by $\tau = ht$ to evaluate constant and linear Taylor terms with interval-enclosed weighted moments, and the Hessian remainder on each box. Exponential Taylor remainders, Gaussian tails,

	Weighted boxes	Dual witness-value enclosure	Width	Seconds
12	16384	$[-1.33876765, -1.33309541]$	0.00567	2.67
	131072	$[-1.33676857, -1.33544680]$	0.00132	15.13
	1048576	$[-1.33631272, -1.33599309]$	0.00032	110.24

se intervals enclose the *dual witness value*, not the restricted optimum. Only their upper endpoints are used as per bounds for that optimum. A separately evaluated restricted policy gives the primal lower bound. Original stopping and all adaptive consumption and portfolio opportunities are retained.

TABLE V

CERTIFIED CURRENT-POLICY EXPENDITURES AND OPTIMAL VALUE OF ACCESS

k	Policy adjustment budget	Optimal access-value interval	Regret target
0.5	[0.01757696, 0.01757697]	[0.06332184, 0.08224255]	< .01
2	[0.01241866, 0.01241867]	[0.04096859, 0.05973218]	< .01
8	[0.00192681, 0.00192682]	[0.00781706, 0.02683234]	< .01

expenditure column evaluates the displayed learned policy. The access columns bound the difference between the o *optimal* values. The lower endpoint pairs a feasible flexible policy with the restricted dual upper; the upper endpoint pairs the flexible upper with a feasible restricted policy. All endpoints are rounded outward from precision records. The cost-two access interval has width below its entire positive gain; at cost eight, the interval still resolves the sign but not a comparably precise magnitude.

TABLE VI

ACCURACY-CALIBRATED OPTIMAL WELFARE LOSSES FROM HIGHER ADJUSTMENT COSTS

Cost change	Optimal welfare loss	Interval width	Width below loss
0.5 $\rightarrow$ 2	[0.01263598, 0.03222764]	0.01959166	Sign resolved
2 $\rightarrow$ 8	[0.02318257, 0.04286880]	0.01968623	Yes
0.5 $\rightarrow$ 8	[0.04553582, 0.06537918]	0.01984335	Yes

interval encloses $V_{k_1} - V_{k_2}$ in the unchanged original economy. Bounds use independent value endpoints, not fitted derivatives in k . The last column compares the width with the lower endpoint of the effect interval.

localization are included. This produces the executed refinement in Table IV; it does not eal to a mesh difference as an error estimate. The final restricted-optimum upper bound 1.33599309 after outward display rounding. The evaluated restricted policy has payoff $[-1.34503942, -1.34503585]$, so the restricted problem itself has an initial optimal-value osure narrower than .00905.

he welfare lower bounds remain strictly positive at all three prices. At $k = 2$ the optimal ss gain lies in $[\text{.04096859}, \text{.05973218}]$; its width is smaller than the lower endpoint. Table VI rately compares the flexible optima at different adjustment prices. The $2 \rightarrow 8$ and $.5 \rightarrow 8$ cts exceed their entire interval widths. The $.5 \rightarrow 2$ comparison resolves the sign but has a er relative uncertainty. No comparison assumes that a displayed policy is an optimizer or rprets the value-ordering theorem as a pointwise ordering of adjustment controls.

der the following minimization problem, which is also used for the external comparison:

$$dX_t = 2a_t dt + \sqrt{.32} dW_t, \quad X_0 = 0, \quad a_t \in [-1, 1]^d, \quad (36)$$

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$$f_d(a) = \frac{1}{d} \sum_{i=1}^d \{a_i^2 + .1 \sin^2 a_i\} + .3 \sin \left(\sum_i a_i \right), \quad (37)$$

$$q_d(x) = \frac{1}{d} \sum_{i=1}^d (x_i - .5)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right), \quad (38)$$

$$V_d^* = \inf_a \mathbb{E} \left[\int_0^1 f_d(a_t) dt + q_d(X_1) \right].$$

ective has genuinely coupled nonconvex actions. No analytic optimal-action or optimal-
bels are supplied to training. At a fixed critic gradient p , the action-dependent Hamilto-

$$H_p(a) = \sum_i f_i(a_i) + .3 \sin z, \quad f_i(s) = \frac{s^2 + .1 \sin^2 s}{d} + 2p_i s, \quad z = \sum_i a_i.$$

is strongly convex with modulus at least $m = 1.8/d$.

LEM 6—Scalar global partition and convex dual bounds: *For any multiplier λ and d], put*

$$D(\lambda) = \sum_i \min_{|s| \leq 1} \{f_i(s) - \lambda s\}.$$

ular interval $z \in [z_0 - r, z_0 + r]$, a lower bound for the full Hamiltonian is

$$D(\lambda) + \lambda z_0 + .3 \sin z_0 - |\lambda + .3 \cos z_0| r - .15 r^2. \quad (39)$$

ible action supplies an upper bound. Each inner convex minimum can itself be bounded
y $q(s_0) - e(s_0)^2/(2m)$, where $q = f_i - \lambda s$ and e is the derivative magnitude in a feasible
direction, or the full derivative magnitude at an interior point. Outward versions of
unds, followed by scalar subdivision and pruning, give a global gap on $[-1, 1]^d$.
ored gradient $\hat{p}$ differs from the true neural gradient by componentwise bounds δ_i , the
action's true-gradient gap is at most its stored-gradient gap plus $4 \sum_i \delta_i$.

ng-point root finding only proposes multipliers and actions. It is not trusted to solve
ex subproblems exactly. Even an inaccurate proposal remains valid after the strong-
y allowance. This structure removes tensor enumeration; it does not establish a
on-free complexity theorem or solve a generic nonseparable nonconvex Hamiltonian.
28-dimensional 10^{-5} calculation uses 339 scalar boxes and 43,392 scalar convex
lems. It replaces neither raw actor training nor the preceding finite tensor safeguard in
ct. At the learned eight- and sixteen-dimensional jets, interval forward differentiation
ored critic weights bounds the ℓ^1 gradient discrepancies below 4.584×10^{-17} and
 10^{-17} , respectively. These allowances are propagated into the action gap. Theorem 2
s the additional state-cover and residual quantities that would be needed for a whole-
uarantee.

Action dim.	Tolerance	Verified gap	Scalar boxes	Convex subproblems	Seconds
8	1e-03	0.000258	33	264	0.299
16	1e-05	6.72e-06	47	376	0.493
16	1e-03	0.000931	37	592	0.476
16	1e-05	4.42e-06	57	912	0.751
32	1e-03	0.00068	85	2720	1.498
32	1e-05	3.52e-06	103	3296	1.991
64	1e-03	0.000647	163	10432	6.041
64	1e-05	3.06e-06	185	11840	5.955
128	1e-03	0.000948	317	40576	19.229
128	1e-05	4.68e-06	339	43392	20.651

rozen jet is used per dimension, at both tolerances. At $d = 8, 16$, these are gradients of retained learned critics at recorded states; displayed gaps include interval neural-derivative allowances. At larger dimensions they are independently generated jets for the same Hamiltonian family. The experiment measures certificate work for these instances, not an average-case scaling law or a whole-state neural PDE residual.

7. AN ABSOLUTE BOUND AND A FROZEN EXTERNAL COMPARISON

7.1. A rigorous lower bound for the unchanged nonconvex family

et $r_0 = 1 + .1 \sin^2(1)$, $\sigma^2 = .32$, and $\lambda_d = \sigma^2 r_0 / (2d)$. Since $\sin^2 a \geq \sin^2(1)a^2$ for $|a| \leq 1$, facing the running cost by $r_0 \|a\|^2 / d - .3$ and allowing unbounded controls gives a relaxation. original terminal function q_d is retained.

PROPOSITION 3—Scalar-integral absolute lower bound: *Let Z be normal with mean 0 and variance σ^2 . Then*

$$V_d^* \geq \frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda_d \log \mathbb{E} \exp \left[-\frac{.2}{\lambda_d} \cos Z \right]. \quad (40)$$

outward enclosure of the positive expectation yields a rigorous numerical lower bound for original objective (38).

he relaxation is an upper/lower-bound device, not a new training problem with a known manufactured value. The terminal cosine, which couples all state coordinates, remains in the ar expectation. Its tails are bounded analytically and its interior is integrated by interval angles.

7.2. Holdout design and exact deployment interpretation

he comparison retains the original actor and critic architectures: two tanh hidden layers of 48 units each, a bounded actor output, and the same terminal-value embedding. NBO separates evaluation and improvement with Adam, three critic steps per actor step, 256 training states, and four successors. SOC-MartNet uses its pinned author solver, RMSprop, 500-output test network, and the inherited $J = 2$, $K = 1$ multiplier setup. These different minimization systems are disclosed rather than described as identical implementations. The committed protocol freezes dimensions 8, 16, 32 and training budgets of 10 and 30 runs. At each dimension, both methods receive the same six validation trials: two seeds

d	Quadrature cells	Relaxed-value enclosure	Original-value lower	Seconds
8	1024	$[-0.08599082, -0.08384626]$	-0.08599082	0.260
8	4096	$[-0.08518701, -0.08465088]$	-0.08518701	0.991
8	16384	$[-0.08498598, -0.08485194]$	-0.08498598	4.506
16	1024	$[-0.20520651, -0.20360335]$	-0.20520651	0.271
16	4096	$[-0.20460665, -0.20420585]$	-0.20460665	1.067
16	16384	$[-0.20445644, -0.20435623]$	-0.20445644	4.212
32	1024	$[-0.28757809, -0.28684340]$	-0.28757809	0.245
32	4096	$[-0.28730394, -0.28712026]$	-0.28730394	0.988
32	16384	$[-0.28723515, -0.28718923]$	-0.28723515	4.172

loses the relaxed value. Its lower endpoint is also a lower bound for the original bounded-control value. The shown by successive enclosures is quadrature uncertainty; it does not include or estimate the structural relaxation gap.

TABLE IX

FROZEN HOLDOUT COMPARISON: TWELVE SEED PAIRS PER PANEL

d	Budget	NBO cost	SOC cost	LQ cost	Median difference	Adjusted p
8	10	0.5812	1.0842	0.5121	-0.4783	0.001465
8	30	0.5785	1.0557	0.5121	-0.4348	0.001465
16	10	0.4663	0.7893	0.1740	-0.1672	1
16	30	0.4630	0.8456	0.1740	-0.2878	0.438
32	10	0.3813	1.2011	0.0854	-0.8691	0.001465
32	30	0.3799	1.0218	0.0854	-0.6962	0.001465

realized cost is better. The median difference is NBO minus SOC-MartNet. The last column is a one-sided sign-test p -value adjusted across the six predeclared dimension–budget panels. The sign test concerns the joint training-and-audit pipeline under independent seed draws, not an outward numerical error guarantee.

se learning-rate multipliers. Selection minimizes mean independent validation cost, fixed tie rule. Twelve different holdout seeds, 720–731, are then evaluated at each on–budget pair. Within each pair, actor and critic starting weights are identical and shes are checked. Framework warmup precedes both clocks, and method order is seed-ized. Training-only and setup-plus-training clocks are retained. The baseline $a_i(t, x) = \frac{1}{1 + 4(1 - t)}[-2(x_i - .5)]$ is specified independently from the associated quadratic and is evaluated on the unchanged nonquadratic objective.

trained network is deployed as a sample-and-hold control on 128 dates and evaluated 96 independent, paired Gaussian paths. In (38), drift under a held action and volatility tant on a slab; the Gaussian transition and running-cost integral are therefore exact in nnetic for *that deployed continuous-time policy*. A 64-date evaluation uses aggregated increments and changes the deployment policy. Its difference from the 128-date result is d an Euler-error bound. Floating-point network evaluation and Monte Carlo uncertainty distinct from the outward deterministic lower bound in Table VIII.

holdout results favor NBO over the author implementation in mean cost in all six panels, r interpretation is not uniform. In dimensions 8 and 32, NBO has lower paired cost twelve seeds at both budgets; each multiplicity-adjusted sign-test value is .001465. In on 16 the adjusted values are 1 and .438, so these panels do not establish a robust el ordering under the declared test. Tripling the budget gives only small improvements mean cost, from .5812 to .5785, .4663 to .4630, and .3813 to .3799 respectively. Most ntly, the independently specified quadratic-feedback baseline has lower mean cost than rned methods in every panel. Its mean costs are .5121, .1740, and .08544 in dimensions

d	Budget	Lower bound	NBO excess	SOC excess	NBO sec.	SOC sec.
8	10	-0.08499	0.6662	1.1692	10.002	10.006
16	30	-0.08499	0.6635	1.1407	30.004	30.008
16	10	-0.20446	0.6707	0.9938	10.002	10.005
16	30	-0.20446	0.6674	1.0500	30.004	30.006
32	10	-0.28724	0.6686	1.4883	10.003	10.007
32	30	-0.28724	0.6672	1.3090	30.003	30.007

cess is the estimated mean policy cost minus the rigorous original-value lower bound. It provides an absolute reference axis but is not itself a deterministic regret enclosure: sampling error and floating-point rollout evaluation are outward certified. Training clocks include completed-update overshoot and exclude separately recorded warmup and setup.

5, and 32. These retained results support the stated NBO–SOC–MartNet comparison at the ten budgets, not a claim that NBO is the best available control rule or that additional training already reached an accurate optimum.

The raw records retain every seed difference, the median and interquartile range, approximate Student intervals, costs for all three rules, path standard errors, training curves, optimizer clocks, checkpoints. Approximate Student intervals are descriptive; the exact sign calculation avoids mixing Gaussian seed-level differences. The source protocol predates holdout execution, while the benchmark family and adapters themselves were developed in earlier revisions. Holdouts do not turn one problem family into evidence of general superiority across economic environments. Two budget points are not a matched-accuracy frontier, and a finite absolute lower bound need not be tight enough to rank methods at a small prescribed regret.

8. METHOD-SPECIFIC TUNING AND AN INDEPENDENT REPLICATION

The preceding holdout answers a precisely frozen comparison, but its scalar learning-rate choice leaves an important design question: does the ordering survive a more consequential method-specific tuning effort? We answer this question with a new experiment, rather than testing new hyperparameters on the already observed holdout or suppressing the earlier results. The objective, diffusion, action box, horizon, equal actor/critic initializations, author solver limit, and 128-date deployed policy are unchanged. The new protocol fixes tuning seeds 1600–1601 and holdout seeds 1820–1831 before execution. None of these is an R9 holdout seed.

8.1. *Prospectively specified equal tuning opportunities*

For each dimension and method, twelve initial configurations and two prospectively specified dimension configurations are each trained for five nominal seconds on two tuning seeds. Thus each method receives 140 nominal training seconds per dimension. Completed-update overshoot, initialization, validation, and holdout clocks are recorded separately. The methods receive equal nominal training opportunity, not equal numbers of gradient operations or an assertion that their hyperparameter spaces have been optimized.

The shared learning-rate multiplier set expands from $\{1/3, 1, 3\}$ to $\{1/3, 1, 3, 9, 27\}$. NBO additionally varies evaluation updates per actor improvement and transition particles. SOC–MartNet additionally varies the adversary learning rate, the number of sinusoidal test outputs, the solver’s J/K parameters, and the initial and updating multiplier parameters. The complete configuration lists appear in the frozen executable protocol; unchanged settings and unsuccessful candidates are not omitted. The pinned author training implementation is unmodified.

d	NBO selected	SOC selected	NBO sec.	SOC sec.
8	3 ($m = 9$)	11 ($m = 3$)	140.15	140.29
16	12 ($m = 3$)	2 ($m = 3$)	140.09	140.18
32	1 ($m = 1$)	3 ($m = 9$)	140.17	140.31

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tion indices are zero based and bind to the complete frozen lists and extension records. m is the actor/critic rate multiplier. Actual seconds sum all 28 tuning fits per method and dimension, including completed-update overshoot but excluding separately retained validation and setup.

order is randomized within tuning seed and candidate, and actor and critic initial hashes are preserved within each holdout pair.

the first twelve configurations, selection minimizes the mean validation cost over the tuning seeds, with an index-based tie rule. A selected multiplier at the current high or low boundary is extended by a factor of three in that direction. If the multiplier is interior, the multiplier slot varies transition-particle count for NBO or adversary rate for SOC. Selection is extended before the second extension. These rules are specified before execution and do not change on the new holdout. Failed tuning trials receive infinite selection score while their raw scores are preserved. Validation uses 1,024 common Brownian paths per tuning seed, and evaluation uses 4,096 new common paths per seed. All recorded outcomes in this experiment are finite, and all 168 tuning trials and 36 holdout pairs are retained.

the selected learning-rate multipliers are interior to the enlarged tested multiplier set. At $d = 8$, NBO selects $m = 9$, while SOC selects $m = 3$ with initial multiplier and multiplier step of 3. At $d = 16$, NBO selects six evaluation steps and sixteen transition particles with $m = 3$; SOC selects its default method-specific parameters with $m = 3$. At $d = 32$, NBO selects $m = 1$ and SOC selects $m = 9$, with the respective default remaining parameters. The large learning-rate boundary identified in the earlier report is therefore tested beyond its old limit. Interiority on this finite set does not establish global tuning optimality or exclude improvements from unsearched interaction. The experiment tests a finite, disclosed method-specific tuning

8.2. Independent holdout and protocol sensitivity

selected configurations are fixed before their twelve new holdout pairs are run at a fixed and nominal training budget. The primary test is again the one-sided exact sign test, with Bonferroni correction across the *three new* predeclared panels. This does not change the panel correction of the earlier experiment. Because the historical evidence motivated the experiment, the new study is a prospective post-review replication, not a retroactive variation of the original benchmark design.

NBO mean differences are $-.23863$, $-.35199$, and $-.07623$ in dimensions 8, 16, and 32, with corresponding paired win counts are 11, 8, and 7 out of twelve, with adjusted sign-test p -values of .00952, .58154, and 1. Only the eight-dimensional panel establishes the declared robust ordering at level .05. In dimension 16, four seeds again favor SOC. In dimension 32, extended tuning yields a substantially smaller mean difference and no significant directional effect, although the earlier frozen experiment favored NBO in all twelve seeds. The improved result is consistent with its selected $m = 9$, which the old scalar grid excluded; this conclusion is not an isolated causal estimate of that hyperparameter because tuning, validation, and holdout seeds all differ.

d	NBO	SOC	LQ	Mean diff.	Median diff.	Wins	Adj. p
8	0.5542	0.7928	0.5125	-0.2386	-0.1967	11/12	0.00952
16	0.4547	0.8067	0.1741	-0.3520	-0.1212	8/12	0.58154
32	0.3824	0.4586	0.0852	-0.0762	-0.0427	7/12	1.00000

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er realized cost is better. Differences are NBO minus SOC, using paired audit paths and twelve independently ed training-and-audit pipelines. The last column applies the prospectively fixed three-panel sign-test correction. are sampled policy evaluations, not outward deterministic certificates. The LQ column retains the independently specified clipped feedback under the unchanged nonconvex objective.

ne independently specified clipped-feedback policy has lower mean realized cost than either ed pipeline in each new panel, as it did in the previous experiment. Its favorable performance not make its cost a lower bound for a minimization problem, and it does not prove global mality. These outcomes distinguish three questions: whether a configured NBO pipeline roves on a configured external solver, whether the ordering is robust to a changed tuning ocol, and whether either neural policy has small absolute error. The new experiment supplies ct evidence for the second question and retains all evidence relevant to the first; Section 9 lies a separate exact-reference experiment for the third. Favorable absolute-accuracy results oother model are not used to erase the sixteen- or thirty-two-dimensional sign-test outcomes .

he evidence pipeline collects with an unconditional completion policy. Every planned cell xplicit status before dependency installation; successful outputs, failed outputs, and missing-metadata are versioned before aggregate validity is checked. Separate immutable commits tify the frozen study source and completed raw results. No branch name alone is used as a ew target, and the 36 new pairs neither replace nor pool selectively with the 72 earlier pairs.

9. FROM IMPLEMENTED ACTOR ERRORS TO ABSOLUTE-ACCURACY FRONTIERS

he original stopped economy establishes economically resolved optimality bounds. A com- entary experiment asks a different question: how much computation is required to attain mmon absolute accuracy when the continuous optimum is independently known? This riment changes neither the economy in Section 2 nor the coupled nonconvex comparison. It s reference-solvable mechanisms with different drift, noise, and control-cost conditioning. exact reference is not replaced by the best realized cost of a competing policy. onsider independent coordinates with

$$\begin{aligned}
 dX_t^j &= (a_j X_t^j + b_j u_t^j) dt + \sigma_j dW_t^j, \\
 J(u) &= \frac{1}{d} \mathbb{E} \left[\int_0^1 \sum_{j=1}^d (q_j (X_t^j)^2 + r_j (u_t^j)^2) dt + \sum_{j=1}^d w_j (X_1^j)^2 \right], \tag{41}
 \end{aligned}$$

re $q_j \geq 0$, $r_j, w_j > 0$, and $b_j > 0$. Controls are unbounded, adapted, and square integrable. action constraints and nonconvex running cost of the external comparison are *not* silently osed on this linear-quadratic problem. Rather, the latter supplies an independently solvable racy experiment. The implemented actor observes $X_{i/N}$ and holds $u_t = -K_i X_{i/N}$ until next decision date. It is a linear output map with time gates, not a continuous-feedback oyment of $-K_i X_t$ and not the width-48 tanh actor used in the external comparison.

$$V^* = \frac{1}{d} \sum_j \left[p_j(0) (X_0^j)^2 + \sigma_j^2 \int_0^1 p_j(t) dt \right] \quad (42)$$

any admissible control satisfies

$$J(u) - V^* = \frac{1}{d} \sum_j \mathbb{E} \int_0^1 r_j \left(u_t^j + \frac{b_j p_j(t)}{r_j} X_t^j \right)^2 dt. \quad (43)$$

division interval $h = 1/N$, let J_h^* be the optimum over held-action rules, with Riccati matrices P_{ij} , optimal gains K_{ij}^* , and positive action curvatures D_{ij} specified in Appendix A. The implemented held-action rule, including a nonlinear neural rule,

$$J(u) - V^* = (J_h^* - V^*) + \frac{1}{d} \sum_{i,j} D_{ij} \mathbb{E} [(u_i^j + K_{ij}^* X_{ih}^j)^2]. \quad (44)$$

Recently, output-error bounds $\mathbb{E}[(u_i^j + K_{ij}^* X_{ih}^j)^2] \leq \varepsilon_{ij}^2$ give a quantitative optimization certificate $d^{-1} \sum_{i,j} D_{ij} \varepsilon_{ij}^2$, in addition to the separately evaluated deployment error $J_h^* - V^*$. Convergence assumption on Adam or on shared neural parameters is used.

This proposition is a specialization of quadratic verification and Bellman completion of squares, a new general Riccati theorem. Its role is to connect an *implemented* policy error to an economic objective. The finite-horizon quadratic Bellman construction requires exactly d outward improvements per coordinate. In this output class, evaluation and improvement are separate operations, and the backward recursion solves the held-action problem rather than merely minimizing a stochastic training loss. The alternative Adam implementation optimizes the same outputs using an independently coded exact-moment objective. Its errors are assessed after d rounded updates; a universal stochastic-optimization convergence theorem is not inferred.

9.1. Independent reference and deployment verification

$c_j = b_j^2 / r_j$, $\lambda_j^2 = a_j^2 + c_j q_j$, and $s = 1 - t$. Define

$$A_j(s) = \cosh(\lambda_j s) + \frac{\sinh(\lambda_j s)}{\lambda_j} (-a_j + c_j w_j),$$

$$B_j(s) = w_j \cosh(\lambda_j s) + \frac{\sinh(\lambda_j s)}{\lambda_j} (q_j + a_j w_j).$$

The continuous Riccati solution is $p_j(t) = B_j(1-t)/A_j(1-t)$, with the continuous limit when $d \rightarrow \infty$. Its integral equals $(\log A_j(1) + a_j)/c_j$. Thus the reference needs no time-stepping solver or discretized paths. Sixty-decimal interval arithmetic encloses the exponentials, logarithms, and divisions. Every coefficient in the certificate is the exact binary64 number stored in the instance. Every gain is the exact stored policy output. Exported interval endpoints are enlarged by one binary64 unit in the outward direction.

Case	B: $N = 16$	B: $N = 32$	B: $N = 128$	Adam (400)
Tracking	0.00839748	0.00409984	0.00100625	0.00409984
Mean-reverting	0.00429055	0.00211098	0.00052088	0.00211101
Ill-conditioned	0.01421242	0.00699643	0.00172851	0.00699643

entry is rounded upward from an interval certificate of the original continuous cost of the stored actor. Columns note quadratic Bellman construction at the stated decision count. Adam uses 32 decisions and 400 updates. These are structured-policy accuracy results, not a substitution for the external neural comparison.

different calculation evaluates the original continuous cost of each held-action policy. Conditional on the state at a decision date,

$$X_{ih+s}^j = \left(e^{a_j s} - b_j K_{ij} \int_0^s e^{a_j v} dv \right) X_{ih}^j + \sigma_j \int_0^s e^{a_j(s-v)} dW_{ih+v}^j.$$

ct second-moment propagation and the integrated running cost therefore determine $J(u)$. Appendix A gives all coefficients and the verification proof. This is not an Euler approximation, a difference between two meshes is not used as an error estimate. In particular, the reported interval $[\underline{J} - \overline{V}, \overline{J} - \underline{V}]$ encloses absolute regret in (41), including deployment error. Stored policies, model coefficients, source hashes, and the two independent evaluators are supplied in replication package.

9.2. Reference-solvable mechanisms and executed accuracy

he prospective protocol specifies three cases. Tracking has $d = 8$, $a = 0$, $b = 2$, $q = 0$, $w = 1$, initial centered state $-.5$, and $\sigma = .5656854249492381$. The mean-reverting case has $d = 8$, $a = -1$, $b = q = w = 1$, $r = .25$, $\sigma = .4$, and initial state one. The ill-conditioned case has $d = 16$ and linearly spaced $a \in [-1, .5]$, $b \in [.5, 2]$, $q \in [.1, 2]$, $\sigma \in [.15, .65]$, $w \in [.5, 2]$, initial states in $[-1, 1]$, with geometrically spaced $r \in [.1, 4]$. The last case contains both stable and unstable uncontrolled coordinates. Exact stored coefficients, rather than rounded points in this description, define each computation.

he quadratic Bellman constructor is evaluated at $N = 4, 8, 16, 32, 64, 128$. Adam uses $N = 32$, initially zero gains, learning rate $.05$, standard moment coefficients $(.9, .999)$, and checkpoints at 0, 25, 100, and 400 updates. The deterministic optimizer configuration is recorded for the executed instances; no tuning search is performed for this reference suite. No holdout set is attached to these analytical test cases. All checkpoints are retained, including those that exceed a stated accuracy threshold.

Table XIII shows that refinement reduces deployment error in every case. At 128 decisions, three absolute regret upper bounds are below $.003$. At 32 decisions, all three are below $.01$. Adam approaches the same 32-decision optimum: increasing its updates cannot eliminate the deployment floor at that fixed decision count. This distinction is observed directly, not inferred from a nearly flat training curve. The $.001$ target is attained on the planned mesh sequence for the mean-reverting case, but not for tracking or the ill-conditioned case; these outcomes remain in the package rather than being removed from the frontier.

the accuracy axis is the same original-cost regret for both policy generators within each mesh. The computation axis includes rejected earlier meshes or checkpoints rather than reporting only the final inexpensive verification. The exact quadratic constructor uses analytic model

Case	Method	.01 (sec.)	.003 (sec.)	.001 (sec.)
Tracking	B	0.023	0.082	—
Tracking	A	0.626	—	—
Mean-reverting	B	0.015	0.053	0.189
Mean-reverting	A	0.641	2.380	—
Ill-conditioned	B	0.109	0.395	—
Ill-conditioned	A	0.741	—	—

includes all earlier generation and certificate checks on the indicated path, but excludes dependency installation and time optimizer construction. B is the quadratic Bellman constructor; A is Adam on gain outputs. A dash indicates the target was not attained among the predeclared checkpoints. Timing is a single-machine descriptive measurement, not a confidence interval or a hardware-independent complexity comparison.

We show that the general tanh and SOC implementations do not exploit; its faster times do not establish a neural-versus-SOC efficiency ranking. Together with the unchanged stopped verification and coupled nonconvex family, the added cases test first-exit verification, adapted nonconvex coupling, mean reversion, and differing control-cost conditioning. They also supply new recursive-utility or game experiments. Those extensions retain their explicit descriptions and complete historical material.

10. RELATION TO RIGOROUS CONTROL AND NEURAL COMPUTATION

The strongest claims should be compared with methods that establish bounds, not only with those that train networks. Theorem 1 is an application of classical stopped verification to upper and lower witnesses. Its role here is to handle the actual liquidation trace without assuming a smooth or boundary-regular optimal value. Theorem 2 supplies a computation-over, derivative, and oracle allowance. Neither theorem claims that classical verification is new from the literature.

Stable, consistent approximations with comparison provide convergence to viscosolutions in the framework of Barles and Souganidis (1991). Controlled Markov-chain approximation is developed systematically by Kushner and Dupuis (2001); adaptive sparse-grid methods address high-dimensional dynamic models in Brumm and Scheidegger (2017). Such methods concern specified approximation schemes and their hypotheses. The finite telescoping in the preserved manuscript names the operator discrepancies needed for a continuous limit; an observed mesh difference does not supply them. The original-payoff certificates should evaluate continuous generators or exact Gaussian moments directly, with separately analytic and arithmetic remainders. They do not assert a convergence rate for the killed-Euler scheme.

Primal-dual numerical control methods also pair feasible decisions with bounds. Brown and Sun (2010) relax nonanticipativity with penalties and establish weak and strong bounds for stochastic dynamic programs. Brown and Smith (2011) use related bounds in portfolio optimization with transaction costs. Our restricted and flexible constructions are deflator witnesses with explicit stopped localization; they are not claims that policy-iteration-bound analysis is new. Theorem 3 adds a verified affine-preference source and a neural-coefficient variance allowance. Its distinction from an information-relaxation bound is that every source integral, variance constant, and face allowance are stated here. Lasserre, Henrion, Prieur, and Trélat (2008) develop occupation-measure and relaxations for polynomial nonlinear optimal control under their stated assumptions.

rolled-exit problem. The explicit source bounds, interval polynomial conversions, Gaussian, and trace inequalities are verified here instead.

Theorem 3, Proposition 5, and Theorem 5 identify the current and preceding model-specific constructions behind the economic bounds. The affine-preference theorem retains costly adjustment through an adapted conditional coefficient, bounds its randomness quantitatively, and meets the declared central-state target. The fitted generator proposition couples a polynomial access to a small localization potential; the restricted theorem cancels every portfolio term before computing a bounded-source expectation. Theorem 6 exploits rank-one action coupling and strong convexity of the remaining scalar problems. It is a structured convex-duality and reach-and-bound construction, not a generic replacement for global nonconvex optimization. Proposition 3 preserves the nonquadratic terminal payoff in a computable relaxation of the exact comparison problem.

Neural policy iteration and neural PDE solvers are important computational antecedents. Berta and Mijatović (2015) analyze continuous-time policy improvement. Kim, Cho, Kim, and Kim (2026) separate iteration, policy-network, and residual errors in entropy-regularized control; Kim, Kim, Kim, and Cho (2026) study interior bounds and the mismatch between localization and closed-loop performance. Lee and Kim (2025) combine policy iteration and generator learning. Cai, Fang, and Zhou (2025) impose martingale conditions without an explicit Hamiltonian infimum. NBO’s raw training comparison is with a pinned implementation of the method, not a locally recreated approximation described as author code. Our certificates also audit a policy produced by another solver. This portability is a property of independent verification, while the training results remain evidence about the specifically recorded NBO and C-MartNet adapters.

Recursive utility, temporal-self equilibria, and games require their own evaluation operators and optimality notions. Their complete equations, assumptions, finite gain bounds, failures, and other numerical results are preserved in the supplement. No new large-scale recursive or game theory result is inferred from the additive-utility certificates above. The stochastic-trace bias certainties and local-optimizer counterexamples are preserved for the same reason: they identify conditions that a neural training loss cannot replace.

11. CONCLUSION

The original continuous preference-adjustment economy now has certified central-state policies at the declared .01 regret tolerance for all three cost coefficients. A feasible learned-and-shed policy supplies each lower bound, while an independent affine-preference dual supplies an upper bound over the original continuous adaptive controls. This closes the numerical target without deleting the original stopping contract or replacing it by a manufactured-payoff problem. The value of access and the welfare cost of more expensive adjustment are resolved at a decision calibrated to the reported effects. At coefficient two the optimal access gain is at least .04096859; raising the coefficient to eight lowers optimal welfare by at least .02318257. The guarantees exceed their corresponding interval widths. Optimal adjustment budgets obey analytical ordering and a valid secant enclosure, but are not equated with the budgets of approximate policies.

The constructive tools also retain their distinct roles: fitted Bernstein witnesses, stopped Gaussian-moment evaluation, market-deflator localization, quantitative output and neural-allowance allowances, and a structured continuous-action oracle. The unchanged coupled comparison problem has an absolute lower reference and a complete frozen holdout record, including unfavorable baseline and sign-test outcomes. Those experiments do not establish a

reference experiments give certified continuous-model accuracy, implemented-output entities, and computation-to-accuracy paths rather than only fixed-budget losses. The tuning study assesses robustness under a second frozen protocol, not by retuning on a holdout. These additional results strengthen the preservation-compatibility link having each certificate attached to its actual model, policy class, and implementation. All manuscripts, equations, negative diagnostics, and raw outcomes remain preserved for examination.

APPENDIX A: CONTINUOUS REFERENCES, HELD-ACTION MOMENTS, AND OUTPUT-ERROR IDENTITIES

Appendix proves Proposition 4 and specifies the complete arithmetic used in Section 9. It is independent of the stopped-economy dual. All quantities here concern the linear-quadratic problem (41); no claim about a constrained nonquadratic optimum is obtained by replacing its solution with this one.

A.1. Continuous verification and the closed reference

We give a scalar proof; summation and division by d yield the stated result. The finite-horizon Riccati equation with $q \geq 0$, $r, w > 0$ has a nonnegative bounded solution. This can also be deduced directly from the displayed reference formula: it is the fractional transformation of the identity by

$$\frac{d}{ds} \begin{pmatrix} A(s) \\ B(s) \end{pmatrix} = \begin{pmatrix} -a & c \\ q & a \end{pmatrix} \begin{pmatrix} A(s) \\ B(s) \end{pmatrix}, \quad (A(0), B(0)) = (1, w), \quad c = b^2/r.$$

The matrix square is $(a^2 + cq)I$, giving the hyperbolic formula in the text. With $P(s) = A(s)^2/B(s)$, differentiation gives $P' = q + 2aP - cP^2$. Nonnegative initial data stay nonnegative, and the negative quadratic term prevents a finite-time upward explosion. Moreover $A' = (-a + cq)A$ and $A(0) = 1$, so A stays positive and

$$\int_0^1 P(s) ds = \frac{\log A(1) + a}{c}.$$

In the specified cases, $a = q = 0$, $A(s) = 1 + cws$, and $B(s) = w$. This branch is identified by its exact limit, not by division by a small interval containing zero.

Let $p(t, x) = p(t)x^2 + \sigma^2 \int_t^1 p(s) ds$, where $p(t) = P(1 - t)$. Direct substitution gives

$$v_t + (ax + bu)v_x + \frac{1}{2}\sigma^2 v_{xx} + qx^2 + ru^2 = r \left(u + \frac{bp(t)}{r}x \right)^2.$$

For a square-integrable adapted control on a finite horizon, the linear state equation has finite second moments. Localizing the Itô integral and passing to the limit as the horizon goes to infinity gives the verification identity (43); equivalently one can use it for bounded stopped controls and use the explicit linear state solution and L^2 martingale representation. The feedback $u = -bp(t)X_t/r$ is admissible, has finite second moments, and the square vanishes. It therefore attains the value in (42). This proves both directions of the inequality rather than using a feasible Riccati policy as a lower bound without justification.

or one interval of length h , put

$$E = e^{ah}, \quad I = \int_0^h e^{as} ds, \quad A_2 = \int_0^h e^{2as} ds, \quad B_2 = \int_0^h e^{as} \left(\int_0^s e^{av} dv \right) ds,$$

$$C_2 = \int_0^h \left(\int_0^s e^{av} dv \right)^2 ds, \quad D_2 = \int_0^h \int_0^s e^{2av} dv ds.$$

$a \neq 0$, these quantities are

$$I = (e^{ah} - 1)/a, \quad A_2 = (e^{2ah} - 1)/(2a),$$

$$B_2 = (A_2 - I)/a, \quad C_2 = (A_2 - 2I + h)/a^2, \quad D_2 = (A_2 - h)/(2a).$$

$a = 0$, they are $I = A_2 = h$, $B_2 = D_2 = h^2/2$, and $C_2 = h^3/3$. Sixty-decimal interval equation, with a separate zero-drift branch, encloses cancellation in these expressions. All real nonzero drift inputs in the finite suite are point intervals bounded away from zero; a division by an interval containing zero is rejected rather than repaired numerically.

Conditional on $X_{ih} = x$ and an action u held through the interval,

$$\mathbb{E}[X_{(i+1)h}^2 \mid x, u] = (Ex + bIu)^2 + \sigma^2 A_2,$$

$$\mathbb{E} \left[\int_{ih}^{(i+1)h} (qX_t^2 + ru^2) dt \mid x, u \right] = qA_2 x^2 + 2qbB_2 xu + (qb^2 C_2 + rh)u^2 + q\sigma^2 D_2.$$

formulas retain the correlation between the held action and the state at the beginning of the interval. They do not approximate the action by a continuous feedback evaluated along the evolving state.

the remaining value after this slab is $P_{i+1}x^2 + z_{i+1}$, define

$$D_i = qb^2 C_2 + rh + P_{i+1}b^2 I^2 > 0,$$

$$L_i = qbB_2 + P_{i+1}EbI, \quad K_i^* = L_i/D_i,$$

$$P_i = qA_2 + P_{i+1}E^2 - L_i^2/D_i,$$

$$z_i = z_{i+1} + q\sigma^2 D_2 + P_{i+1}\sigma^2 A_2, \quad P_N = w, \quad z_N = 0.$$

Bellman minimization is exact completion of the square. Nonnegativity of costs implies $D_i > 0$, so the strict inequality for D_i follows from $rh > 0$. It applies to the entire unbounded state space, not merely to a stationary point. These N backward steps are closed exact evaluation and improvement operations for quadratic values and linear held-action outputs.

For any adapted held-action rule, its conditional Bellman excess at stage i is $D_i(u_i + K_i^* X_{ih})^2$. Averaging the conditional expectations gives

$$J(u) - J_h^* = \frac{1}{d} \sum_{i,j} D_{ij} \mathbb{E}[(u_i^j + K_{ij}^* X_{ih}^j)^2].$$

Combining this equality with the continuous verification result proves (44). In particular, for an adapted linear actor $u_i^j = -K_{ij}^* X_{ih}^j$, the optimization error is weighted by the *deployed*

A.3. Independent interval policy evaluation

erifier does not reuse the backward value as the value of a stored floating-point policy.
 starting with $m_0 = x_0^2$, it propagates

$$\begin{aligned} m_{i+1} &= (E - bIK_i)^2 m_i + \sigma^2 A_2, \\ \Delta J_i &= \{q(A_2 - 2bK_i B_2 + b^2 K_i^2 C_2) + rhK_i^2\} m_i + q\sigma^2 D_2, \\ J(K) &= \sum_i \Delta J_i + w m_N. \end{aligned}$$

efficient and stored gain is imported as the exact binary64 value that defines the deployed
 nd actor. The arithmetic is outward at 60 decimal digits. Consequently the policy-cost
 and the closed-reference interval enclose two independently computed original-model
 es. Subtraction in interval arithmetic yields an absolute-regret interval; a negative upper
 t is an error and is not clipped to zero. Conversion to machine-readable binary64
 ts expands the lower endpoint toward $-\infty$ and the upper endpoint toward $+\infty$ by one
 e output files hash the little-endian binary64 gain array, binding every certificate to its

ifferentiable Adam objective is a separate torch implementation of the moment equations
 64. Its agreement with the interval evaluator is checked to a small conversion tolerance,
 agreement is only a software cross-check. The interval equations and continuous-
 e proof, not the tolerance in that test, establish the certificate. Additional tests check the
 ormula against a separately integrated Riccati equation, test the scalar tracking formula,
 nfinite gains, and verify that a deliberately perturbed actor changes both its policy hash
 . Timings and arithmetic intervals are kept separate: timing fluctuations cannot change
 a stored policy is certified at the prescribed tolerance.

APPENDIX B: PROOF OF THE AFFINE-PREFERENCE DUAL

appendix gives the analytic inequalities behind the new upper bound and identifies
 which finite interval calculations enter the proof. The frozen fitting objectives are not
 uted. The only fitted objects subsequently used are the exact stored values of y_0 and the
 eference drifts, whose admissible ranges are checked independently.

B.1. The supporting-plane inequality

$= u - 1, l = \log y$, and

$$L = \text{proj}_{[-\log(.8), -\log(.05)]}(l/u), \quad c = e^{-L}, \quad E = e^{Lr}.$$

umption maximizer is c , including both constrained branches. Write $\chi = 1$ on the
 branch and zero on a constrained branch; on a box intersecting a switching surface its

$$\begin{aligned}
\Psi &= -E/r - yc, & \beta &= E(1 - Lr)/r^2, \\
(S_0)_l &= -yc + .01y, \\
(S_0)_{uu} &= -\frac{E\{(Lr - 1)^2 + 1\}}{r^3} + \chi \frac{ycL^2}{u}, \\
(S_0)_{ul} &= -\chi \frac{ycL}{u}, \\
(S_0)_u &= -yc + \chi \frac{yc}{u} + .01y.
\end{aligned} \tag{45}$$

the interior branch $yc = E$. The envelope is continuously differentiable across the consumption constraint; the displayed one-sided second derivatives enclose its almost-everywhere Hessian. This is sufficient both for supporting inequalities and for the integral Taylor remainder below.

no lower-consumption constraint is active on $u \in [1.5, 2.5]$, $y \in [.2, 12]$, because $12/1.5 < -\log(.05)$. Moreover $L < 2$ throughout that rectangle. On its interior consumption branch,

$$\partial_l(S_0)_{uu} = \partial_u \beta_l = \frac{EL(2 - L)}{u^2} \geq 0. \tag{46}$$

on the upper-consumption branch $(S_0)_{uu}$ is independent of l and $\beta_l = 0$. The jump in $(S_0)_{uu}$ on increasing l enters the interior branch is nonnegative. When increasing u leaves that branch, β_l jumps upward to zero. Thus $(S_0)_{uu}$ is nondecreasing in l , and β_l is nondecreasing including the switching boundary.

The following enclosures are computed by the interval program on complete interval covers, by point sampling:

$$\sup_{1.5 \leq u \leq 2.3} (S_0)_{uu}(u, 12) < -.0862, \tag{47}$$

$$\sup_{2.3 \leq u \leq 2.5} \beta(u, 12) < -.9687, \tag{48}$$

$$\beta(2.2, 12) > -.9573. \tag{49}$$

The first cover has 1,600 intervals and the second has 2,000. All decimal endpoints and analytic constants are enclosed at 70-digit interval precision; all subsequent binary operations and interval hulls are rounded outward. Full endpoints and the finite cover prescription are stored in `agent_primitive_checks.json` and its generating program.

By (46), the first inequality proves concavity on $[1.5, 2.3]$ for every y in the verification range. For $u \in [2.3, 2.5]$ and $m \in [2, 2.2]$, the difference $\beta(u, y) - \beta(m, y)$ is nondecreasing since β_l is nondecreasing in u . Its maximum therefore occurs at $y = 12$. At that endpoint, concavity gives $\beta(m, 12) \geq \beta(2.2, 12)$, and the last two inequalities in (49) make the difference strictly negative. Integrating the derivative between m and u now proves (15) on both sides of the switching boundary. This argument does not infer a two-dimensional supporting plane from concavity at a single point without controlling the cross derivative.

The same program covers $m \in [2, 2.2]$ by 1,000 intervals and uses $(S_0)_l < 0$, $\beta_l \leq 0$ to check the two y endpoints. It obtains

$$-6.989 < S_0(m, y) < -1.2443, \quad |\beta(m, y)| < .9712. \tag{50}$$

B.2. An adapted solution of the affine auxiliary problem

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This subsection starts at $(0, 2, y_0)$ and uses the full-line auxiliary processes. All sources are in U , with bounded coefficients, and all controls are bounded. Fubini's theorem, optional expectation, and optional stopping on bounded rectangles are legitimate by Gaussian moment bounds. With

$$B_s = \int_s^1 e^{-\rho(t-s)} \mathbb{E}[\beta_t | \mathcal{F}_s] dt + e^{-\rho(1-s)} b_T,$$

independence of future Brownian increments gives $\mathbb{E}[\beta_t | \mathcal{F}_s] = \mathbb{E}[\beta_t | Y_s]$, even when control observes the full correlated Brownian history. In particular, B_s is an adapted function of Y and $\mathbb{E}B_s = b(s)$.

$$U_t - m(t) = \int_0^t (\theta_s - \bar{\theta}_s) ds + .05 Z_t^u.$$

Equation (16), followed by conditional Fubini, shows that its control-dependent part is

$$\mathbb{E} \int_0^1 e^{-\rho s} \{(\theta_s - \bar{\theta}_s) B_s - k\theta_s^2/2\} ds.$$

Pointwise maximizer $\theta_s = \text{proj}_{[-.2, .2]}(B_s/k)$ is adapted and bounded. It gives a well-defined auxiliary U because it is a function of the exogenous process Y . Thus this maximization requires neither perfect foresight nor a new assumption about the optimizer of the original problem. Its exact value yields

$$\begin{aligned} A_k(0, 2, y_0) &= e^{-\rho} \phi(m(1)) + \int_0^1 e^{-\rho t} \{\mathbb{E}S_0(m(t), \hat{Y}_t) - k\bar{\theta}_t^2/2\} dt + C \\ &\quad + \int_0^1 e^{-\rho s} \mathbb{E}g_k(B_s, \bar{\theta}_s) ds. \end{aligned} \tag{51}$$

For the stochastic source term, the joint Gaussian covariance is

$$\text{Cov}(.05 Z_t^u, \log Y_t) = (.05)(-.3)(-.25)t = .00375t.$$

Integration by parts consequently gives exactly the term C in (22). The clipping assumption switches are continuous Lipschitz maps; their weak derivatives, rather than pointwise classical derivatives at isolated switches, suffice for this identity. Since $\beta_t \leq 0$, in particular, treating the two noises as independent would discard a real term of the economy.

B.3. The conditional-variance allowance

Function H_k is continuously differentiable, with $H'_k(q) = \text{proj}_{[-.2, .2]}(q/k)$. Its derivative is Lipschitz. The same bound applies to the derivative of $q \mapsto g_k(q, a)$ for fixed a . Therefore

$$\mathbb{E}g_k(B_s, \bar{\theta}_s) \leq g_k(b(s), \bar{\theta}_s) + \frac{\text{Var}(B_s)}{2k}. \tag{52}$$

ance .09s. The Gaussian Poincare inequality gives

$$\text{Var}(B_s) \leq .09s \mathbb{E}|f'_s(l_s)|^2.$$

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completeness, with an orthonormal Hermite expansion $f = \sum_{n \geq 0} c_n H_n$ under a Gaussian variance v , one has $\text{Var } f = \sum_{n \geq 1} c_n^2$ and $v \mathbb{E}|f'|^2 = \sum_{n \geq 1} n c_n^2$. Polynomial approximation bounds the inequality to the bounded Lipschitz functions used here. No unknown mixing or concentration constant enters the calculation.

Differentiation of the conditional Gaussian transition is a translation derivative, so

$$f'_s(l_s) = \int_s^1 e^{-\rho(t-s)} \mathbb{E}[\partial_t \beta_t | Y_s] dt.$$

Additional Jensen and Minkowski give

$$\text{Var}(B_s) \leq .09s \left\{ \int_s^1 e^{-\rho(t-s)} \sqrt{\mathbb{E}|\partial_t \beta_t|^2} dt \right\}^2 \leq .09s D^2 C_\rho (1-s)^2. \quad (53)$$

Next verify the constant in this inequality. On the interior branch and before clipping, for $[2, 2.2]$,

$$|\partial_l \beta(m, e^l)|^2 = e^{2(1-1/m)l} \frac{l^2}{m^4} \leq \frac{1}{16} e^{(12/11)l} l^2 \quad (l > 0).$$

$l \leq 0$, the upper-consumption constraint makes the derivative zero. At a clipped y face the derivative is also zero almost everywhere. The positive right-hand side can therefore be integrated over the entire untruncated Gaussian as an upper bound. Its exact exponential moment

$$\mathbb{E}[e^{al} l^2] = e^{a\mu + a^2 v/2} \{(\mu + av)^2 + v\}.$$

$y_0 \in [1.2, 2.4]$, the means $\mu = M - .025t$ are positive. The expression is increasing in both μ and v on the relevant range. Using $\mu \leq M$ and $v \leq .09$ gives (24). Finally,

$$\int_0^1 e^{-\rho s} s C_\rho (1-s)^2 ds \leq \int_0^1 s(1-s)^2 ds = 1/12.$$

Combining (51), (52), and (53) proves (23). The allowance is positive and remains present at any quadrature resolution.

B.4. Transfer from the auxiliary problem to the actual contract

et

$$W(t, u, x, y) = y(x - .5) + .1 \log(.5) + A_k(t, u, y).$$

Accounting identity for Yx cancels every original adaptive portfolio choice. The maximization defining Ψ bounds every original consumption. Inequality (15) bounds the remaining source as long as (u, y) lies in the verification rectangle. Dynamic programming for the explicitly defined affine auxiliary problem then makes discounted W plus accumulated original payoff a martingale up to the first relevant face, for any original policy. One can establish the same argument directly from conditional Fubini and the adapted selector above; no differentiability of

form source-error corrections, give the equivalent classical verification argument. The terminal tangent obeys $Q \geq \phi$ everywhere because ϕ is concave. Write $F = A_k - Q$. The feasible auxiliary choice $\theta = 0$, discounted Ito applied to the affine function Q gives

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$$, u, y) \geq \mathbb{E} \int_t^1 e^{-\rho(r-t)} \{S_0(m(r), \hat{Y}_r) + \beta(m(r), \hat{Y}_r)(U_r - m(r)) - \rho Q(U_r)\} dr.$$

starting $u \in [1.5, 2.5]$, one has

$$\mathbb{E}|U_r - m(r)| \leq .7 + .05 = .75, \quad \mathbb{E}|Q(U_r)| \leq .0008 + .008(.75) = .0068.$$

iently (50) implies

$$y) \geq \{-7.1 - 1.1(.75) - .04(.0068)\}(1 - t) = -7.925272(1 - t) > -8(1 - t). \quad (54)$$

a uniform bound for the required starting rectangle; it is not a claim that the affine l is bounded on the entire u line.

$\in [.5, 2]$ and $y \geq .2$,

$$y(x - .5) + .1 \log(.5) - .1 \log x \geq 0,$$

differentiation in x , with equality at $x = .5$. Thus $W \geq G - 8h$ on actual wealth exits $W \geq G$ at $h = 0$. For the preference localization, $.2(120) + .00125(120)^2 = 42$ bounds the generator for either sign and every original θ . For log-deflator localization, $dl = .3dZ^x$ and $22.33 = .025(22) + .045(22)^2$ bounds either sign. Each term in (18) is a negative supermartingale potential that contributes at least $8h$ on its artificial face. Their sum therefore raises the continuation value to at least G at every artificial face. The already known global upper $V_k \leq G$, for $k \geq .5$, applies beyond that face. Stopped verification holds and this fallback after it prove (19). All constants correspond to the original discount, correlation, action bounds, and liquidation fee.

APPENDIX C: VALIDATED EVALUATION AND IMPLEMENTATION ERROR

C.1. Interval quadrature of the dual

Frozen references have sixteen time slabs. A slab $t \in [j/16, (j+1)/16]$ is transformed to $[\sqrt{j/16}, \sqrt{(j+1)/16}]$ and subdivided into 4, 8, or 16 intervals. The standard Gaussian is approximated by 64, 128, or 256 intervals on $[-6, 6]$, respectively. Thus the three meshes use 65,536, 131,072, and 262,144 source boxes. All square-root endpoints are enclosed before binary64 rounding.

Approximate the reference mean and log-deflator are

$$m(v^2) = 2 + \mu_j + \bar{\theta}_j(v^2 - j/16), \quad l(v, z) = M - .025v^2 - .3vz,$$

$\bar{\theta}_j = \sum_{i < j} \bar{\theta}_i/16$ is an interval-enclosed sum of exact deployment constants. The source integral in (v, z) follows from (45) and the chain rule. The entire image box is enclosed, with the union of all possible consumption branches. The program asserts that the covered Gaussian is inside the y clipping faces for each of the fitted references.

Let $w_v = 2ve^{-.04v^2}$ and $w_z = (2\pi)^{-1/2}e^{-z^2/2}$, the source integral uses the exact interval-enclosed zeroth, first-centered, and second-centered moments of both weights. The exponential

aining terms analytically. The Gaussian weight uses its degree-110 exponential series with analytic absolute remainder on $[-6, 6]$. Antiderivatives of each polynomial term are evaluated at interval endpoints. No floating-point Gaussian quadrature weight is assumed exact in the interval. 30

r_v, r_z are box radii, a valid second-order remainder bound is

$$\frac{1}{2} \|S_{vv}\|_{\infty} M_{v,2} M_{z,0} + \|S_{vz}\|_{\infty} r_v r_z M_{v,0} M_{z,0} + \frac{1}{2} \|S_{zz}\|_{\infty} M_{v,0} M_{z,2}.$$

interval first-centered moments retain their signs; constant and linear terms are not discarded even though all weighted boxes were symmetric. The source is C^1 with locally bounded one-sided derivatives across an active-consumption switch, so the integral Taylor formula with the Hessian remains valid there.

Gaussian Chernoff bound gives $\Pr(|Z| > 6) \leq 2e^{-18}$. The source tail lies between $1/\Pr(|Z| > 6)$ and zero at each time; the coefficient tail has magnitude at most $1.1 \Pr(|Z| > 6)$. For the covariance tail,

$$|\partial_t \beta| \leq 12^{6/11} \log(12)/4 < 2.5$$

the nonzero-derivative region, so its omitted contribution lies in $[-.00375(2.5) \Pr(|Z| > 6), 0]$. The signs are retained because $\beta_t \leq 0$. These corrections account for the full unbounded Hessian, not just the sampled central region.

For $b(s)$, interval source-coefficient integrals are accumulated backwards. The remaining portion of the current time cell ranges independently from zero to its entire discounted mass. Multiplying by the interval for $e^{\rho s}$ encloses $b(s)$ throughout that cell. The convexity of $g_k(q, \bar{\theta})$ ensures the maximum over its resulting q interval occur at an endpoint. At an uncertain saturation threshold the unconstrained quadratic conjugate supplies a safe upper bound. This calculation supplies the control-gap integral in (23).

Every elementary binary64 addition, subtraction, multiplication, division, and pairwise sum is rounded outward. Analytic constants and weighted moments use 70-digit interval arithmetic. The following objective, the source-quadrature enclosure, the policy-payoff enclosure, and the optimality test have different meanings and are recorded in different files. The acceptance test subtracts the policy's lower endpoint from the optimal-value upper endpoint with outward rounding. The comparison with .01 is then a test of the original-policy regret, not of a residual norm.

C.2. Proof of the feasible-output allowance

PROOF OF PROPOSITION 2: Couple the two policies to the same preference Brownian motion. Wealth is deterministic, decreasing, and remains in $(.5, 1.25]$ because of the stated budget and consumption constraints. The difference of preference paths at time t is at most $|\varrho_s - \tilde{\vartheta}_s| ds$. The terminal wealth difference is at most $e^{.02} \delta_c$.

In the event that neither policy exits through preference before time one, $u \in [1.2, 2.8]$. For $u \in [.7, .8]$, put $r = u - 1$ and $L = -\log c$. Since $Lr < 1$, the running utility satisfies

$$|\partial_c \ell| = c^{-u} < 3, \quad 0 \leq \partial_u \{c^{1-u}/(1-u)\} = e^{Lr}(1-Lr)/r^2 \leq 1/r^2 \leq 25.$$

The terminal preference derivative has magnitude at most .032, while the terminal wealth derivative is at most .2. Finally, $\frac{k}{2} |\theta^2 - \tilde{\theta}^2| \leq .2k |\theta - \tilde{\theta}|$. Integrating these inequalities and using count factors at most one gives the first two terms of (25).

the independent evaluator. The union thus has probability at most $4e^{-72}$. On any stopped policy has magnitude below 6 before the adjustment cost, that cost contributes at most 1.1 + .02k. Bounding the payoff differences by the expression $4e^{-72}$ gives term of (25). The argument never replaces the stopped economy with a reflected or payoff. *Q.E.D.*

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C.3. What the finite execution establishes

Three cost instances pass the .01 acceptance test. The interval upper bounds apply to original adaptive controls; the lower bounds apply to the stored polished time policies. Distinction is essential to an optimality certificate. The computations do not identify a optimal feedback surface, establish a rate for the neural optimizer, or prove that a fixed maximization converges to exact value as its quadrature mesh tends to zero. Its nonvanishing errors are displayed explicitly. The finite target and the economic inequalities reported in the text follow from the executed upper and lower bounds alone.

APPENDIX D: THE PRECEDING FITTED CONSTRUCTION, RETAINED FOR COMPARISON

The following section is preserved verbatim from the completed R9 paper. Its policies, witnesses, tables, and statements that the .01 target was not met describe that preceding construction. The current R10 certificate is Theorem 3 and Table I. This dated preservation is an alternative current claim.

APPENDIX E: CONSTRUCTIVE CERTIFICATES IN THE ORIGINAL ECONOMY

Initial numerical bounds in this section concern $(t, u, x) = (0, 2, 1.25)$. A central-state certificate is not advertised as a uniform certificate for every state in D . The policy used here is trained in a time-control neural subclass. The retained lattice-trained, state-feedback policies, their finite corrections, and their previous continuous bounds remain separately in the supplement.

E.1. Fitting an upper witness without importing an optimal value

$\tau = 1 - t$ and restrict the *verification domain*, not the economic domain, to $u \in [1.5, 2.6]$.

$$F(h, u) = h \sum_{i=0}^{12} \sum_{j=0}^{14} \beta_{ij} T_i(2h - 1) T_j \left(2 \frac{u - 1.5}{1.1} - 1 \right).$$

Coefficients are obtained from a method-of-lines initializer and residual least squares. The solver's collocation residual is not the certificate. For $w = G + F$, the consumption and wealth maximizers are .8 and .8. Maximizing the remaining wealth term over $x \in [.5, 2]$ gives the residual majorant

$$\mathcal{R}_k F = -F_h + .00125 F_{uu} - .04 F + H_k \{ F_u - .04(u - 2) \} + \frac{.8^{1-u}}{1 - u} \quad (55)$$

$$+ .0008(u - 2)^2 + .00195 + .00352 - .04 - .004 \log 2, \quad (56)$$

$$H_k(q) = \max_{|\theta| \leq .2} (q\theta - k\theta^2/2).$$

lower coefficients are enclosed at 70-digit interval precision and converted to Bernstein coefficients. The utility term is replaced by a polynomial with a proved uniform remainder below 7×10^{-9} . De Casteljau subdivision uses outward rounding at every elementary step. The positive-saturation, negative-saturation, and interior branches of H_k are all included whenever the certified q range permits them. Outward thresholds also cover a cell touching a switching face. The same calculation proves $F/h \geq -8$.

PROPOSITION 5—Localized fitted upper witness: *Suppose $F(0, u) = 0$, $F/h \geq -8$, and $\epsilon \leq \epsilon$ on the verification rectangle. For $k \geq .5$, let*

$$Z(h, u) = 8h\{e^{-120(u-1.5)+42h} + e^{-160(2.6-u)+64h}\}, \quad (57)$$

$$U(t, u, x) = G(u, x) + F(1-t, u) + Z(1-t, u) + \epsilon_+ C_\rho(1-t). \quad (58)$$

$U(0, 2, 1.25) \leq U(0, 2, 1.25)$. No reflecting boundary or truncation of the original policy introduced at $u = 1.5$ or 2.6 .

The exponential terms bound continuation when an artificial verification face is reached. The locally valid upper witness G is used beyond that face. At the central state the localization is less than 1.219×10^{-7} , rather than an eight-unit initial gap. Uniform residual section and localization are different terms and are reported separately.

E.2. A learned feasible policy and independent stopped evaluation

The actor has one time input, a twelve-unit tanh layer, and two outputs. At sixteen fixed nodes, a sigmoid constrains adjustment to $[0, .2]$, and a differentiable budget projection constrains consumption to $[\cdot 05, .8]$. The portfolio is zero. Training uses the original expected off, Gaussian quadrature, and no optimal-value or greedy-action labels. The deployed controls are the resulting frozen binary64 constants on each time slab; there is no claim that this control class exhausts the state-feedback opportunities in (5).

The budget projection enforces

$$\sum_{j=0}^{15} c_j \int_{j/16}^{(j+1)/16} e^{-\cdot 02r} dr < 1.25 - \cdot 5e^{-\cdot 02}. \quad (59)$$

As $x_t = e^{-\cdot 02t} \{1.25 - \int_0^t e^{-\cdot 02r} c_r dr\}$ is deterministic. Since consumption exceeds $\cdot 02(1.25)$, wealth decreases and its interval-enclosed terminal value remains strictly above $\cdot 5$. Wealth never exits. The remaining preference exit is controlled by a Brownian maximal inequality, not by a retized boundary check.

An evaluator independent of the training quadrature approximates utility by a degree-90 polynomial in $u - 2$, with a uniform analytic remainder on the original preference interval. Free Gaussian moments are evaluated by recurrence, integrated over outward time rectangles, and projected for the stopped event using a maximal probability bound and Cauchy–Schwarz. The final wealth, adjustment expenditure, polynomial approximation, time integration, exact-exit section, and arithmetic are all enclosed. There is no Euler, action-grid, or spatial-interpolation in this route. Those omissions are consequences of the deployed policy and analytic evaluation, not uncomputed entries assigned zero.

Table E.I reports the actual original-economy bounds. Table E.II makes the constructive remainder explicit. The last row gives substantially more precise policy evaluation than optimality

k	Policy payoff enclosure	Optimal-value upper	Regret upper
0.5	$[-1.27324522, -1.27322724]$	-1.16618072	0.10706450
2	$[-1.29504316, -1.29502986]$	-1.18870455	0.10633861
8	$[-1.32818098, -1.32817412]$	-1.22254633	0.10563464

points and upper bounds are rounded outward. Each regret upper is the independent optimal-value upper the lower endpoint of the learned policy's payoff interval. All controls in the optimal-value bound are continuous. The .01 full-policy target is not met in any row.

TABLE E.II
EXECUTED REFINEMENT FOR THE $k = 2$ CERTIFICATE

Time boxes	Payoff width	Sec.	Bernstein boxes	Residual upper	Sec.	Regret upper
1024	0.000209	0.363	36	0.00477	0.079	0.110237
4096	5.25e-05	0.614	57	0.00195	0.116	0.107396
16384	1.33e-05	1.618	86	0.000894	0.186	0.106339

es refine the independent policy evaluator. Bernstein boxes refine the already fitted upper witness. The two columns measure the respective integration and subdivision stages; complete timings, including coefficient construction, are in the raw records. A shrinking evaluation or residual allowance need not eliminate the approximation gap between the chosen upper witness and policy.

tion. The fitted witness is independent of the policy and valid over the full original and action ranges. Its loss of wealth-specific information is economically consequential. Late wealth-sensitive linear-programming pilot is retained with its computation log and candidate diagnostics; an unvalidated collocation objective is not used in either table.

APPENDIX F: VERIFICATION AND COMPUTATIONAL ERROR

F.1. Selection, stopped comparison, and the neural bridge

OF PROPOSITION 1: For a fixed jet the three controls enter additively. The control objective has derivative $c^{-u} - v_x$ and second derivative $-uc^{-u-1} < 0$. If $v_x \leq 0$ it is increasing; otherwise its unconstrained maximizer is $v_x^{-1/u}$. The adjustment objective is $\theta^2/2$, which is strictly concave for $k > 0$. The portfolio objective is $h_1p + h_2p^2/2$. A positive h_2 gives the projected vertex. A nonnegative h_2 gives a maximum at an endpoint, giving the linear and constant cases. Compactness and the stated tie rule give a feasible selection. These are control-space statements at a fixed jet, not parameter-space statements. *Q.E.D.*

OF THEOREM 1: Localize the process on interior compact sets and apply discounted formula to an upper witness on each time slab. The discounted terminal witness plus expected reward equals the initial witness, the expected integral of $R_a U$, and the expected discounted forward jumps. The latter sum is nonpositive. Boundary comparison gives b_U . Nonnegativity of discounting and the deterministic bound $\tau \leq T$ therefore imply

$$J_k^a(t, s) \leq U(t, s) + b_U + \epsilon_U \mathbb{E} \int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T-t)\epsilon_U.$$

integrability permits removal of localization. Taking the supremum over all admissible policies proves the optimal-value bound; attainment is unnecessary. Apply the same identity to L

ing the lower bound. Subtraction proves the regret statement. The process is stopped at the economic exit; there is no use of an unproved interior trace of the optimal value. *Q.E.D.*

PROOF OF THEOREM 2: The cover and representative intervals imply $|R_\pi v| \leq e$ and $0 \leq q$ everywhere. Hence $\sup_a R_a v = R_\pi v + \Gamma \leq e + q$ and $R_\pi v \geq -e$. Apply Theorem 1 with $L = v$, $b_U = b_L = b$, $\epsilon_U = e + q$, and $\epsilon_L = e$. A derivative bound with respect to the chosen norm gives the two moduli by integration along segments, or along a specified path inside cell. In a nonconvex cell, a verified path-length radius must replace the Euclidean radius. The shing-allowance assertion follows directly from the bound and does not require a compact of network parameters. *Q.E.D.*

The finite counterpart follows from positivity of the continuation operators: propagate the equation error with one factor δ , propagate the optimality error with one factor δ , and add the validate gain. Backward induction from terminal error b gives (8). The full state-dependent action, correction rule, and recursive-operator versions are reproduced in the preservation element. Unlike this finite identity, a continuous interpretation additionally needs either verified operator discrepancy terms or an independent continuous verification route. The present final-payoff calculation uses the latter.

F.2. Arithmetic convention

The mathematical constants specifying the economy are exact decimals. Stored network rights, policy constants, and fitted coefficients are interpreted as exact binary64 numbers. Instant formation and polynomial coefficient conversion use 70-digit interval arithmetic; linear optimization bounds use at least 50 digits. Vectorized interval addition, subtraction, multiplication, division by a strictly positive interval, and pairwise summation round both endpoints outwards with the adjacent binary64 number. An interval square may be enclosed by binary interval multiplication; this is conservative even when its lower endpoint is negative. Transcendental library approximation is treated as a rigorous interval merely because its output was subsequently rounded once. Exponentials in vectorized routines are replaced by Taylor polynomials with analytic remainders; interval logarithms, trigonometric constants, and square roots are formed in the multiprecision layer.

All derivative and residual claims concern the mathematical function represented by the stored constants, not the real-time floating-point implementation of an arbitrary neural policy. The final time controls are themselves stored constants, so their continuous implementation is unambiguous. External-network rollouts are explicitly floating-point performance estimates. This distinction is retained in every numerical table.

APPENDIX G: ORIGINAL-ECONOMY FITTED WITNESSES

G.1. Global fallback and localized upper bound

For $G = -.02(u - 2)^2 + .1 \log x$, consumption and portfolio selection give $c = .8$ and $p = .8$. The adjustment gain is bounded above by $.000512/k \leq .001024$ for $k \geq .5$. The preference part ρG is at most $.000512$. Consumption utility is maximized at the upper endpoint $u = 2.8$ of the stated domain. The portfolio drift and diffusion contributions at $p = .8$ are $.00352$. Minimizing $-.08/x - .004 \log x$ over $[.5, 2]$ gives $-.04 - .004 \log 2$. Consequently

$$\sup_a R_a G \leq \frac{.8^{-1.8}}{-1.8} + .000512 + .001024 + .00195 + .00352 - .04 - .004 \log 2 < -.8659.$$

OF PROPOSITION 5: The derivatives of $G + F$ in x coincide with those of G . Thus c and p selection applies. The residual after maximizing over wealth is precisely (56). $8h \exp\{\lambda(u - u_0) + \kappa h\}$, where $h = 1 - t$,

$$(\partial_t + \mathcal{L}^a - \rho)z = 8e^{\lambda(u-u_0)+\kappa h} \{-1 + h(-\kappa + \lambda\theta + .00125\lambda^2 - \rho)\}.$$

ices $(|\lambda|, \kappa) = (120, 42), (160, 64)$ satisfy $\kappa = .2|\lambda| + .00125\lambda^2$. Both terms therefore positive killed generators for every admissible control. Moreover $(\partial_t - \rho)C_\rho(1 - t) =$ allows that U has nonpositive residual inside the verification domain. $F = 1$, $F = Z = C_\rho = 0$, and $U = G$. At an actual wealth exit, $F \geq -8h$ implies $U \geq G -$ At either artificial preference face, one summand of Z is at least $8h$, so $U \geq G$. Stop first economic exit or an artificial face. If the artificial face occurs first, the remaining original-payoff is bounded by G under every admissible continuation, by the global fallback verified. Concatenated stopped verification gives the asserted bound without modifying the dynamics or opportunity set. *Q.E.D.*

G.2. Polynomial source enclosure and refinement

$u - 1 = A + Bz$, where $A = 1.05$, $B = .55$, and $|z| \leq 1$. Let $D = (A^2 - B^2)^{1/2}$ and $A + D < 1$. The uniformly convergent expansion

$$\frac{1}{A + Bz} = \frac{1}{D} \left[1 + 2 \sum_{n=1}^{\infty} (-q)^n T_n(z) \right]$$

ication error at degree 16 bounded by $2q^{17}/\{D(1 - q)\}$. If $L = -\log(.8)$, approximate e^{LBz} by e^{LA} times the degree-eight Taylor polynomial of e^{LBz} . Its remainder is at most $(LB)^9/9!$. Multiplying the two enclosures, with the relevant sup-norm bounds, gives uniform error below 2.117×10^{-9} for $-.8^{-(u-1)}/(u - 1)$. This error is added to the upper bound, not fitted away.

I_k branches are $-.2q - .02k$ for $q \leq -.2k$, $q^2/(2k)$ for $|q| \leq .2k$, and $.2q - .02k$ for $q \geq .2k$. On each box, interval Bernstein coefficients enclose q and every potentially applicable polynomial. A box touching a switching threshold includes both adjoining branches. Comparison uses outward threshold intervals; binary64 representation of $.2k$ cannot exclude branch. A tensor polynomial lies between its minimum and maximum Bernstein coefficients. The de Casteljau subdivision preserves this enclosure and is repeated on boxes whose upper bound exceeds the prescribed ceiling. The independent check of $F/h \geq -8$ uses the same null property.

fixed polynomial whose true residual lies strictly below a ceiling, Bernstein refinement verifies that ceiling, up to the explicitly accounted source and arithmetic allowances. A certification statement about a fixed candidate with slack. It is not a theorem that fitting, a wealth-independent ansatz, or arbitrary Adam training produces approximants close to the optimal value. The executed ceiling sequence and all visited-cell counts are recorded. The separate wealth-sensitive linear-programming pilot has not received this validation and contributes no upper endpoint to the reported results.

the projected policy has constant c_j, θ_j on each of sixteen slabs, $p = 0$, and $0 \leq \theta_j \leq .2$. Training chooses these constants, but the subsequent evaluator reads only the frozen policy. The exact wealth formula in the main text and the outward budget check show $x_1 > .5$. Since $\theta_j \leq .025$ and $x_0 = 1.25$, wealth is decreasing throughout the horizon and never reaches the lower boundary. Neither Brownian correlation nor a wealth discretization affects this conclusion in $p = 0$.

Without preference stopping, $u_t = 2 + \mu(t) + .05W_t^u$, with $\mu(t) = \int_0^t \theta_r dr \in [0, .2]$. An lower-boundary exit requires $W_t^u \geq 12$ at some $t \leq 1$; a lower exit requires $W_t^u \leq -16$ and is contained in $\{\sup_{t \leq 1} |W_t^u| \geq 12\}$. The exponential martingale maximal inequality on each gives

$$p_{\text{exit}} \leq 2e^{-72} < 1.077 \times 10^{-31}.$$

The original preference diffusion is still stopped at exact first exit. Its rare-exit correction is not zero.

At $r = u - 2$ and $L_j = -\log c_j$. Before exit, $|r| \leq .8$ and utility is $-e^{L_j} e^{L_j r} / (1 + r)$. Replace the reciprocal by $\sum_{m=0}^{80} (-r)^m$ and the exponential by its degree-ten Taylor polynomial. The metric remainder is at most $.8^{81} / .2$; the exponential remainder is at most $e^{.8L_j} (.8L_j)^{11} / 11!$. The product bound, including the reciprocal and exponential suprema, yields a uniform error for degree-90 polynomial P_j . This bound holds on the whole original preference interval, not just at training quadrature nodes.

For a Gaussian random variable with mean μ and variance v , its raw moments satisfy $M_0 = 1$, $M_1 = \mu$, and

$$M_n = \mu M_{n-1} + (n-1)v M_{n-2}.$$

Since $\mathbb{E}P_j(u_t - 2)$ is evaluated by this recurrence with $v = .0025t$, and the terminal quadratic preference settlement has an exact Gaussian expectation. On each time rectangle the intervals for μ, v , discounting, and the polynomial coefficients enclose the entire integrand. Outward bounds of interval range times interval length therefore enclose its integral. These ranges are the quadrature-error estimates inferred from two discretizations. The three resolutions are independent enclosures of the same integral.

To compare this free-Gaussian polynomial calculation with the stopped economic payoff, let E be the event of preference exit. Off E the two paths agree through T . On E , the omitted polynomial tail is bounded using

$$\mathbb{E}[\mathbf{1}_E |P_j(u_t - 2)|] \leq \sqrt{p_{\text{exit}}} \|P_j(u_t - 2)\|_2, \quad \|P_j\|_2 \leq \sum_m |b_{jm}| \sqrt{M_{2m}}.$$

Gaussian moments are bounded uniformly by evaluating their nonnegative coefficient formulas at $\mu = .2$ and $v = .0025$. The same argument bounds the free terminal settlement, the actual exit settlement has magnitude below 8.1. Discounted adjustment cost omitted after exit is at most $.02k \mathbf{1}_E$. One valid total stopping allowance is thus

$$\sqrt{p_{\text{exit}}} \left\{ \max_j \|P_j\|_2 + \|G(u_1, x_1)\|_2 \right\} + p_{\text{exit}} (.02k + 8.1).$$

The uniform utility approximation allowance is added separately, multiplied by $C_\rho(1)$. At $p = 0$ the stopping allowance is below 5.388×10^{-16} ; the utility approximation allowance is below 1.151×10^{-7} before multiplying by $C_\rho(1)$. The larger remaining width comes from the explicitly enclosed time integration, not from ignoring a boundary event.

by at most $.02p_{\text{exit}}$. Its interval, and the strictly positive wealth-feasibility margin, are at full precision. Tight budget intervals describe these deployed policies, not optimal-identification.

APPENDIX I: ECONOMIC COMPARISON AND THE RESTRICTED DUAL

I.1. Cost ordering and value enclosures

OF THEOREM 4: For each policy, $J_k(\pi) = A(\pi) - kB(\pi)$ is affine and nonincreasing because $B(\pi) \geq 0$. The supremum is convex and nonincreasing. The two approximate inequality constraints give

$$A_1 - k_1 B_1 \geq A_2 - k_1 B_2 - \varepsilon_1, \quad A_2 - k_2 B_2 \geq A_1 - k_2 B_1 - \varepsilon_2.$$

and dividing by $k_2 - k_1 > 0$ gives (26). Zero allowances give budget ordering for optimizers at distinct prices. If π_k is optimal at k , feasibility at k_+ implies $V(k_+) \geq (k_+ - k)B(\pi_k)$; feasibility at k_- implies $V(k_-) \geq V(k) + (k - k_-)B(\pi_k)$. Substitute appropriate value-enclosure endpoints and intersect with the primitive budget range to obtain *Q.E.D.*

access lower bound follows from $V_k \geq J_k$ and $V_R \leq \bar{V}_R$. Its nonnegativity follows because the restricted policy set is contained in the flexible policy set. Likewise $V_k - V_R \leq \bar{V}_R$. These inequalities compare two optimal values even though the feasible policies giving the lower bounds need not be optimal.

I.2. Cancellation of the portfolio and stopped localization

OF THEOREM 5: For Y as specified, quadratic covariation gives

$$d(Y_t x_t) = (.04Y_t x_t - Y_t c_t)dt + \text{local martingale}.$$

Yx portfolio drift cancels the $-.06pYx$ covariation term for every adaptive p . Including the discount term, and the preference quadratic yields

$$_{,p}W_0 = \frac{c^{1-u}}{1-u} - yc + .01y - .00005 - .004 \log(.5) + .0008(u-2)^2 \leq S(u, y).$$

no W_0 mixed $u-x$ or $u-y$ derivative. Correlation is retained in the transition law of W_0 used to evaluate the source potential.

source lower bound use the feasible consumption $.35$: its utility on $u \in [1.5, 2.6]$ is bounded below by $-2/\sqrt{.35}$, while $-.35y \geq -4.2$. The remaining terms sum to a nonnegative on this rectangle. Thus $S > -7.581 > -8$. For an upper bound discard $-yc$, maximize $c = .8, u = 2.6$, use $.01y \leq .12$, and bound the remaining terms by $.0033$. This gives $.698 < 0$. The clipped extension is bounded in $[-8, 0]$, so $-8h \leq F_R \leq 0$.

discounted Feynman–Kac potential satisfies the conditional expectation identity for pricing source integral. Consequently $e^{-\rho r} F_R(h - r, U_r, Y_r) + \int_0^r e^{-\rho s} \tilde{S}(U_s, Y_s) ds$ is a martingale. Stopping this identity on a bounded verification rectangle is legitimate because the source is bounded. Inside the rectangle $\tilde{S} = S$, so adding the identity to discounted Itô decomposition for W_0 cancels its residual majorant. This argument does not require an assumed smooth optimal value, or differentiability of the clipped source at its outer faces.

For $l = \log Y$, $dl = -.025dt - .3dZ^x$. Each exponential localization term with $|\lambda| = 22$ nonpositive killed generator because $22.33 = .025(22) + .045(22)^2$ dominates either sign of the log drift. As before, the factor $8h$ supplies at least $8h$ at its associated artificial face. For $x \in [.5, .38]$ and $y \geq .2$,

$$W_0 - G = y(x - .5) + .1 \log(.5/x) \geq 0 :$$

expression is zero at $.5$ and has derivative $y - .1/x \geq 0$. At an actual wealth exit the source is at least $-8h$, so $W_0 + F_R + Z_R \geq g$. At a terminal time it is at least G . At any artificial y face a localization term pays for the entire possible source deficit, giving continuation at least G . Actual preference exit cannot precede these inner preference faces. Invoke global fallback $V_R \leq G$ after an artificial exit and apply the stopped identity before it. This covers (34) over all original adaptive consumption and portfolio controls. The value of Y_0 is an auxiliary witness parameter, not an imposed wealth price or policy restriction. *Q.E.D.*

1.3. Validated evaluation of the restricted witness

Let z_1, z_2 be independent standard Gaussians and put $r = v^2$. At the central state,

$$u = 2 + .05v\{-.25z_1 + \sqrt{.9375}z_2\}, \quad l = \log(1.55) - .025v^2 - .3vz_1.$$

Source expectation is integrated with weight $2ve^{-.04v^2}\varphi(z_1)\varphi(z_2)$ on $v \in [0, 1]$. The box $z_2 \in [-6, 6]$ lies strictly inside the stated u, y verification rectangle under this transformation. Omitted probability is at most $4e^{-18}$. Since the extended source lies in $[-8, 0]$, omission contributes an interval $[-8C_\rho(1)4e^{-18}, 0]$, whose width is less than 4.778×10^{-7} . The initial localization addition is less than 2.200×10^{-9} .

For completeness, the Hessian used by the evaluator is explicit. Let $c = \text{proj}_{[.05, .8]} e^{-l/u}$, $-\log c, y = e^l$, and let χ be one for an interior consumption solution and zero for a strictly trained solution. Across a switching box use $\chi \in [0, 1]$. With $Q = c^{1-u}/(1-u)$,

$$\begin{aligned} Q_u &= \frac{e^{L(u-1)}\{1 - L(u-1)\}}{(u-1)^2}, & Q_{uu} &= -\frac{e^{L(u-1)}[\{L(u-1) - 1\}^2 + 1]}{(u-1)^3}, \\ S_u &= Q_u + .0016(u-2), & S_l &= -yc + .01y, \\ S_{uu} &= Q_{uu} + \chi ycL^2/u + .0016, & S_{ul} &= -\chi ycL/u, \\ S_{ll} &= -yc + \chi yc/u + .01y. \end{aligned}$$

The envelope is continuously differentiable across clipping surfaces; its gradient is locally Lipschitz and these piecewise second derivatives give a valid integral Taylor bound across such a face. Differentiate the displayed (u, l) transformation by the chain rule to obtain a Hessian in (z_1, z_2) . Every intermediate expression is enclosed on the entire box. Bounds on the absolute Hessian entries give the usual one-half diagonal and full mixed-term Taylor remainder.

The constant and linear Taylor terms are integrated exactly for their polynomial approximation of the weight. Gaussian weighted moments through order two use 111 terms of $e^{-z^2/2}$ and remainder bounded on $|z| \leq 6$ by $e^{18}18^{111}/111!$. Time moments use thirteen terms of $e^{-.04v^2}$ and their analytic remainder. Source exponentials are evaluated with degree-40 Taylor polynomials on arguments in $[-3, 3]$, with error at most $e^33^{41}/41!$. All antiderivatives, remainders,

; intersecting their enclosures with the nonnegative half-line is valid. Mixed moments
 ervatively bounded by cell-radius products times probability mass.
 successive box counts are 16,384, 131,072, and 1,048,576. Their intervals enclose the
 itial *dual witness*. Only an upper endpoint is used for reservation restricted optimum.
 ing the final upper endpoint with the independently certified restricted-policy lower
 true optimal-value interval of width less than .009047. The distinct width below .000320
 s integration of the witness, not the primal–dual gap.

APPENDIX J: CONTINUOUS ACTION AND ABSOLUTE-VALUE CERTIFICATION

J.1. The scalar global action bound and interval neural jets

OF THEOREM 6: For any action with $\sum_i a_i = z$ and any real λ ,

$$\sum_i f_i(a_i) = \sum_i \{f_i(a_i) - \lambda a_i\} + \lambda z \geq D(\lambda) + \lambda z.$$

nd derivative of $.3 \sin z$ is bounded below by $-.3$. For $z = z_0 + e$, $|e| \leq r$, its quadratic
 minorant gives

$$\lambda z + .3 \sin z \geq \lambda z_0 + .3 \sin z_0 + (\lambda + .3 \cos z_0)e - .15e^2,$$

mples (39). Each scalar f_i has second derivative $(2 + .2 \cos 2s)/d \geq 1.8/d = m$. Thus
 λs is m -strongly convex. Its tangent quadratic at any feasible s_0 gives a lower bound
 $q'(s_0)^2/(2m)$. At an endpoint, only a feasible descent direction is relevant: an outward-
 minimization direction contributes zero, while an inward-pointing direction contributes
 vative magnitude. This follows either by minimizing the tangent quadratic over the
 or directly from the sign of the linear term. The stated lower bound remains valid if the
 c minimizer lies beyond the opposite endpoint, because relaxing the displacement can
 ver it.

um of these scalar lower bounds is no larger than $D(\lambda)$. A feasible action gives a
 upper bound by direct interval evaluation. The scalar interval $[-d, d]$ covers every
 action sum. Subdivision, pruning only a box whose lower bound exceeds a feasible
 und, and taking the minimum surviving lower bound preserve a global enclosure. Root
 affects efficiency but not validity. Finally $|H_p(a) - H_{\hat{p}}(a)| \leq 2 \sum_i \delta_i$ on the action box.
 e candidate upper and true minimum lower can each worsen by at most this amount,
 e $4 \sum_i \delta_i$ gap correction. *Q.E.D.*

n affine layer with stored weights, interval automatic differentiation propagates both
 ns and spatial derivatives. A tanh layer is evaluated as $(e^{2z} - 1)/(e^{2z} + 1)$ with a
 positive denominator and derivative $1 - \tanh^2 z$, all in multiprecision intervals. The
 embedding is differentiated as its mathematical formula, including the exact $\sqrt{d}$, not as
 usly rounded autograd constant. This gives a componentwise interval for the derivative
 presented network. Subtracting the stored floating-point gradient gives the δ_i . The two
 tests use the inherited eight- and sixteen-dimensional learned checkpoints and the same
 e action certificates. The summed errors are below 4.584×10^{-17} and 7.877×10^{-17}
 vely. These are local jet checks, not whole-state residual bounds.

PROOF OF PROPOSITION 3: For $|a| \leq 1$, $\sin(a)/a \geq \sin 1$, with the continuous value at 1. Consequently $a^2 + .1 \sin^2 a \geq r_0 a^2$, where $r_0 = 1 + .1 \sin^2 1$. Also $.3 \sin(\sum_i a_i) \geq -.3$. Replacing running cost by $r_0 \|a\|^2/d - .3$ decreases every original control's cost; allowing all $\mathbb{R}^d$ decreases the infimum further. The original terminal payoff q_d is unchanged. The relaxed HJB equation is

$$v_t + \frac{\sigma^2}{2} \Delta v - \frac{d}{r_0} |\nabla v|^2 - .3 = 0, \quad v(1, x) = q_d(x), \quad \sigma^2 = .32.$$

$\lambda = \sigma^2 r_0 / (2d)$ and use the logarithmic heat representation. At $X_0 = 0$ it gives

$$v(0, 0) = -.3 - \lambda \log \mathbb{E} \exp\{-q_d(\sigma Z)/\lambda\}, \quad Z \sim N(0, I_d).$$

Bounded cosine and coercive quadratic terminal terms give finite positive expectations. The Gaussian heat solution is smooth before the terminal date and has at most quadratic growth. Standard localization of its verification identity, or truncation followed by monotone Gaussian integration, validates the relaxed lower bound without assuming a smooth solution of the original bounded-control problem.

Complete the square for the quadratic terminal component. Its Gaussian normalization concentrates

$$\frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3.$$

Under the tilted Gaussian law the scalar sum divided by $\sqrt{d}$ has mean $m = 2\sqrt{d}/(r_0 + 4)$ and variance $s^2 = \sigma^2 r_0 / (r_0 + 4)$. Orthogonal coordinates disappear, leaving

$$v(0, 0) = \frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda \log \mathbb{E} \exp\{-(.2/\lambda) \cos(m + sZ_1)\}.$$

This is the representation in the proposition. An upper enclosure for the positive scalar expectation gives a lower enclosure for v , because $-\lambda \log$ is decreasing. That lower endpoint is before a lower bound for the original value. *Q.E.D.*

For the scalar integral let $\eta = .2/\lambda$ and $R = \lceil \sqrt{2(\eta + 50)} \rceil$. The omitted Gaussian contribution is no larger than $2e^{\eta - R^2/2}$. Interval ranges of the positive integrand times cell lengths close the integral on $[-R, R]$. The successive 1,024, 4,096, and 16,384-cell calculations add analytic tail allowance and then apply the decreasing logarithm with outward rounding. Error widths measure integration uncertainty in the relaxed value. Relaxing the action bound under the running sine can leave a large structural gap even when this quadrature width is small.

APPENDIX K: REPLICATION AND SCOPE LEDGER

The central economic certificate is independent of the inherited killed-Euler discretization. The restricted upper has a verified continuous-generator residual, an explicit source-polynomial remainder, a stopped-trace inequality, and exponentially bounded artificial-face continuation. The restricted lower has exact deterministic wealth, interval Gaussian polynomial moments, an explicit utility approximation remainder, outward time integration, and an exact-exit probability section. The restricted upper cancels continuous portfolio controls algebraically and evaluates

external experiment has a different status. Each declared sample-and-hold policy is a continuous control in the stated additive-noise dynamics. Real-time network evaluations are for holding slab, but the stored path estimates use floating-point network evaluations and a Monte Carlo sample. The rigorous lower bound supplies an absolute reference; subtracting a sampled mean does not create an outward policy-regret certificate. Paired path errors, variation, completed-update clock overshoot, setup, and warmup are recorded separately. A sign test concerns the frozen joint training-and-audit pipeline under independent laws. It does not certify an underlying PDE or imply that the six panels represent six different problem families.

A new table is generated from raw JSON or path records; display endpoints are rounded conservatively in direction. The frozen external protocol, all tuning trials, all 72 holdout pairs, endpoints, and all path arrays are retained. The validation program checks coverage of the panels, matching initialization hashes, pinned author revision, finite outputs, refinement, primitive feasibility, the positive access lower bounds, and the explicitly false flexible division flags. The complete R8, R4, and delivered R5 textual and numerical material is in the preservation supplement, and earlier source files remain byte-for-byte unchanged. A failed pilot nor a historical manufactured-payoff test is used as new original-economy evidence.

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PART II: COMPLETE R11 PRESERVATION SUPPLEMENT

This supplement preserves the complete preceding R10 paper and its entire preservation supplement without deleting equations, proofs, numerical tables, or unsuccessful calculations. The current manuscript is ECTA_R11. Earlier precision and comparative-performance statements are dated results, not new claims. The R11 revision adds a frozen method-specific tuning study, independent holdout evidence, exact continuous reference calculations, implemented-output error identities, and executed absolute-accuracy frontiers. All historical source files remain unchanged.

READING MAP

The R11 manuscript contains the original economy, all R10 main results and proofs, the method-specific tuning study, the full absolute-accuracy calculations, and their verification details. A separate point-by-point response identifies the changes against every finding in the latest R9 referee report. The immutable source, result, build, and review identities are recorded in the revision directory.

The following pages reproduce the complete R10 manuscript. They are followed by the complete R10 supplement, which already preserves R9, R8, and earlier development material. In particular, recursive utility, sophisticated temporal selves, games, stochastic traces, action safe-sets, unsuccessful earlier certificates, and all retained external experiments remain available. These earlier documents must be read with their original version labels. R11 does not turn an earlier failed tolerance or an exploratory numerical result into a current certificate merely by reproducing it.

PART I: COMPLETE R10 MANUSCRIPT

prove on the preceding wealth-independent witness, whose $k = 2$ regret witness and its complete proofs remain available; the new certificate is evaluation precision as optimality precision. The restricted economy's bound is retained, together with its independently evaluated feasible restricted-value interval narrower than .00905. Combining the two wishes the welfare inequalities above.

addresses the inner optimization needed for verification. In a cou- family, all nonconvex coupling depends on the sum of the actions. h-and-bound calculation, together with strongly convex scalar dual e full continuous action box without a tensor action grid. Executed low 10^{-5} through 128 action dimensions. At the eight- and sixteen- test states, interval differentiation of the actual network weights also floating-point critic gradient. This is an implemented bridge between n action certificate, not an assumption that an optimizer's stationary

family is given an absolute accuracy axis. A pointwise running-cost of control bounds admit a Cole–Hopf representation that retains the ninal payoff. Orthogonal Gaussian coordinates integrate analytically, integral with an outward enclosure. The resulting number is a lower original minimization problem. Relative performance is evaluated out experiment: three dimensions, two training budgets, twelve seed nning-rate search budgets, warmed initialization, randomized method dently specified feedback rule. The pinned author implementation of ified.

monotone approximation, and primal–dual numerical methods supply he stopped Itô argument is not claimed as a new general verification ing neural evaluation and improvement claimed as a new algorithm on is the constructive combination of original-contract verification, per and lower bounds, computable neural-jet allowances, and a non- n audit, with each error quantity matched to an executed calculation. objects theorem by theorem with neighboring methods. Recursive util- l selves, dynamic games, stochastic-trace identities, counterexamples, g manuscripts are preserved in the supplement rather than being used claims of the present application.

2. THE ECONOMIC ENVIRONMENT

, where u denotes relative risk aversion and x denotes wealth. An c, θ, p) in the compact set

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

, discounting is $\rho = .04$, and the open state domain is $D = (1.2, 2.8) \times \inf\{r \geq t : S_r \notin D\}$. For $k > 0$, running utility and settlement are

$$= \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$= -.02(u-2)^2 + .1 \log x, \quad g(t, u, x) = G(u, x) - 8(1-t). \quad (4)$$

and optimal value are

$$\int_t^\tau e^{-\rho(r-t)} \ell_k(S_r, a_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \Big], \quad V_k = \sup_\pi J_k^\pi. \quad (5)$$

value V_R restricts θ to zero and retains *all* admissible consumption and is independent of k . The cost experiment holds the controls, transition law, event, utility normalization, and consumption units fixed. A state-dependent utility would generally define another economy.

ent fee is a liquidation covenant. There is no continuation consumption or allowing exit. At the lower wealth boundary the largest admissible wealth at the upper boundary, $p = 0$ eliminates the wealth diffusion and permits another a silently viable rectangle nor a theorem requiring uniform controlled- can be substituted for this contract. The verification below requires only with test functions on the trace at which the process is actually stopped.

tion v , put

$$[(.02 + .06p)x - c]v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$^av + \ell_k - \rho v. \quad (7)$$

is retained. Its sign is not inferred solely from the negative shock correla-

-Complete selection at a fixed jet: *For the generator in (7), a global can be selected componentwise. Consumption is $\text{proj}_{[.05, .8]}(v_x^{-1/u})$ when otherwise. Adjustment is $\text{proj}_{[-.2, .2]}(v_u/k)$. Let $h_1 = .06xv_x - .0025xv_{ux}$ If $h_2 < 0$, the portfolio is $\text{proj}_{[-.5, .8]}(-h_1/h_2)$; if $h_2 \geq 0$, an endpoint $p^2/2$ is selected, with a fixed tie rule.*

cludes convex and zero-curvature portfolio objectives and nonpositive . It applies to the continuous generator, not to the interpolated, killed-Euler . It says nothing about global optimization over shared neural parameters.

retain differentiation through the controlled successor state. A present in a game, are also frozen during unilateral improvement. This or loss from lowering itself by changing the object being evaluated. It network optimization convex. The ideal policy-iteration compactness are retained in the supplement; no such argument is invoked to certify ory.

eguard is a distinct algorithm. For $\mathcal{T}_n^a w = r_n^a + \delta P_n^a w$, positive sub- ρ^Δ , define the policy-evaluation residual $e_n = \|v_n - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$ and ement gap $\eta_n = \|\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$. If the terminal error mparison yields

$$\|V_{h,n}^* - V_{h,n}^\pi\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

ope and threshold-correction proof are reproduced in the supplement. e fraction of decisions changed while leaving the cost of a complete y unchanged. In particular, the retained 125-action tensor audit is not ifier of Section 6. Its floating-point envelope is not relabeled as an bound.

1. A quantitative bridge from a fitted network

)/ ρ , with $C_0(h) = h$.

missible witnesses: *Witnesses are bounded, continuous on the stopped erior of each time slab. Localized Itô identities pass to the stopping ity. At slab interfaces an upper witness has nonpositive forward jumps onnegative forward jumps. The implemented policy has a well-defined*

one-sided comparison: *Suppose Assumption 1 holds, $U \geq g - b_U$ and d trace, $\sup_a R_a U \leq \epsilon_U$, and $R_\pi L \geq -\epsilon_L$, where all four allowances*

$$L - b_L - C_\rho(T - t)\epsilon_L, \quad V_k \leq U + b_U + C_\rho(T - t)\epsilon_U, \quad (9)$$

$$U - L + b_U + b_L + C_\rho(T - t)(\epsilon_U + \epsilon_L). \quad (10)$$

quire an attained optimizer or a smooth optimal value.

nd-oracle certificate for neural output: *Let a continuous neural witness policy π satisfy Assumption 1, with stopped-trace error at most b . Let*

cluding jet, inner-solve, and arithmetic allowances. Set $e = \max_j (e_j + \omega_j)$. Then

$$0 \leq V_k - J_k^\pi \leq 2b + C_\rho(T - t)(2e + q). \quad (11)$$

NBO: R12 PRESERVATION SUPPLEMENT

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derivative bounds M_j, N_j and cell radii h_j permit $\omega_j = M_j h_j$ and hence of learned outputs for which these verified allowances vanish has perspective of whether its parameters converge.

ve a posteriori connection to network errors, not a claim that an empirical controls a supremum. Interval differentiation or weight-based derivative cell moduli. They may be conservative, and covering a high-dimensional costly. Section 6 executes the jet-to-action part for actual learned critics. tion instead fits an independent polynomial witness; it does not falsely ed a whole-state neural critic merely because its action audit passed at

CONSTRUCTIVE CERTIFICATE FOR THE FLEXIBLE ECONOMY

erical target is a regret bound of .01 for the original problem (5), not for n. The certificate below attains this target at $(0, 2, 1.25)$ for each of the es. It uses a feasible policy to bound value from below and a separate, rket-deflator construction to bound the optimum from above. The latter ted consumption, portfolio choice, and preference adjustment. Neither a timal-value label enters its construction.

An affine source that preserves the original opportunities

$- 2)^2$, $h = 1 - t$, and let Y satisfy (28), with its initial level $y_0 > 0$ now parameter. Define

$$S_0(u, y) = \Psi(u, y) + .01y - .004\log(.5), \quad (12)$$

$$\beta(u, y) = \partial_u \Psi(u, y), \quad (13)$$

$$m(t) = 2 + \int_0^t \bar{\theta}_s ds, \quad 0 \leq \bar{\theta}_s \leq .2, \quad (14)$$

al consumption conjugate in (30). The reference drift $\bar{\theta}$ is deterministic. It pper witness, not a restriction on the decision maker's adjustment policy. and

$$= \phi(m(1)) + \phi'(m(1))(u - m(1)), \quad b_T = \phi'(m(1)).$$

$(\bar{y}), \bar{y}) + \beta(m(r), \hat{y})(u - m(r))$. Consider the full-line auxiliary control

$$= \sup_{|\theta| \leq .2} \mathbb{E}_{t,u,y}^{50} \left[\int_t^1 e^{-\rho(r-t)} \{ \hat{S}_r(U_r, Y_r) - k\theta_r^2/2 \} dr + e^{-\rho(1-t)} Q(U_1) \right]. \quad (16)$$

Z_r^u , and the auxiliary control remains adapted. The affine source makes an explicitly defined conditional expectation, which can be bounded neural control problem. Put

$$) = 8h \{ e^{-120(u-1.5)+42h} + e^{-120(2.5-u)+42h} \}, \quad (17)$$

$$) = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (18)$$

preference dual and a computable initial upper bound: For $k \geq .5$, deterministic $\bar{\theta} \in [0, .2]$, the original flexible optimum satisfies

$$\leq .75y_0 + .1 \log(.5) + A_k(0, 2, y_0) + Z_u(1, 2) + Z_y(1, y_0). \quad (19)$$

put $\beta_t = \beta(m(t), \hat{Y}_t)$ and

$$b(s) = \int_s^1 e^{-\rho(t-s)} \mathbb{E} \beta_t dt + e^{-\rho(1-s)} b_T, \quad (20)$$

$$(q, a) = H_k(q) - qa + ka^2/2, \quad (21)$$

$$C = .00375 \int_0^1 te^{-\rho t} \mathbb{E} \left[\partial_t \beta(m(t), \hat{Y}_t) \right] dt, \quad (22)$$

cludes the clipping map and $H_k(q) = \max_{|a| \leq .2} (qa - ka^2/2)$. Then

$$e^{-\rho} \phi(m(1)) + \int_0^1 e^{-\rho t} \left\{ \mathbb{E} S_0(m(t), \hat{Y}_t) - k\bar{\theta}_t^2/2 \right\} dt + C + \int_0^1 e^{-\rho s} g_k(b(s), \bar{\theta}_s) ds + \frac{.09D^2}{24k}, \quad (23)$$

$$.5a^2 \text{---} \{(M + .09a)^2 + .09\}, \quad a = 12/11, \quad M = \log y_0. \quad (24)$$

conditional mean without this allowance would generally produce an invalid variance term C also matters: it retains the original correlation between shocks and is nonpositive here. It is evaluated, not silently discarded or of measure.

NBO: R12 PRESERVATION SUPPLEMENT

$S_0 \geq -7.1$ and $|\beta| \leq 1.1$ imply $A_k - Q \geq -7.925272h$ on the verification of zero adjustment in the auxiliary problem. Hence the factors $8h$ in (18) at artificial faces. On actual wealth exits, the uncorrected affine witness the liquidation settlement. The global fallback $V_k \leq G$ covers continuation. These facts connect (23) to the original stopped economy rather than to a economy.

Policy generation, verified refinement, and the stopping rule

Policies start from the retained one-input, twelve-unit neural time-control L-BFGS-B step polishes their sixteen consumption and adjustment outputs off and the same exact budget projection. This is an explicit enlargement of the image to the feasible time-control outputs; it is not described as additions or as a theorem that the neural optimizer converges. Consumption and adjustments are rounded inward relative to the *exact decimal* action bounds, and feasibility is rounded to their exact binary rational values. The finite-node training objective is a not an asserted untruncated Gaussian expectation of singular CRRA utility. A polynomial-moment evaluator in Appendix G then encloses each deployed policy under exact first preference exit. The full network histories, original outputs, and output-optimization records remain separate.

The policy is fitted with Gaussian quadrature, but its fitted objective is never used. The policy uses $t = v^2$, subdivides each of the sixteen time slabs, and covers the coordinate on $[-6, 6]$. Constant and linear Taylor terms are integrated with weighted moments. Hessian ranges enclose the entire remainder, including the consumption constraint. Gaussian tails, the terminal tangent, the conditional-localization, and outward arithmetic are separately included. The small interval tests establishing (15) precede every certificate execution.

The policy is simple. Given a policy lower bound L and an upper bound U from the policy with tolerance ε only when $U - L \leq \varepsilon$. Otherwise refine the policy to the explicit unachieved target at the declared work limit. Soundness does not imply convergence. Nor does soundness imply that an arbitrary tolerance can be relaxed. For the three declared .01 instances, the finite executions produce a termination certificate.

The bounds are approximately -1.26279687 , -1.28530724 , and -1.31820708 . As feasible-policy lower endpoints, they give regret upper bounds .00987440, .006896 after outward display rounding. The central target is attained without a fine grid, an Euler-error assumption, or an assumption that the policy is in the neural class. The preceding wealth-independent witness and its larger gaps

row .01. Each upper bound covers the full original continuous control set and stopping evaluates the specified feasible time-control policy. Endpoints are rounded outward; gaps These are central-state certificates, not uniform claims over every initial condition.

TABLE II

WORK NEEDED TO CROSS THE ABSOLUTE-ERROR TARGET

ce boxes	Regret upper	Verification seconds	.01 met
4,096	0.01027899	5.46	No
16,384	0.00995838	11.73	Yes
65,536	0.00987440	27.54	Yes
4,096	0.01013162	5.25	No
16,384	0.00980424	11.30	Yes
65,536	0.00971727	27.06	Yes
4,096	0.01037189	5.29	No
16,384	0.01005494	11.48	No
65,536	0.00996896	26.79	Yes

evaluated once at the finest stated precision. Time includes that evaluation, dual
 dual verification through each row, on one CPU thread. It excludes inherited network
 ts, and setup. This is an absolute-accuracy *verification* frontier, not an all-in training
 external matched-accuracy superiority claim. Raw stage clocks are retained.

TABLE III

SEPARATE UPPER-BOUND ALLOWANCES AT $k = 2$

Source remainder	Control gap	Variance allowance	Localization
0.000339131	0.000063959	0.000081080	0.000000247
0.000082675	0.000041416	0.000081080	0.000000247
0.000020398	0.000032100	0.000081080	0.000000247

for allowance on the covered Gaussian region. Analytic Gaussian tails, interval weight
 the covariance interval are included in the full records. The variance allowance does
 ding quadrature boxes. No row is obtained by treating a mesh difference as a rigorous
 error estimate.

C, where their loss of wealth information can be compared with the

tative bridge from implemented controls to the certificate

the stored deployment constants themselves. A further perturbation
 ness to control evaluation error, rather than inferring robustness from a

$$)| \leq (3 + .2e^{.02})\delta_c + (25 + .2k + .032)\delta_\theta + 8(14.1 + .02k)e^{-72}. \quad (25)$$

dated output error bounded by the right side is a **PRESERVATION SUPPLEMENT**
y's regret certificate. Feasibility must still be checked separately; small
l error alone does not rule out early wealth exit.

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es the output allowance is zero: their binary64 constants are treated as exact
Proposition 2 instead gives an explicit interface for a budget-preserving
perturbed outputs. Together with the neural-jet bound of Theorem 2, it
plementation errors to certified quantities without attributing an unproved
te to Adam, RMSprop, or L-BFGS-B.

CERTIFYING THE VALUE OF PREFERENCE ADJUSTMENT

olicy write $J_k(\pi) = A(\pi) - kB(\pi)$, where

$$= \mathbb{E}^\pi \int_t^\tau e^{-\rho(r-t)} \theta_r^2 / 2 \, dr, \quad 0 \leq B(\pi) \leq .02C_\rho(T-t).$$

mon to every k .

curacy-calibrated cost comparisons: *The optimal value is convex and
If optimizers exist, their adjustment budgets are nonincreasing across
generally, let π_i be ε_i -optimal at k_i , with $k_2 > k_1$. Then*

$$B(\pi_2) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \quad (26)$$

)), the budget of an optimizer at k , for $k_- < k < k_+$, lies in the primitive
ted with

$$\left[\frac{V(k) - \overline{V}(k_+)}{k_+ - k}, \frac{\overline{V}(k_-) - \underline{V}(k)}{k - k_-} \right]. \quad (27)$$

erns discounted adjustment expenditures, not a pointwise ordering of
 V_u/k is not a comparative-static proof. The new full-value enclosures
e strict welfare losses across the cost grid. At $k = 2$, (27) also gives the
sure [.00386376, .01960529], conditional on existence of an optimizer. Its
nt is informative, while its upper endpoint is still the primitive bound. The
nditures in Table V describe the specified learned-and-polished policies,
nal budget.

deduced from the original shocks. It does not restrict portfolio information, traded asset. For

$$u, x, y) = y(x - .5) + .1 \log(.5) - .02(u - 2)^2, \quad (54)$$

or of Yx cancel exactly. With $\theta = 0$, define

$$\max_{.5 \leq c \leq .8} \left\{ \frac{c^{1-u}}{1-u} - yc \right\}, \quad (29)$$

$$(u, y) + .01y - .00005 - .004 \log(.5) + .0008(u - 2)^2. \quad (30)$$

12], one has $-8 < S < 0$. Extend S to a bounded function $\tilde{S} \in [-8, 0]$ on the other faces. Let

$$F_R(h, u, y) = \mathbb{E}_{u, y} \int_0^h e^{-.04r} \tilde{S}(U_r, Y_r) dr, \quad (31)$$

and Y follows (28). Thus $-8h \leq F_R \leq 0$.

Restricted-economy upper bound: Let Z_R be the sum of the following potentials:

$$= 8h \{ e^{-200(u-1.5)+50h} + e^{-240(2.6-u)+72h} \}, \quad (32)$$

$$= 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (33)$$

$$) \leq W_0(2, 1.25, 1.55) + F_R(1, 2, 1.55) + Z_R(1, 2, 1.55). \quad (34)$$

all adaptive original consumption and portfolio controls with $\theta = 0$. For any flexible policy with certified payoff lower bound $\underline{J}_k$ and any $\epsilon > 0$,

$$, 2, 1.25) - V_R(0, 2, 1.25) \geq \max\{0, \underline{J}_k - \bar{V}_R\}. \quad (35)$$

has an explicit two-Gaussian representation. We transform time by $t \mapsto \tau$ and linear Taylor terms with interval-enclosed weighted moments, and exponential Taylor remainders, Gaussian tails, and Gaussian remainders. This produces the executed refinement in Table IV; it does not replace the error estimate. The final restricted-optimum upper bound is 0.03585, so the restricted problem itself has an initial optimal-value of 0.00905.

the *dual witness value*, not the restricted optimum. Only their upper endpoints are used as optimum. A separately evaluated restricted policy gives the primal lower bound. Original g and all adaptive consumption and portfolio opportunities are retained.

TABLE V

ED CURRENT-POLICY EXPENDITURES AND OPTIMAL VALUE OF ACCESS

Policy adjustment budget	Optimal access-value interval	Regret target
[0.01757696, 0.01757697]	[0.06332184, 0.08224255]	< .01
[0.01241866, 0.01241867]	[0.04096859, 0.05973218]	< .01
[0.00192681, 0.00192682]	[0.00781706, 0.02683234]	< .01

evaluates the displayed learned policy. The access columns bound the difference between the lower endpoint pairs a feasible flexible policy with the restricted dual upper; the upper flexible upper with a feasible restricted policy. All endpoints are rounded outward from cost-two access interval has width below its entire positive gain; at cost eight, the interval still resolves the sign but not a comparably precise magnitude.

TABLE VI

ILIBRATED OPTIMAL WELFARE LOSSES FROM HIGHER ADJUSTMENT COSTS

Optimal welfare loss	Interval width	Width below loss
[0.01263598, 0.03222764]	0.01959166	Sign resolved
[0.02318257, 0.04286880]	0.01968623	Yes
[0.04553582, 0.06537918]	0.01984335	Yes

— V_{k_2} in the unchanged original economy. Bounds use independent value endpoints, not k . The last column compares the width with the lower endpoint of the effect interval.

bounds remain strictly positive at all three prices. At $k = 2$ the optimal [0.096859, .05973218]; its width is smaller than the lower endpoint. Table VI the flexible optima at different adjustment prices. The $2 \rightarrow 8$ and $.5 \rightarrow 8$ entire interval widths. The $.5 \rightarrow 2$ comparison resolves the sign but has a fainty. No comparison assumes that a displayed policy is an optimizer or ordering theorem as a pointwise ordering of adjustment controls.

GLOBAL ACTION CERTIFICATION WITHOUT A TENSOR GRID

ing minimization problem, which is also used for the external comparison:

$$_t = 2a_t dt + \sqrt{.32} dW_t, \quad X_0 = 0, \quad a_t \in [-1, 1]^d, \quad (36)$$

$$) = \frac{1}{d} \sum_{i=1}^d \{a_i^2 + .1 \sin^2 a_i\} + .3 \sin \left(\sum_i a_i \right), \quad (37)$$

to training. At a fixed critic gradient p , the action-dependent Hamilto-

$$f_i(s) = \frac{s^2 + .1 \sin^2 s}{d} + 2p_i s, \quad z = \sum_i a_i.$$

with modulus at least $m = 1.8/d$.

global partition and convex dual bounds: *For any multiplier λ and*

$$D(\lambda) = \sum_i \min_{|s| \leq 1} \{f_i(s) - \lambda s\}.$$

$z_0 - r, z_0 + r]$, a lower bound for the full Hamiltonian is

$$\lambda) + \lambda z_0 + .3 \sin z_0 - |\lambda + .3 \cos z_0| r - .15 r^2. \quad (39)$$

es an upper bound. Each inner convex minimum can itself be bounded $(2m)$, where $q = f_i - \lambda s$ and e is the derivative magnitude in a feasible full derivative magnitude at an interior point. Outward versions of scalar subdivision and pruning, give a global gap on $[-1, 1]^d$. differs from the true neural gradient by componentwise bounds δ_i , the dient gap is at most its stored-gradient gap plus $4 \sum_i \delta_i$.

ding only proposes multipliers and actions. It is not trusted to solve exactly. Even an inaccurate proposal remains valid after the strong- is structure removes tensor enumeration; it does not establish a y theorem or solve a generic nonseparable nonconvex Hamiltonian. 10^{-5} calculation uses 339 scalar boxes and 43,392 scalar convex neither raw actor training nor the preceding finite tensor safeguard in eight- and sixteen-dimensional jets, interval forward differentiation nts bounds the ℓ^1 gradient discrepancies below 4.584×10^{-17} and ely. These allowances are propagated into the action gap. Theorem 2 tate-cover and residual quantities that would be needed for a whole-

ABSOLUTE BOUND AND A FROZEN EXTERNAL COMPARISON

ous lower bound for the unchanged nonconvex family

, $\sigma^2 = .32$, and $\lambda_d = \sigma^2 r_0 / (2d)$. Since $\sin^2 a \geq \sin^2(1) a^2$ for $|a| \leq 1$, by $r_0 \|a\|^2 / d - .3$ and allowing unbounded controls gives a relaxation. action q_d is retained.

1e-03	0.00068	85	2720	1.498	NBO: R12 PRESERVATION SUPPLEMENT	57
1e-05	3.52e-06	103	3296	1.991		
1e-03	0.000647	163	10432	6.041		
1e-05	3.06e-06	185	11840	5.955		
1e-03	0.000948	317	40576	19.229		
1e-05	4.68e-06	339	43392	20.651		

dimension, at both tolerances. At $d = 8, 16$, these are gradients of retained learned critics at varied gaps include interval neural-derivative allowances. At larger dimensions they are jets for the same Hamiltonian family. The experiment measures certificate work for these s, not an average-case scaling law or a whole-state neural PDE residual.

TABLE VIII
LOWER BOUNDS FOR THE ORIGINAL NONCONVEX CONTROL PROBLEM

Quadrature cells	Relaxed-value enclosure	Original-value lower	Seconds
1024	$[-0.08599082, -0.08384626]$	-0.08599082	0.260
4096	$[-0.08518701, -0.08465086]$	-0.08518701	1.081
16384	$[-0.08498598, -0.08485194]$	-0.08498598	4.506
1024	$[-0.20520651, -0.20360335]$	-0.20520651	0.271
4096	$[-0.20460665, -0.20420585]$	-0.20460665	1.067
16384	$[-0.20445644, -0.20435623]$	-0.20445644	4.212
1024	$[-0.28757809, -0.28684340]$	-0.28757809	0.245
4096	$[-0.28730394, -0.28712026]$	-0.28730394	0.988
16384	$[-0.28723515, -0.28718923]$	-0.28723515	4.172

value. Its lower endpoint is also a lower bound for the original bounded-control value. The conservative enclosures is quadrature uncertainty; it does not include or estimate the structural relaxation gap.

–Scalar-integral absolute lower bound: *Let Z be normal with mean variance $\sigma^2 r_0/(r_0 + 4)$. Then*

$$\sigma r_0/4 + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda_d \log \mathbb{E} \exp \left[-\frac{.2}{\lambda_d} \cos Z \right]. \tag{40}$$

of the positive expectation yields a rigorous numerical lower bound for (38).

an upper/lower-bound device, not a new training problem with a known The terminal cosine, which couples all state coordinates, remains in the s tails are bounded analytically and its interior is integrated by interval

The median difference is NBO minus SOC-MartNet. The last column is a one-sided across the six predeclared dimension–budget panels. The sign test concerns the joint line under independent seed draws, not an outward numerical error guarantee.

TABLE X

UTE REFERENCE EXCESS AND ACTUAL TRAINING CLOCKS					
	Lower bound	NBO excess	SOC excess	NBO sec.	SOC sec.
	-0.08499	0.6662	1.1692	10.002	10.006
	-0.08499	0.6635	1.1407	30.004	30.008
	-0.20446	0.6707	0.9938	10.002	10.005
	-0.20446	0.6674	1.0500	30.004	30.006
	-0.28724	0.6686	1.4883	10.003	10.007
	-0.28724	0.6672	1.3090	30.003	30.007

policy cost minus the rigorous original-value lower bound. It provides an absolute deterministic regret enclosure: sampling error and floating-point rollout evaluation are clocks include completed-update overshoot and exclude separately recorded warmup and setup.

holdout design and exact deployment interpretation

s the original actor and critic architectures: two tanh hidden layers unded actor output, and the same terminal-value embedding. NBO and improvement with Adam, three critic steps per actor step, 256 successors. SOC-MartNet uses its pinned author solver, RMSprop, rk, and the inherited $J = 2$, $K = 1$ multiplier setup. These different disclosed rather than described as identical implementations.

ol freezes dimensions 8, 16, 32 and training budgets of 10 and 30 ion, both methods receive the same six validation trials: two seeds multipliers. Selection minimizes mean independent validation cost, elve different holdout seeds, 720–731, are then evaluated at each Within each pair, actor and critic starting weights are identical and Framework warmup precedes both clocks, and method order is seed- y and setup-plus-training clocks are retained. The baseline $a_i(t, x) = 1 + 4(1 - t)\}$ is specified independently from the associated quadratic on the unchanged nonquadratic objective.

s deployed as a sample-and-hold control on 128 dates and evaluated paired Gaussian paths. In (38), drift under a held action and volatility e Gaussian transition and running-cost integral are therefore exact in *deployed continuous-time policy*. A 64-date evaluation uses aggregated changes the deployment policy. Its difference from the 128-date result is ound. Floating-point network evaluation and Monte Carlo uncertainty outward deterministic lower bound in Table VIII.

tain every seed difference, the median and interquartile range, approximate
 s for all three rules, path standard errors, training curves, optimizer clocks,
 approximate Student intervals are descriptive; the exact sign calculation avoids
 ed-level differences. The source protocol predates holdout execution, while
 y and adapters themselves were developed in earlier revisions. Holdout
 e problem family into evidence of general superiority across economic
 dget points are not a matched-accuracy frontier, and a finite absolute lower
 ht enough to rank methods at a small prescribed regret.

ATION TO RIGOROUS CONTROL AND NEURAL COMPUTATION

as should be compared with methods that establish bounds, not only with
 tworks. Theorem 1 is an application of classical stopped verification to
 er witnesses. Its role here is to handle the actual liquidation trace without
 h or boundary-regular optimal value. Theorem 2 supplies a computation-
 e, and oracle allowance. Neither theorem claims that classical verification
 literature.

consistent approximations with comparison provide convergence to vis-
 e framework of Barles and Souganidis (1991). Controlled Markov-chain
 oped systematically by Kushner and Dupuis (2001); adaptive sparse-grid
 -dimensional dynamic models in Brumm and Scheidegger (2017). Such
 fied approximation schemes and their hypotheses. The finite telescoping
 ed manuscript names the operator discrepancies needed for a continuous
 d mesh difference does not supply them. The original-payoff certificates
 continuous generators or exact Gaussian moments directly, with separately
 d arithmetic remainders. They do not assert a convergence rate for the
 scheme.

rical control methods also pair feasible decisions with bounds. Brown,
 0) relax nonanticipativity with penalties and establish weak and strong
 e dynamic programs. Brown and Smith (2011) use related bounds in
 n with transaction costs. Our restricted and flexible constructions are
 esses with explicit stopped localization; they are not claims that policy-
 alysis is new. Theorem 3 adds a verified affine-preference source and a
 t variance allowance. Its distinction from an information-relaxation bound
 enalty is that every source integral, variance constant, and face allowance
 erre, Henrion, Prieur, and Trélat (2008) develop occupation-measure and
 or polynomial nonlinear optimal control under their stated assumptions.
 r deterministic convergence theorem for the present logarithmic, stochastic,

ded-source expectation. Theorem 6 exploits rank-one action coupling the remaining scalar problems. It is a structured convex-duality and uction, not a generic replacement for global nonconvex optimization. e nonquadratic terminal payoff in a computable relaxation of the exact

and neural PDE solvers are important computational antecedents. (15) analyze continuous-time policy improvement. Kim, Cho, Kim, iteration, policy-network, and residual errors in entropy-regularized and Cho (2026) study interior bounds and the mismatch between top performance. Lee and Kim (2025) combine policy iteration and ng, and Zhou (2025) impose martingale conditions without an explicit BO’s raw training comparison is with a pinned implementation of the recreated approximation described as author code. Our certificates duced by another solver. This portability is a property of independent ning results remain evidence about the specifically recorded NBO and

oral-self equilibria, and games require their own evaluation operators their complete equations, assumptions, finite gain bounds, failures, and are preserved in the supplement. No new large-scale recursive or game l from the additive-utility certificates above. The stochastic-trace bias izer counterexamples are preserved for the same reason: they identify aining loss cannot replace.

9. CONCLUSION

us preference-adjustment economy now has accepted central-state l regret tolerance for all three cost coefficients. A feasible learned-and- ach lower bound, while an independent affine-preference dual supplies original continuous adaptive controls. This closes the numerical target al stopping contract or replacing it by a manufactured-payoff problem. d the welfare cost of more expensive adjustment are resolved at a e reported effects. At coefficient two the optimal access gain is at he coefficient to eight lowers optimal welfare by at least .02318257. heir corresponding interval widths. Optimal adjustment budgets obey d a valid secant enclosure, but are not equated with the budgets of

also retain their distinct roles: fitted Bernstein witnesses, stopped ation, market-deflator localization, quantitative output and neural- d a structured continuous-action oracle. The unchanged coupled com- solute lower reference and a complete frozen holdout record, including sign-test outcomes. Those experiments do not establish a universally or an external matched-accuracy dominance theorem. All preceding

$\log y$, and

$$= \text{proj}_{[-\log(.8), -\log(.05)]}(l/u), \quad c = e^{-L}, \quad E = e^{Lr}.$$

maximizer is c , including both constrained branches. Write $\chi = 1$ on the
 zero on a constrained branch; on a box intersecting a switching surface its
 $[0, 1]$. The derivatives required by the certifier are

$$\begin{aligned} \Psi &= -E/r - yc, & \beta &= E(1 - Lr)/r^2, \\ (S_0)_l &= -yc + .01y, \\ (S_0)_{uu} &= -\frac{E\{(Lr - 1)^2 + 1\}}{r^3} + \chi \frac{ycL^2}{u}, \\ (S_0)_{ul} &= -\chi \frac{ycL}{u}, \\ (S_0)_u &= -yc + \chi \frac{yc}{u} + .01y. \end{aligned} \tag{41}$$

th $yc = E$. The envelope is continuously differentiable across the con-
 the displayed one-sided second derivatives enclose its almost-everywhere
 cient both for supporting inequalities and for the integral Taylor remainder

ption constraint is active on $u \in [1.5, 2.5]$, $y \in [.2, 12]$, because
 .05). Moreover $L < 2$ throughout that rectangle. On its interior consump-

$$\partial_l(S_0)_{uu} = \partial_u \beta_l = \frac{EL(2 - L)}{u^2} \geq 0. \tag{42}$$

ption branch $(S_0)_{uu}$ is independent of l and $\beta_l = 0$. The jump in $(S_0)_{uu}$
 ters the interior branch is nonnegative. When increasing u leaves that
 ard to zero. Thus $(S_0)_{uu}$ is nondecreasing in l , and β_l is nondecreasing in
 hing boundary.

losures are computed by the interval program on complete interval covers,
 ;:

$$\sup_{1.5 \leq u \leq 2.3} (S_0)_{uu}(u, 12) < -.0862, \tag{43}$$

$$\sup_{2.3 \leq u \leq 2.5} \beta(u, 12) < -.9687, \tag{44}$$

β] and $m \in [2, 2.2]$, the difference $\beta(u, y) - \beta(m, y)$ is nondecreasing using in u . Its maximum therefore occurs at $y = 12$. At that endpoint, $\geq \beta(2.2, 12)$, and the last two inequalities in (45) make the difference of the derivative between m and u now proves (15) on both sides of u . It does not infer a two-dimensional supporting plane from concavity at a single point, but using the cross derivative.

ers $m \in [2, 2.2]$ by 1,000 intervals and uses $(S_0)_l < 0$, $\beta_l \leq 0$ to check that β attains

$$-9 < S_0(m, y) < -1.2443, \quad |\beta(m, y)| < .9712. \quad (46)$$

bounds $-7.1 \leq S_0 \leq 0$ and $|\beta| \leq 1.1$ are used for tails and localization. After clipping y .

An adapted solution of the affine auxiliary problem

start at $(0, 2, y_0)$ and use the full-line auxiliary processes. All sources are bounded coefficients, and all controls are bounded. Fubini's theorem, and optional stopping on bounded rectangles are legitimate by Gaussian increments.

$$\mathcal{B}_s = \int_s^1 e^{-\rho(t-s)} \mathbb{E}[\beta_t \mid \mathcal{F}_s] dt + e^{-\rho(1-s)} b_T,$$

future Brownian increments gives $\mathbb{E}[\beta_t \mid \mathcal{F}_s] = \mathbb{E}[\beta_t \mid Y_s]$, even when β is not adapted to the Brownian history. In particular, B_s is an adapted function of Y .

$$U_t - m(t) = \int_0^t (\theta_s - \bar{\theta}_s) ds + .05 Z_t^u.$$

owed by conditional Fubini, shows that its control-dependent part is

$$\mathbb{E} \int_0^1 e^{-\rho s} \{(\theta_s - \bar{\theta}_s) B_s - k \theta_s^2 / 2\} ds.$$

er $\theta_s = \text{proj}_{[-.2, .2]}(B_s/k)$ is adapted and bounded. It gives a well-defined process θ since it is a function of the exogenous process Y . Thus this maximization problem has no foresight nor a new assumption about the optimizer of the original problem, which yields

$$e^{-\rho} \phi(m(1)) + \int_0^1 e^{-\rho t} \{\mathbb{E} S_0(m(t), \hat{Y}_t) - k \bar{\theta}_t^2 / 2\} dt + C$$

A.3. The conditional-variance allowance

continuously differentiable, with $H'_k(q) = \text{proj}_{[-.2, .2]}(q/k)$. Its derivative
 same bound applies to the derivative of $q \mapsto g_k(q, a)$ for fixed a . Therefore

$$\mathbb{E}g_k(B_s, \bar{\theta}_s) \leq g_k(b(s), \bar{\theta}_s) + \frac{\text{Var}(B_s)}{2k}. \quad (48)$$

write $B_s = f_s(l_s)$. The law of l_s is Gaussian with mean $M - .025s$ and
 Gaussian Poincare inequality gives

$$\text{Var}(B_s) \leq .09s \mathbb{E}|f'_s(l_s)|^2.$$

with an orthonormal Hermite expansion $f = \sum_{n \geq 0} c_n H_n$ under a Gaussian
 $\text{Var} f = \sum_{n \geq 1} c_n^2$ and $v\mathbb{E}|f'|^2 = \sum_{n \geq 1} n c_n^2$. Polynomial approximation
 y to the bounded Lipschitz functions used here. No unknown mixing or
 it enters the calculation.

ne conditional Gaussian transition is a translation derivative, so

$$f'_s(l_s) = \int_s^1 e^{-\rho(t-s)} \mathbb{E}[\partial_t \beta_t | Y_s] dt.$$

and Minkowski give

$$\leq .09s \left\{ \int_s^1 e^{-\rho(t-s)} \sqrt{\mathbb{E}|\partial_t \beta_t|^2} dt \right\}^2 \leq .09s D^2 C_\rho (1-s)^2. \quad (49)$$

onstant in this inequality. On the interior branch and before clipping, for

$$|\beta(m, e^l)|^2 = e^{2(1-1/m)l} \frac{l^2}{m^4} \leq \frac{1}{16} e^{(12/11)l} l^2 \quad (l > 0).$$

consumption constraint makes the derivative zero. At a clipped y face
 zero almost everywhere. The positive right-hand side can therefore be
 fire untruncated Gaussian as an upper bound. Its exact exponential moment

$$\mathbb{E}[e^{al^2}] = e^{a\mu + a^2 v/2} \{(\mu + av)^2 + v\}.$$

$$, u, x, y) = y(x - .5) + .1 \log(.5) + A_k(t, u, y).$$

or Yx cancels every original adaptive portfolio choice. The maximization of the original consumption. Inequality (15) bounds the remaining source of the verification rectangle. Dynamic programming for the explicitly bounded problem then makes discounted W plus accumulated original payoff a first relevant face, for any original policy. One can establish the same conditional Fubini and the adapted selector above; no differentiability of the source is needed. Smooth approximations of the bounded Lipschitz source, with corrections, give the equivalent classical verification argument.

obeys $Q \geq \phi$ everywhere because ϕ is concave. Write $F = A_k - Q$. Every choice $\theta = 0$, discounted Ito applied to the affine function Q gives

$$^{\rho(r-t)} \{S_0(m(r), \hat{Y}_r) + \beta(m(r), \hat{Y}_r)(U_r - m(r)) - \rho Q(U_r)\} dr.$$

2.5], one has

$$.7 + .05 = .75, \quad \mathbb{E}|Q(U_r)| \leq .0008 + .008(.75) = .0068.$$

s

$$1(.75) - .04(.0068)\}(1-t) = -7.925272(1-t) > -8(1-t). \quad (50)$$

for the required starting rectangle; it is not a claim that the affine function is the entire u line.

.2,

$$y(x - .5) + .1 \log(.5) - .1 \log x \geq 0,$$

with equality at $x = .5$. Thus $W \geq G - 8h$ on actual wealth exits for the preference localization, $.2(120) + .00125(120)^2 = 42$ bounds either sign and every original θ . For log-deflator localization, $dl = .33 = .025(22) + .045(22)^2$ bounds either sign. Each term in (18) is a martingale potential that contributes at least $8h$ on its artificial face. Their sum gives the continuation value to at least G at every artificial face. The already known $V_k \leq G$, for $k \geq .5$, applies beyond that face. Stopped verification after it prove (19). All constants correspond to the original discount, the bounds, and liquidation fee.

$$2 + \mu_j + \bar{\theta}_j(v^2 - j/16), \quad l(v, z) = M_{v,0} - 0.25v^2 + 3z^2$$

16 is an interval-enclosed sum of exact deployment constants. The source flows from (41) and the chain rule. The entire image box is enclosed, with the consumption branches. The program asserts that the covered Gaussian y clipping faces for each of the fitted references.

v^2 and $w_z = (2\pi)^{-1/2}e^{-z^2/2}$, the source integral uses the exact interval-centered, and second-centered moments of both weights. The exponential weight retains terms through degree twelve in $-.04v^2$ and bounds the analytically. The Gaussian weight uses its degree-110 exponential series with remainder on $[-6, 6]$. Antiderivatives of each polynomial term are evaluated. No floating-point Gaussian quadrature weight is assumed exact in the

dii, a valid second-order remainder bound is

$$M_{v,2}M_{z,0} + \|S_{vz}\|_\infty r_v r_z M_{v,0}M_{z,0} + \frac{1}{2}\|S_{zz}\|_\infty M_{v,0}M_{z,2}.$$

ered moments retain their signs; constant and linear terms are not discarded. All boxes were symmetric. The source is C^1 with locally bounded one-sided derivative-consumption switch, so the integral Taylor formula with the Hessian are.

off bound gives $\Pr(|Z| > 6) \leq 2e^{-18}$. The source tail lies between zero at each time; the coefficient tail has magnitude at most $1.1\Pr(|Z| > 6)$ tail,

$$|\partial_t \beta| \leq 12^{6/11} \log(12)/4 < 2.5$$

ative region, so its omitted contribution lies in $[-.00375(2.5)\Pr(|Z| > 6)]$ retained because $\beta_l \leq 0$. These corrections account for the full unbounded sampled central region.

source-coefficient integrals are accumulated backwards. The remaining at time cell ranges independently from zero to its entire discounted mass. interval for $e^{\rho s}$ encloses $b(s)$ throughout that cell. The convexity of $g_k(q, \theta)$ over its resulting q interval occur at an endpoint. At an uncertain saturation obtained quadratic conjugate supplies a safe upper bound. This calculation is an integral in (23).

binary64 addition, subtraction, multiplication, division, and pairwise sum is analytic constants and weighted moments use 70-digit interval arithmetic. The source-quadrature enclosure, the policy-payoff enclosure, and the optimality meanings and are recorded in different files. The acceptance test subtracts

constraints. The difference of preference paths at time t is at most
 nal wealth difference is at most $e^{.02}\delta_c$.
 er policy exits through preference before time one, $u \in [1.2, 2.8]$. For
 and $L = 66 \log c$. Since $Lr < 1$, the running utility satisfies

$$0 \leq \partial_u \{c^{1-u}/(1-u)\} = e^{Lr}(1-Lr)/r^2 \leq 1/r^2 \leq 25.$$

derivative has magnitude at most .032, while the terminal wealth
 nally, $\frac{k}{2}|\theta^2 - \tilde{\theta}^2| \leq .2k|\theta - \tilde{\theta}|$. Integrating these inequalities and using
 ne gives the first two terms of (25).
 ce exit has probability at most $2e^{-72}$ by the same maximal inequality
 valuator. The union thus has probability at most $4e^{-72}$. On any stopped
 de below 6 before the adjustment cost, that cost contributes at most
 magnitude below 8.1. Each payoff therefore has absolute value at
 ing the payoff difference on the exit union by twice this quantity gives
 e argument never replaces the stopped economy with a reflected or
Q.E.D.

B.3. What the finite execution establishes

s pass the .01 acceptance test. The interval upper bounds apply to
 controls; the lower bounds apply to the stored polished time policies.
 ial to an optimality certificate. The computations do not identify a
 surface, establish a rate for the neural optimizer, or prove that a fixed
 s to exact value as its quadrature mesh tends to zero. Its nonvanishing
 explicitly. The finite target and the economic inequalities reported in
 the executed upper and lower bounds alone.

PRECEDING FITTED CONSTRUCTION, RETAINED FOR COMPARISON

is preserved verbatim from the completed R9 paper. Its policies,
 nd statements that the .01 target was not met describe that preceding
 R10 certificate is Theorem 3 and Table I. This dated preservation is
 claim.

CONSTRUCTIVE CERTIFICATES IN THE ORIGINAL ECONOMY

ounds in this section concern $(t, u, x) = (0, 2, 1.25)$. A central-state
 ed as a uniform certificate for every state in D . The policy used here
 -control neural subclass. The retained lattice-trained, state-feedback
 corrections, and their previous continuous bounds remain separately
 nt.

$$-F_h + .00125F_{uu} - .04F + H_k\{F_u - .04(u-2)\} + \frac{.8^{1-u}}{1-u} \quad (51)$$

$$+ .0008(u-2)^2 + .00195 + .00352 - .04 - .004 \log 2, \quad (52)$$

$$\max_{|\theta| \leq .2} (q\theta - k\theta^2/2).$$

inal utility, not a payoff manufactured to make F exact.

are enclosed at 70-digit interval precision and converted to Bernstein
y term is replaced by a polynomial with a proved uniform remainder below
steljau subdivision uses outward rounding at every elementary step. The
egative-saturation, and interior branches of H_k are all included whenever
permits them. Outward thresholds also cover a cell touching a switching
ulation proves $F/h \geq -8$.

-Localized fitted upper witness: *Suppose $F(0, u) = 0$, $F/h \geq -8$, and
ication rectangle. For $k \geq .5$, let*

$$u) = 8h\{e^{-120(u-1.5)+42h} + e^{-160(2.6-u)+64h}\}, \quad (53)$$

$$x) = G(u, x) + F(1-t, u) + Z(1-t, u) + \epsilon_+ C_\rho(1-t). \quad (54)$$

$U(0, 2, 1.25)$. *No reflecting boundary or truncation of the original policy
.5 or 2.6.*

ms bound continuation when an artificial verification face is reached. The
witness G is used beyond that face. At the central state the localization
 1.219×10^{-7} , rather than an eight-unit initial gap. Uniform residual
ation are different terms and are reported separately.

learned feasible policy and independent stopped evaluation

time input, a twelve-unit tanh layer, and two outputs. At sixteen fixed
d constrains adjustment to $[0, .2]$, and a differentiable budget projection
on to $[\cdot 05, .8]$. The portfolio is zero. Training uses the original expected
adrature, and no optimal-value or greedy-action labels. The deployed
ing frozen binary64 constants on each time slab; there is no claim that this
hausts the state-feedback opportunities in (5).

unds are rounded outward. Each regret upper is the independent optimal-value upper of the learned policy’s payoff interval. All controls in the optimal-value bound are continuous. The .01 full-policy target is not met in any row.

TABLE D.II
EXECUTED REFINEMENT FOR THE $k = 2$ CERTIFICATE

off width	Sec.	Bernstein boxes	Residual upper	Sec.	Regret upper
.000209	0.363	36	0.00477	0.079	0.110237
5.25e-05	0.614	57	0.00195	0.116	0.107396
1.33e-05	1.618	86	0.000894	0.186	0.106339

ent policy evaluator. Bernstein boxes refine the already fitted upper witness. The two respective integration and subdivision stages; complete timings, including coefficient w records. A shrinking evaluation or residual allowance need not eliminate the ximation gap between the chosen upper witness and policy.

enforces

$$\sum_{j=0}^{15} c_j \int_{j/16}^{(j+1)/16} e^{-.02r} dr < 1.25 - .5e^{-.02}. \tag{55}$$

$\int_0^t e^{-.02r} c_r dr \}$ is deterministic. Since consumption exceeds .02(1.25), interval-enclosed terminal value remains strictly above .5. Wealth never erence exit is controlled by a Brownian maximal inequality, not by a ck.

ent of the training quadrature approximates utility by a degree-90 a uniform analytic remainder on the original preference interval. Free evaluated by recurrence, integrated over outward time rectangles, and event using a maximal probability bound and Cauchy–Schwarz. The nt expenditure, polynomial approximation, time integration, exact-exit are all enclosed. There is no Euler, action-grid, or spatial-interpolation e omissions are consequences of the deployed policy and analytic ed entries assigned zero.

actual original-economy bounds. Table D.II makes the constructive ist row gives substantially more precise policy evaluation than optimal- d witness is independent of the policy and valid over the full original Its loss of wealth-specific information is economically consequential. ve linear-programming pilot is retained with its computation log and stics; an unvalidated collocation objective is not used in either table.

LEM 1: Localize the process on interior compact sets and apply discounted upper witness on each time slab. The discounted terminal witness plus equals the initial witness, the expected integral of $R_a U$, and the expected reward jumps. The latter sum is nonpositive. Boundary comparison givesativity of discounting and the deterministic bound $\tau \leq T$ therefore imply

$$U(t, s) + b_U + \epsilon_U \mathbb{E} \int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T-t)\epsilon_U.$$

permits removal of localization. Taking the supremum over all admissible terminal-value bound; attainment is unnecessary. Apply the same identity to L and policy. Its forward jumps are nonnegative, $g \geq L - b_L$, and $R_\pi L \geq -\epsilon_L$. and. Subtraction proves the regret statement. The process is stopped at the U . There is no use of an unproved interior trace of the optimal value. *Q.E.D.*

LEM 2: The cover and representative intervals imply $|R_\pi v| \leq e$ and $0 \leq \sup_a R_a v = R_\pi v + \Gamma \leq e + q$ and $R_\pi v \geq -e$. Apply Theorem 1 with $b = b$, $\epsilon_U = e + q$, and $\epsilon_L = e$. A derivative bound with respect to the chosen v modulo by integration along segments, or along a specified path inside each cell, a verified path-length radius must replace the Euclidean radius. The assertion follows directly from the bound and does not require a compact set. *Q.E.D.*

part follows from positivity of the continuation operators: propagate the one factor δ , propagate the optimality error with one factor δ , and add the forward induction from terminal error b gives (8). The full state-dependent rule, and recursive-operator versions are reproduced in the preservation of this finite identity, a continuous interpretation additionally needs either redundancy terms or an independent continuous verification route. The present verification uses the latter.

E.2. Arithmetic convention

constants specifying the economy are exact decimals. Stored network constants, and fitted coefficients are interpreted as exact binary64 numbers. and polynomial coefficient conversion use 70-digit interval arithmetic; bounds use at least 50 digits. Vectorized interval addition, subtraction, division by a strictly positive interval, and pairwise summation round both

the floating-point implementation of an arbitrary neural policy. The themselves stored constants, so their continuous implementation is network rollouts are explicitly floating-point performance estimates. d in every numerical table.

DIX F: ORIGINAL-ECONOMY FITTED WITNESSES

1. *Global fallback and localized upper bound*

$+ .1 \log x$, consumption and portfolio selection give $c = .8$ and $p = .8$. unded above by $.000512/k \leq .001024$ for $k \geq .5$. The preference part 12. Consumption utility is maximized at the upper endpoint $u = 2.8$ he portfolio drift and diffusion contributions at $p = .8$ are $.00352$. $.04 \log x$ over $[\cdot 5, 2]$ gives $-.04 - .004 \log 2$. Consequently

$$.000512 + .001024 + .00195 + .00352 - .04 - .004 \log 2 < -.8659.$$

fication gives $V_k \leq G$ globally on the original domain, including as a artificial verification face.

ON 4: The derivatives of $G + F$ in x coincide with those of G . Thus n applies. The residual after maximizing over wealth is precisely (52). $\rangle + \kappa h\}$, where $h = 1 - t$,

$$)z = 8e^{\lambda(u-u_0)+\kappa h}\{-1 + h(-\kappa + \lambda\theta + .00125\lambda^2 - \rho)\}.$$

$(20, 42), (160, 64)$ satisfy $\kappa = .2|\lambda| + .00125\lambda^2$. Both terms therefore nerators for every admissible control. Moreover $(\partial_t - \rho)C_\rho(1 - t) =$ nonpositive residual inside the verification domain.

$= 0$, and $U = G$. At an actual wealth exit, $F \geq -8h$ implies $U \geq G -$ preference face, one summand of Z is at least $8h$, so $U \geq G$. Stop first artificial face. If the artificial face occurs first, the remaining original- l by G under every admissible continuation, by the global fallback l stopped verification gives the asserted bound without modifying the opportunity set. *Q.E.D.*

2. *Polynomial source enclosure and refinement*

, where $A = 1.05$, $B = .55$, and $|z| \leq 1$. Let $D = (A^2 - B^2)^{1/2}$ and uniformly convergent expansion

$$\frac{1}{A + Bz} = \frac{1}{D} \left[1 + 2 \sum_{n=1}^{\infty} (-q)^n T_n(z) \right]$$

A box touching a switching threshold includes both adjoining branches. Outward threshold intervals; binary64 representation of $.2k$ cannot exclude or polynomial lies between its minimum and maximum Bernstein coefficient-subdivision preserves this enclosure and in repeated refinements support the prescribed ceiling. The independent check of $F/h \geq -8$ uses the same

minial whose true residual lies strictly below a ceiling, Bernstein refinement at ceiling, up to the explicitly accounted source and arithmetic allowances. A statement about a fixed candidate with slack. It is not a theorem that n -independent ansatz, or arbitrary Adam training produces approximants minimal value. The executed ceiling sequence and all visited-cell counts are wealth-sensitive linear-programming pilot has not received this validation per endpoint to the reported results.

§: INDEPENDENT EVALUATION OF THE LEARNED TIME POLICY

Policy has constant c_j, θ_j on each of sixteen slabs, $p = 0$, and $0 \leq \theta_j \leq .2$. These constants, but the subsequent evaluator reads only the frozen policy. Formula in the main text and the outward budget check show $x_1 > .5$. Since $.25$, wealth is decreasing throughout the horizon and never reaches the inner Brownian correlation nor a wealth discretization affects this conclusion

the stopping, $u_t = 2 + \mu(t) + .05W_t^u$, with $\mu(t) = \int_0^t \theta_r dr \in [0, .2]$. An exit requires $W_t^u \geq 12$ at some $t \leq 1$; a lower exit requires $W_t^u \leq -16$ and is $\mathcal{O}_{t \leq 1} \{|W_t^u| \geq 12\}$. The exponential martingale maximal inequality on each

$$p_{\text{exit}} \leq 2e^{-72} < 1.077 \times 10^{-31}.$$

the diffusion is still stopped at exact first exit. Its rare-exit correction is not

$c_j = -\log c_j$. Before exit, $|r| \leq .8$ and utility is $-e^{L_j} e^{L_j r} / (1+r)$. Replace $\sum_{m=0}^{\infty} (-r)^m$ and the exponential by its degree-ten Taylor polynomial. The error is at most $.8^{81}/.2$; the exponential remainder is at most $e^{.8L_j} (.8L_j)^{11}/11!$. Including the reciprocal and exponential suprema, yields a uniform error for polynomial P_j . This bound holds on the whole original preference interval, not future nodes.

random variable with mean μ and variance v , its raw moments satisfy $M_0 = 1$,

$$M_n = \mu M_{n-1} + (n-1)vM_{n-2}.$$

is evaluated by this recurrence with $v = .0025t$, and the terminal quadratic has an exact Gaussian expectation. On each time rectangle the intervals

$$\|P_j\|_2 \leq \sqrt{p_{\text{exit}}} \|P_j(u_t - 2)\|_2, \quad \|P_j\|_2 \leq \sum_m |b_{jm}| \sqrt{M_{2m}}.$$

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are bounded uniformly by evaluating their nonnegative coefficient $p_{\text{exit}} = .0025$. The same argument bounds the free terminal settlement, which has magnitude below 8.1. Discounted adjustment cost omitted after the valid total stopping allowance is thus

$$\left\{ \max_j \|P_j\|_2 + \|G(u_1, x_1)\|_2 \right\} + p_{\text{exit}}(.02k + 8.1).$$

Approximation allowance is added separately, multiplied by $C_\rho(1)$. At this point, the total allowance is below 5.388×10^{-16} ; the utility approximation allowance is negligible. The remaining width comes from the integration, not from ignoring a boundary event.

The budget is exactly $\sum_j (\theta_j^2/2) \int_{j/16}^{(j+1)/16} e^{-.04t} dt$. Stopping can reduce this budget. Its interval, and the strictly positive wealth-feasibility margin, are preserved. Tight budget intervals describe these deployed policies, not optimal-

APPENDIX H: ECONOMIC COMPARISON AND THE RESTRICTED DUAL

H.1. Cost ordering and value enclosures

4: For each policy, $J_k(\pi) = A(\pi) - kB(\pi)$ is affine and nonincreasing in k . The supremum is convex and nonincreasing. The two approximate value functions

$$J_k \geq A_2 - k_1 B_2 - \varepsilon_1, \quad A_2 - k_2 B_2 \geq A_1 - k_2 B_1 - \varepsilon_2.$$

for $k_2 - k_1 > 0$ gives (26). Zero allowances give budget ordering for all prices. If π_k is optimal at k , feasibility at k_+ implies $V(k_+) \geq V(k)$. Feasibility at k_- implies $V(k_-) \geq V(k) + (k - k_-)B(\pi_k)$. Substitute enclosure endpoints and intersect with the primitive budget range to obtain $V(k_-) \geq V(k_+)$. *Q.E.D.*

It follows from $V_k \geq J_k$ and $V_R \leq \bar{V}_R$. Its nonnegativity follows from the fact that the policy set is contained in the flexible policy set. Likewise $V_k - V_R \leq 0$. We compare two optimal values even though the feasible policies need not be optimal.

$$-yc + .01y - .00005 - .004\log(.5) + .0008(u-2)^2 \leq S(u, y).$$

and $u-x$ or $u-y$ derivative. Correlation is retained in the transition law of the source potential.

bound use the feasible consumption .35: its utility on $u \in [1.5, 2.6]$ is $2/\sqrt{.35}$, while $-.35y \geq -4.2$. The remaining terms sum to a nonnegative multiple. Thus $S > -7.581 > -8$. For an upper bound discard $-yc$, maximize 2.6 , use $.01y \leq .12$, and bound the remaining terms by .0033. This gives the clipped extension is bounded in $[-8, 0]$, so $-8h \leq F_R \leq 0$.

synman-Kac potential satisfies the conditional expectation identity for the integral. Consequently $e^{-\rho r} F_R(h-r, U_r, Y_r) + \int_0^r e^{-\rho s} \tilde{S}(U_s, Y_s) ds$ is a martingale. This identity on a bounded verification rectangle is legitimate because of the boundedness of $\tilde{S}$. Inside the rectangle $\tilde{S} = S$, so adding the identity to discounted Itô increments its residual majorant. This argument does not require an assumed terminal value, or differentiability of the clipped source at its outer faces.

ation with $\theta = 0$ uses $\kappa = .00125\lambda^2$, giving 50 and 72 for $|\lambda| = 200$ and $l = -.025dt - .3dZ^x$. Each exponential localization term with $|\lambda| = 22$ is a generator because $22.33 = .025(22) + .045(22)^2$ dominates either sign. Therefore, the factor $8h$ supplies at least $8h$ at its associated artificial face.

$y \geq .2$,

$$W_0 - G = y(x - .5) + .1\log(.5/x) \geq 0 :$$

at .5 and has derivative $y - .1/x \geq 0$. At an actual wealth exit the source is $W_0 + F_R + Z_R \geq g$. At a terminal time it is at least G . At any artificial exit term pays for the entire possible source deficit, giving continuation value g . A preference exit cannot precede these inner preference faces. Invoke $g \leq G$ after an artificial exit and apply the stopped identity before it. This is the original adaptive consumption and portfolio controls. The value of Y_0 is an indicator, not an imposed wealth price or policy restriction. *Q.E.D.*

H.3. Validated evaluation of the restricted witness

independent standard Gaussians and put $r = v^2$. At the central state,

$$v\{-.25z_1 + \sqrt{.9375}z_2\}, \quad l = \log(1.55) - .025v^2 - .3vz_1.$$

on is integrated with weight $2ve^{-.04v^2}\varphi(z_1)\varphi(z_2)$ on $v \in [0, 1]$. The box is strictly inside the stated u, y verification rectangle under this transformation. The probability is at most $4e^{-18}$. Since the extended source lies in $[-8, 0]$, omission

$$1)^2}, \qquad Q_{uu} = -\frac{1}{(u-1)^3},$$

$$(u-2), \qquad S_l = -yc + .01y,$$

$$L^2/u + .0016, \qquad S_{ul} = -\chi ycL/u,$$

$$/u + .01y.$$

ously differentiable across clipping surfaces; its gradient is locally
 wise second derivatives give a valid integral Taylor bound across such a
 displayed (u,l) transformation by the chain rule to obtain a Hessian in
 diate expression is enclosed on the entire box. Bounds on the absolute
 usual one-half diagonal and full mixed-term Taylor remainder.

Taylor terms are integrated exactly for their polynomial approximation
 weighted moments through order two use 111 terms of $e^{-z^2/2}$ and
 $|\leq 6$ by $e^{18}18^{111}/111!$. Time moments use thirteen terms of $e^{-.04v^2}$
 nder. Source exponentials are evaluated with degree-40 Taylor poly-
 $[-3,3]$, with error at most $e^33^{41}/41!$. All antiderivatives, remainders,
 ms are rounded outward. The two diagonal second moments are non-
 ir enclosures with the nonnegative half-line is valid. Mixed moments
 ed by cell-radius products times probability mass.

unts are 16,384, 131,072, and 1,048,576. Their intervals enclose the
 . Only an upper endpoint is used to bound the restricted optimum.
 er endpoint with the independently certified restricted-policy lower
 interval of width less than .009047. The distinct width below .000320
 he witness, not the primal–dual gap.

CONTINUOUS ACTION AND ABSOLUTE-VALUE CERTIFICATION

scalar global action bound and interval neural jets

5: For any action with $\sum_i a_i = z$ and any real λ ,

$$f_i(a_i) = \sum_i \{f_i(a_i) - \lambda a_i\} + \lambda z \geq D(\lambda) + \lambda z.$$

$3\sin z$ is bounded below by $-.3$. For $z = z_0 + e$, $|e| \leq r$, its quadratic

$$\sin z \geq \lambda z_0 + .3\sin z_0 + (\lambda + .3\cos z_0)e - .15e^2,$$

scalar f_i has second derivative $(2 + .2\cos 2s)/d \geq 1.8/d = m$. Thus
 y convex. Its tangent quadratic at any feasible s_0 gives a lower bound
 an endpoint, only a feasible descent direction is relevant: an outward-
 ection contributes zero, while an inward-pointing direction contributes

ing the minimum surviving lower bound preserve a global enclosure. Root
 accuracy but not validity. Finally $|H_p(a) - H_{\hat{p}}(a)| \leq 2 \sum_i \delta_i$ on the action box.
 upper and true minimum lower can each worsen by at most this amount,
 o correction.

with stored weights, interval automatic differentiation propagates both
 al derivatives. A tanh layer is evaluated as $(e^{2z} - 1)/(e^{2z} + 1)$ with a
 minator and derivative $1 - \tanh^2 z$, all in multiprecision intervals. The
 is differentiated as its mathematical formula, including the exact $\sqrt{d}$, not as
 autograd constant. This gives a componentwise interval for the derivative
 work. Subtracting the stored floating-point gradient gives the δ_i . The two
 inherited eight- and sixteen-dimensional learned checkpoints and the same
 ificates. The summed errors are below 4.584×10^{-17} and 7.877×10^{-17}
 the local jet checks, not whole-state residual bounds.

A lower bound retaining the nonquadratic terminal payoff

DITION 3: For $|a| \leq 1$, $\sin(a)/a \geq \sin 1$, with the continuous value at
 $^2 + .1 \sin^2 a \geq r_0 a^2$, where $r_0 = 1 + .1 \sin^2 1$. Also $.3 \sin(\sum_i a_i) \geq -.3$.
 st by $r_0 \|a\|^2/d - .3$ decreases every original control's cost; allowing all
 infimum further. The original terminal payoff q_d is unchanged.
 equation is

$$\Delta v - \frac{d}{r_0} |\nabla v|^2 - .3 = 0, \quad v(1, x) = q_d(x), \quad \sigma^2 = .32.$$

and use the logarithmic heat representation. At $X_0 = 0$ it gives

$$) = -.3 - \lambda \log \mathbb{E} \exp\{-q_d(\sigma Z)/\lambda\}, \quad Z \sim N(0, I_d).$$

and coercive quadratic terminal terms give finite positive expectations. The
 on is smooth before the terminal date and has at most quadratic growth.
 of its verification identity, or truncation followed by monotone Gaussian
 the relaxed lower bound without assuming a smooth solution of the original
 lem.

re for the quadratic terminal component. Its Gaussian normalization con-

$$\frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3.$$

sian law the scalar sum divided by $\sqrt{d}$ has mean $m = 2\sqrt{d}/(r_0 + 4)$ and
 $(r_0 + 4)$. Orthogonal coordinates disappear, leaving

$$+ \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda \log \mathbb{E} \exp\{-(.2/\lambda) \cos(m + sZ_1)\}.$$

ce and then apply the decreasing logarithm with outward rounding.egration uncertainty in the relaxed value. Relaxing the action boundeave a large structural gap even when this quadrature width is small.

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ENDIX J: REPLICATION AND SCOPE LEDGER

certificate is independent of the inherited killed-Euler discretization.ified continuous-generator residual, an explicit source-polynomialce inequality, and exponentially bounded artificial-face continuation.ect deterministic wealth, interval Gaussian polynomial moments, anion remainder, outward time integration, and an exact-exit probabilityupper cancels continuous portfolio controls algebraically and evaluatesstic potential with Taylor, Gaussian-tail, and localization allowances.ish the still-uncomputed operator discrepancy budget for the old finite

ent has a different status. Each declared sample-and-hold policy is aol in the stated additive-noise dynamics. Real-arithmetic transitions aret the stored path estimates use floating-point network evaluations and a. The rigorous lower bound supplies an absolute reference; subtractingoes not create an outward policy-regret certificate. Paired path errors,-update clock overshoot, setup, and warmup are recorded separately.erns the frozen joint training-and-audit pipeline under independentertify an underlying PDE or imply that the six panels represent sixilies.

erated from raw JSON or path records; display endpoints are roundedon. The frozen external protocol, all tuning trials, all 72 holdout pairs,th arrays are retained. The validation program checks coverage of theinitialization hashes, pinned author revision, finite outputs, refinementility, the positive access lower bounds, and the explicitly false flexibleomplete R8, R4, and delivered R5 textual and numerical material ison supplement, and earlier source files remain byte-for-byte unchanged.a historical manufactured-payoff test is used as new original-economy

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READING MAP

contains the affine-preference dual, its complete verification proof, the estimates below .01, the quantitative output-error allowance, the economic model, and every main R9 theorem or its explicitly preserved appendix counterpart. The joint response to the R9 referee accompanies the paper in the R10 revision.

This supplement reproduces the completed R9 paper, including its older .10634 flexible-format version, followed by the complete R9 supplement, which preserves R8, R4, numerical material, recursive utility, sophisticated temporal selves, games, and other utilities, finite safeguards, and historical diagnostics. Historical claims must be labeled with original version labels. The R10 current status is determined by its own the-ory/source/result/build manifest, not by the older precision flags reproduced in the R9 paper.

PART I: COMPLETED R9 PAPER, PRESERVED IN FULL

ing a stronger independently specified feedback baseline. valuation, optimality, computational cost, and the economic shes.

Neural pol80 iteration, Endogenous preferences, Recur-onal economics.

1. INTRODUCTION

or economic analysis only when its error is small relative requirement is distinct from a small training loss. It is also learning algorithm. When a decision maker can change a gain may reflect either the adjustment technology or an hout that technology. When termination is an economic st exit can dominate both effects.

perators (NBO): separated policy evaluation and feasible neural networks and assessed by independent verification. onsumption, portfolio choice, stochastic risk aversion, analysis retains its original running utility, continuous and first-exit liquidation covenant. Neither reflecting the g the running payoff is used to establish the economic

y positive, verified lower bound on the value of access to initial state and adjustment-cost coefficient $k = 2$, access 1994 in the model's fixed utility units. The corresponding and .00781212 at $k = 8$. These are comparisons with the merely with an arbitrarily chosen no-adjustment policy. flexible economy's lower bound. An auxiliary market-economy's upper bound while allowing every original icy. A scalar state used to construct the witness does not

ree parts. First, smooth one-sided witnesses bound the ning a smooth optimal value or regularity of the boundary certification are separate operations. A polynomial fitted converted to outward Bernstein coefficients and refined

policy surface.
 inner optimization needed for verification. In a cou-
 nvex coupling depends on the sum of the actions.
 culation, together with strongly convex scalar dual
 action box without a tensor action grid. Executed
 h 128 action dimensions. At the eight- and sixteen-
 al differentiation of the actual network weights also
 ic gradient. This is an implemented bridge between
 te, not an assumption that an optimizer’s stationary

a absolute accuracy axis. A pointwise running-cost
 admit a Cole–Hopf representation that retains the
 nogonal Gaussian coordinates integrate analytically,
 outward enclosure. The resulting number is a lower
 zation problem. Relative performance is evaluated
 three dimensions, two training budgets, twelve seed
 budgets, warmed initialization, randomized method
 feedback rule. The pinned author implementation of

mation, and primal–dual numerical methods supply
 argument is not claimed as a new general verification
 ation and improvement claimed as a new algorithm-
 ctive combination of original-contract verification,
 unds, computable neural-jet allowances, and a non-
 a error quantity matched to an executed calculation.
 y theorem with neighboring methods. Recursive util-
 games, stochastic-trace identities, counterexamples,
 e preserved in the supplement rather than being used
 ent application.

ECONOMIC ENVIRONMENT

es relative risk aversion and x denotes wealth. An
 mpact set

$$\times [-.2, .2] \times [-.5, .8].$$

$$.1 \log x, \quad g(t, u, x) = G(u, x) - 8(1 - t). \quad (4)$$

e

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$$S_r, a_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \Big], \quad V_k = \sup_{\pi} J_k^\pi. \quad (5)$$

θ to zero and retains *all* admissible consumption and c . The cost experiment holds the controls, transition law, randomization, and consumption units fixed. A state-dependent experiment defines another economy.

the continuation covenant. There is no continuation consumption or lower wealth boundary the largest admissible wealth process, $p = 0$ eliminates the wealth diffusion and permits an empty rectangle nor a theorem requiring uniform controlled-stopping for this contract. The verification below requires only the trace at which the process is actually stopped.

$$c]v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$(7)$$

sign is not inferred solely from the negative shock correlation

tion at a fixed jet: *For the generator in (7), a global componentwise. Consumption is $\text{proj}_{[.05,.8]}(v_x^{-1/u})$ when u is $\text{proj}_{[-.2,.2]}(v_u/k)$. Let $h_1 = .06xv_x - .0025xv_{ux}$ and portfolio is $\text{proj}_{[-.5,.8]}(-h_1/h_2)$; if $h_2 \geq 0$, an endpoint is chosen with a fixed tie rule.*

and zero-curvature portfolio objectives and nonpositive continuous generator, not to the interpolated, killed-Euler approximation of global optimization over shared neural parameters.

algorithm. For $J_n^a w = r_n^a + \delta P_n^a w$, positive sub-policy-evaluation residual $e_n = \|v_n - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$ and $\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$. If the terminal error

$$\gamma^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} (2e_j + \eta_j). \quad (8)$$

l-correction proof are reproduced in the supplement. The policy changes changed while leaving the cost of a complete particular, the retained 125-action tensor audit is not 5. Its floating-point envelope is not relabeled as an

bridge from a fitted network

$= h$.

: Witnesses are bounded, continuous on the stopped slab. Localized Itô identities pass to the stopping faces an upper witness has nonpositive forward jumps jumps. The implemented policy has a well-defined

erison: Suppose Assumption 1 holds, $U \geq g - b_U$ and $J \leq \epsilon_U$, and $R_\pi L \geq -\epsilon_L$, where all four allowances

$$-t)\epsilon_L, \quad V_k \leq U + b_U + C_\rho(T-t)\epsilon_U, \quad (9)$$

$$+ C_\rho(T-t)(\epsilon_U + \epsilon_L). \quad (10)$$

optimizer or a smooth optimal value.

te for neural output: Let a continuous neural witness assumption 1, with stopped-trace error at most b . Let

er its parameters converge.

nection to network errors, not a claim that an empirical
m. Inter-84 differentiation or weight-based derivative
y may be conservative, and covering a high-dimensional
executes the jet-to-action part for actual learned critics.
an independent polynomial witness; it does not falsely
neural critic merely because its action audit passed at

CERTIFICATES IN THE ORIGINAL ECONOMY

section concern $(t, u, x) = (0, 2, 1.25)$. A central-state
m certificate for every state in D . The policy used here
al subclass. The retained lattice-trained, state-feedback
and their previous continuous bounds remain separately

tness without importing an optimal value

ution domain, not the economic domain, to $u \in [1.5, 2.6]$.

$$\sum_j \beta_{ij} T_i (2h - 1) T_j \left(2 \frac{u - 1.5}{1.1} - 1 \right).$$

ethod-of-lines initializer and residual least squares. The
the certificate. For $w = G + F$, the consumption and
minimizing the remaining wealth term over $x \in [.5, 2]$ gives

$$u - .04F + H_k \{F_u - .04(u - 2)\} + \frac{.8^{1-u}}{1 - u} \quad (12)$$

$$+ .00195 + .00352 - .04 - .004 \log 2, \quad (13)$$

on when an artificial verification face is reached. The
yond that face. At the central state the localization
r than an eight-unit initial gap. Uniform residual
ms and are reported separately.

y and independent stopped evaluation

-unit tanh layer, and two outputs. At sixteen fixed
ent to $[0, .2]$, and a differentiable budget projection
portfolio is zero. Training uses the original expected
imal-value or greedy-action labels. The deployed
onstants on each time slab; there is no claim that this
ack opportunities in (5).

$$e^{-.02r} dr < 1.25 - .5e^{-.02}. \quad (16)$$

deterministic. Since consumption exceeds $.02(1.25)$,
terminal value remains strictly above $.5$. Wealth never
controlled by a Brownian maximal inequality, not by a

ng quadrature approximates utility by a degree-90
ic remainder on the original preference interval. Free
rence, integrated over outward time rectangles, and
ximal probability bound and Cauchy–Schwarz. The
olynomial approximation, time integration, exact-exit
There is no Euler, action-grid, or spatial-interpolation
consequences of the deployed policy and analytic
ed zero.

omy bounds. Table II makes the constructive refine-
ally more precise policy evaluation than optimality

Bernstein boxes	Residual upper	Sec.	Regret upper
36	0.00477	0.079	0.110237
57	0.00195	0.116	0.107396
86	0.000894	0.186	0.106339
86			

ator. Bernstein boxes refine the already fitted upper witness. The two
ation and subdivision stages; complete timings, including coefficient
inking evaluation or residual allowance need not eliminate the
etween the chosen upper witness and policy.

ependent of the policy and valid over the full original
ealth-specific information is economically consequential.
rogramming pilot is retained with its computation log and
alidated collocation objective is not used in either table.

VALUE OF PREFERENCE ADJUSTMENT

$= A(\pi) - k B(\pi)$, where

$$\theta_r^2/2\,dr,\qquad 0\leq B(\pi)\leq .02C_\rho(T-t).$$

d cost comparisons: *The optimal value is convex and
st, their adjustment budgets are nonincreasing across
e ε_i -optimal at k_i , with $k_2 > k_1$. Then*

$$) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}.$$
(17)

an optimizer at k , for $k_- < k < k_+$, lies in the primitive

$$\frac{\overline{V}(k_+)}{-k}, \frac{\overline{V}(k_-) - \underline{V}(k)}{k - k_-} \Bigg].$$
(18)

adjustment expenditures, not a pointwise ordering of
mparative-static proof. The central full-value enclosures

$$y) + .1 \log(.5) - .02(u - 2)^-,$$

exactly. With $\theta = 0$, define

$$\left. \begin{aligned} &yc \} \\ &.00005 - .004 \log(.5) + .0008(u - 2)^2. \end{aligned} \right\} \quad (21)$$

$S < 0$. Extend S to a bounded function $\tilde{S} \in [-8, 0]$ on the remaining faces. Let

$$y \int_0^h e^{-.04r} \tilde{S}(U_r, Y_r) dr, \quad (22)$$

Thus $-8h \leq F_R \leq 0$.

Upper bound: Let Z_R be the sum of the following

$$+ 50h + e^{-240(2.6-u)+72h} \}, \quad (23)$$

$$+ 22.33h + e^{-22 \log(12/y) + 22.33h} \}. \quad (24)$$

$$.55) + F_R(1, 2, 1.55) + Z_R(1, 2, 1.55). \quad (25)$$

final consumption and portfolio controls with $\theta = 0$. Policy with certified payoff lower bound $\underline{J}_k$ and any

$$2, 1.25) \geq \max\{0, \underline{J}_k - \bar{V}_R\}. \quad (26)$$

two-Gaussian representation. We transform time by terms with interval-enclosed weighted moments, box. Exponential Taylor remainders, Gaussian tails, the executed refinement in Table III; it does not estimate. The final restricted-optimum upper bound

ent budget	Access gain: lower	Access gain: enclosure
3159557]	0.06274788	[0.06274788, 0.17885870]
5505498]	0.04094994	[0.04094994, 0.15633486]
8262028]	0.00781212	[0.00781212, 0.12249308]

learned policy. The access columns bound the difference between the a feasible flexible policy with the restricted dual upper; the upper ble restricted policy. The positive lower endpoints are informative d access intervals are not narrow estimates of the welfare magnitude.

y rounding. The evaluated restricted policy has payoff e restricted problem itself has an initial optimal-value

ctly positive at all three prices. At $k = 2$, the quadrature more than two orders of magnitude narrower than the establishes a minimum economically meaningful gain ed policy is optimal. It does not identify the exact access magnitude, or an optimal pointwise policy ordering.

CERTIFICATION WITHOUT A TENSOR GRID

problem, which is also used for the external comparison:

$$\bar{2} dW_t, \quad X_0 = 0, \quad a_t \in [-1, 1]^d, \tag{27}$$

$$1 \sin^2 a_i \} + .3 \sin \left(\sum_i a_i \right), \tag{28}$$

$$6)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right), \tag{29}$$

$$(a_t) \, dt + q_d(X_1) \Big].$$

onconvex actions. No analytic optimal-action or optimal- t a fixed critic gradient p , the action-dependent Hamilto-

$$f_i(s) = \frac{s^2 + .1 \sin^2 s}{d} + 2p_i s, \quad z = \sum_i a_i.$$

t least $m = 1.8/d$.

and convex dual bounds: *For any multiplier λ and*

$$\min_{|s| \leq 1} \{f_i(s) - \lambda s\}.$$

lower bound for the full Hamiltonian is

$$z_0 - |\lambda + .3 \cos z_0| r - .15 r^2. \quad (30)$$

d. *Each inner convex minimum can itself be bounded*
 $f_i - \lambda s$ and e is the derivative magnitude in a feasible
magnitude at an interior point. Outward versions of
on and pruning, give a global gap on $[-1, 1]^d$.
the neural gradient by componentwise bounds δ_i , the
ost its stored-gradient gap plus $4 \sum_i \delta_i$.

es multipliers and actions. It is not trusted to solve
inaccurate proposal remains valid after the strong-
oves tensor enumeration; it does not establish a
ve a generic nonseparable nonconvex Hamiltonian.
uses 339 scalar boxes and 43,392 scalar convex
training nor the preceding finite tensor safeguard in
n-dimensional jets, interval forward differentiation
gradient discrepancies below 4.584×10^{-17} and

owances are propagated into the action gap. Theorem 2
d residual quantities that would be needed for a whole-
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D AND A FROZEN EXTERNAL COMPARISON

ound for the unchanged nonconvex family

d $\lambda_d = \sigma^2 r_0 / (2d)$. Since $\sin^2 a \geq \sin^2(1)a^2$ for $|a| \leq 1$,
 $-.3$ and allowing unbounded controls gives a relaxation.
ained.

absolute lower bound: *Let Z be normal with mean*
 $+ 4)$. *Then*

$$+ 4/r_0) - .3 - \lambda_d \log \mathbb{E} \exp \left[-\frac{.2}{\lambda_d} \cos Z \right]. \quad (31)$$

expectation yields a rigorous numerical lower bound for

ound device, not a new training problem with a known
ine, which couples all state coordinates, remains in the
ed analytically and its interior is integrated by interval

and exact deployment interpretation

l actor and critic architectures: two tanh hidden layers
output, and the same terminal-value embedding. NBO
ement with Adam, three critic steps per actor step, 256
OC-MartNet uses its pinned author solver, RMSprop,
herited $J = 2$, $K = 1$ multiplier setup. These different
er than described as identical implementations.
mensions 8, 16, 32 and training budgets of 10 and 30
thods receive the same six validation trials: two seeds

O excess	SOC excess	NBO sec.	SOC sec.
0.6662	1.1692	10.002	10.006
0.6635	1.1407	30.004	30.008
0.6707	0.9938	10.002	10.005
0.6674	1.0500	30.004	30.006
0.6686	1.4883	10.003	10.007
0.6672	1.3090	30.003	30.007

NBO: R12 PRESERVATION SUPPLEMENT

the rigorous original-value lower bound. It provides an absolute enclosure: sampling error and floating-point rollout evaluation are omitted-update overshoot and exclude separately recorded warmup and setup.

tion minimizes mean independent validation cost, rollout seeds, 720–731, are then evaluated at each actor and critic starting weights are identical and warmup precedes both clocks, and method order is seed-training clocks are retained. The baseline $a_i(t, x) =$ specified independently from the associated quadratic and nonquadratic objective.

sample-and-hold control on 128 dates and evaluated paths. In (29), drift under a held action and volatility reduction and running-cost integral are therefore exact in *real-time policy*. A 64-date evaluation uses aggregated rollout policy. Its difference from the 128-date result is joint network evaluation and Monte Carlo uncertainty is a stochastic lower bound in Table VI.

author implementation in mean cost in all six panels, in dimensions 8 and 32, NBO has lower paired cost multiplicity-adjusted sign-test value is .001465. In dimensions 8 and 32, so these panels do not establish a robust improvement. Tripling the budget gives only small improvements in mean cost: .663 to .4630, and .3813 to .3799 respectively. Most quadratic-feedback baseline has lower mean cost than the rollout costs are .5121, .1740, and .08544 in dimensions 8 and 32. The stated NBO–SOC–MartNet comparison at the

pared with methods that establish bounds, not only with Theorem 1 is an application of classical stopped verification to the role here is to handle the actual liquidation trace without regular optimal value. Theorem 2 supplies a computational advantage. Neither theorem claims that classical verification

approximations with comparison provide convergence to viscosity solutions (Crandall and Souganidis (1991). Controlled Markov-chain approximation is used originally by Kushner and Dupuis (2001); adaptive sparse-grid approximation of dynamic models in Brumm and Scheidegger (2017). Such approximation schemes and their hypotheses. The finite telescoping approximation of the operator discrepancies needed for a continuous approximation does not supply them. The original-payoff certificates approximate the operators or exact Gaussian moments directly, with separately approximated remainders. They do not assert a convergence rate for the

methods also pair feasible decisions with bounds. Brown, et al. (2010) pair optimality with penalties and establish weak and strong convergence. Brown and Smith (2011) use related bounds in portfolio optimization. A no-adjustment construction is a market-deflator witness to a small localization; it is not a claim that policy-plus-upper-bound is optimal. Priour, Prieur, and Trélat (2008) develop occupation-measure approximation of nonlinear optimal control under their stated assumptions. The convergence theorem for the present logarithmic, stochastic, bounded-source bounds, interval polynomial conversions, Gaussian approximation here instead.

Justify the model-specific constructions responsible for the approximation. It couples a fitted generator witness to a small localization approximation of the portfolio term before computing a bounded-source expectation. The coupling and strong convexity of the remaining scalar approximation, optimality and branch-and-bound construction, not a generic approximation of optimization. Proposition 3 preserves the nonquadratic approximation of the exact comparison problem.

PDE solvers are important computational antecedents. Approximation of continuous-time policy improvement. Kim, Cho, Kim, et al. (2010) approximate policy-network, and residual errors in entropy-regularized

imum value of access to preference adjustment in so by combining a feasible learned policy with an restricted optimum, rather than by assuming either nomy is enclosed within .00905 at the central state. ore accurate than its present .10634 regret bound at remains a distinct numerical target.

onal ways to improve and audit these calculations: an-moment evaluation, market-deflator localization, erative continuous-action oracle. The coupled com- ute lower bound and a frozen multi-budget, multi- use results sharpen the connection between neural hile leaving every inherited derivation, experiment, ther review.

TION AND COMPUTATIONAL ERROR

comparison, and the neural bridge

ed jet the three controls enter additively. The con- $_x$ and second derivative $-uc^{-u-1} < 0$. If $v_x \leq 0$ it maximizer is $v_x^{-1/u}$. The adjustment objective is $k > 0$. The portfolio objective is $h_1p + h_2p^2/2$. A nonnegative h_2 gives a maximum at an endpoint, Compactness and the stated tie rule give a feasible space statements at a fixed jet, not parameter-space *Q.E.D.*

process on interior compact sets and apply discounted h time slab. The discounted terminal witness plus ess, the expected integral of R_aU , and the expected ter sum is nonpositive. Boundary comparison gives and the deterministic bound $\tau \leq T$ therefore imply

$$e^{-\rho(r-t)}dr \leq U(t,s) + b_U + C_\rho(T-t)\epsilon_U.$$

positivity of the continuation operators: propagate the optimality error with one factor δ , and add the terminal error b gives (8). The full state-dependent operators are reproduced in the preservation layer, a continuous interpretation additionally needs either an independent continuous verification route. The present paper.

Arithmetic convention

By using the economy are exact decimals. Stored network coefficients are interpreted as exact binary64 numbers. Coefficient conversion use 70-digit interval arithmetic; at least 50 digits. Vectorized interval addition, subtraction, multiplication, positive interval, and pairwise summation round both to a binary64 number. An interval square may be enclosed by a conservative even when its lower endpoint is negative. Interval division is treated as a rigorous interval merely because its denominator is positive. Exponentials in vectorized routines are replaced by interval logarithms, trigonometric constants, and a precision layer. The paper concerns the mathematical function represented by the stored coefficients, a point implementation of an arbitrary neural policy. The stored constants, so their continuous implementation is not. Outputs are explicitly floating-point performance estimates. Numerical table.

ORIGINAL-ECONOMY FITTED WITNESSES

Global and localized upper bound

Consumption and portfolio selection give $c = .8$ and $p = .8$. The utility is $.000512/k \leq .001024$ for $k \geq .5$. The preference part of the utility is maximized at the upper endpoint $u = 2.8$.

$\frac{1}{2} + \kappa(-\kappa + \lambda_0 + .00125\lambda - \rho)\}.$

satisfy $\kappa = .2|\lambda| + .00125\lambda^2$. Both terms therefore
 y admissible control. Moreover $(\partial_t - \rho)C_\rho(1 - t) =$
 ual inside the verification domain.

At an actual wealth exit, $F \geq -8h$ implies $U \geq G -$
 one summand of Z is at least $8h$, so $U \geq G$. Stop first
 the artificial face occurs first, the remaining original-
 ry admissible continuation, by the global fallback
 tion gives the asserted bound without modifying the
Q.E.D.

Source enclosure and refinement

$.5$, $B = .55$, and $|z| \leq 1$. Let $D = (A^2 - B^2)^{1/2}$ and
 rgent expansion

$$\left[1 + 2 \sum_{n=1}^{\infty} (-q)^n T_n(z)\right]$$

by $2q^{17}/\{D(1 - q)\}$. If $L = -\log(.8)$, approximate
 aylor polynomial of e^{LBz} . Its remainder is at most
 closures, with the relevant sup-norm bounds, gives
 $^{-9}$ for $-.8^{-(u-1)}/(u - 1)$. This error is added to the

$\leq -.2k$, $q^2/(2k)$ for $|q| \leq .2k$, and $.2q - .02k$ for
 eefficients enclose q and every potentially applicable
 itching threshold includes both adjoining branches.
 ervals; binary64 representation of $.2k$ cannot exclude
 between its minimum and maximum Bernstein coeffi-
 this enclosure and is repeated on boxes whose upper
 The independent check of $F/h \geq -8$ uses the same

al lies strictly below a ceiling, Bernstein refinement
 plicitly accounted source and arithmetic allowances.
 ixed candidate with slack. It is not a theorem that
 z, or arbitrary Adam training produces approximants

at some $t \leq 1$; a lower exit requires $W_t^u \leq -16$ and is
The exponential martingale maximal inequality on each

$$2e^{-72} < 9677 \times 10^{-31}.$$

l stopped at exact first exit. Its rare-exit correction is not

ore exit, $|r| \leq .8$ and utility is $-e^{L_j} e^{L_j r} / (1+r)$. Replace
e exponential by its degree-ten Taylor polynomial. The
the exponential remainder is at most $e^{8L_j} (.8L_j)^{11} / 11!$.
rocal and exponential suprema, yields a uniform error for
and holds on the whole original preference interval, not

mean μ and variance v , its raw moments satisfy $M_0 = 1$,

$$M_{n-1} + (n-1)vM_{n-2}.$$

s recurrence with $v = .0025t$, and the terminal quadratic
ussian expectation. On each time rectangle the intervals
nial coefficients enclose the entire integrand. Outward
length therefore enclose its integral. These ranges are
d from two discretizations. The three resolutions are
egral.

nomial calculation with the stopped economic payoff,
ff E the two paths agree through T . On E , the omitted

$$\|P_j(u_t-2)\|_2, \qquad \|P_j\|_2 \leq \sum_m |b_{jm}| \sqrt{M_{2m}}.$$

uniformly by evaluating their nonnegative coefficient
e same argument bounds the free terminal settlement,
ude below 8.1. Discounted adjustment cost omitted after
stopping allowance is thus

$$\left. + \|G(u_1,x_1)\|_2 \right\} + p_{\text{exit}}(.02k+8.1).$$

y, $J_k(\pi) = A(\pi) - kB(\pi)$ is affine and nonincreasing, convex and nonincreasing. The two approximate

$$_1, \quad A_2 - k_2 B_2 \geq A_1 - k_2 B_1 - \varepsilon_2. \quad \text{NBO: R12 PRESERVATION SUPPLEMENT}$$

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es (17). Zero allowances give budget ordering for
s optimal at k , feasibility at k_+ implies $V(k_+) \geq$
implies $V(k_-) \geq V(k) + (k - k_-)B(\pi_k)$. Substitute
and intersect with the primitive budget range to obtain
Q.E.D.

$V_k \geq J_k$ and $V_R \leq \overline{V}_R$. Its nonnegativity follows
ned in the flexible policy set. Likewise $V_k - V_R \leq$
o optimal values even though the feasible policies
ptimal.

portfolio and stopped localization

cified, quadratic covariation gives

$$-Y_t c_t) dt + \text{local martingale.}$$

pYx covariation term for every adaptive p . Including
e quadratic yields

$$.05 - .004 \log(.5) + .0008(u - 2)^2 \leq S(u, y).$$

ive. Correlation is retained in the transition law of
l.

ble consumption .35: its utility on $u \in [1.5, 2.6]$ is
 ≥ -4.2 . The remaining terms sum to a nonnegative
1 > -8 . For an upper bound discard $-yc$, maximize
nd bound the remaining terms by .0033. This gives
bounded in $[-8, 0]$, so $-8h \leq F_R \leq 0$.

l satisfies the conditional expectation identity for
ly $e^{-\rho r} F_R(h - r, U_r, Y_r) + \int_0^r e^{-\rho s} \tilde{S}(U_s, Y_s) ds$ is

$\geq g$. At a terminal time it is at least G . At any artificial
 r the entire possible source deficit, giving continuation
 it cannot precede these inner preference faces. Invoke
 official exit and apply the stopped identity before it. This
 98 consumption and portfolio controls. The value of Y_0 is an
 imposed wealth price or policy restriction. *Q.E.D.*

valuation of the restricted witness

Gaussians and put $r = v^2$. At the central state,

$$\overline{375} z_2\}, \quad l = \log(1.55) - .025v^2 - .3vz_1.$$

with weight $2ve^{-.04v^2}\varphi(z_1)\varphi(z_2)$ on $v \in [0, 1]$. The box
 ated u, y verification rectangle under this transformation.
⁻¹⁸. Since the extended source lies in $[-8, 0]$, omission
 $[-8, 0]$, whose width is less than 4.778×10^{-7} . The initial
 $\times 10^{-9}$.

l by the evaluator is explicit. Let $c = \text{proj}_{[.05, .8]} e^{-l/u}$,
 for an interior consumption solution and zero for a strictly
 g box use $\chi \in [0, 1]$. With $Q = c^{1-u}/(1-u)$,

$$Q_{uu} = -\frac{e^{L(u-1)}[\{L(u-1)-1\}^2+1]}{(u-1)^3},$$

$$S_l = -yc + .01y,$$

$$S_{ul} = -\chi ycL/u,$$

ntiable across clipping surfaces; its gradient is locally
 derivatives give a valid integral Taylor bound across such a
) transformation by the chain rule to obtain a Hessian in
 on is enclosed on the entire box. Bounds on the absolute
 diagonal and full mixed-term Taylor remainder.
 are integrated exactly for their polynomial approximation
 ments through order two use 111 terms of $e^{-z^2/2}$ and

tion bound and interval neural jets

n with $\sum_i a_i = z$ and any real λ ,

NBO: R12 PRESERVATION SUPPLEMENT

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$$t_i) - \lambda a_i\} + \lambda z \geq D(\lambda) + \lambda z.$$

l below by $-.3$. For $z = z_0 + e$, $|e| \leq r$, its quadratic

$$\sin z_0 + (\lambda + .3 \cos z_0)e - .15e^2,$$

ond derivative $(2 + .2 \cos 2s)/d \geq 1.8/d = m$. Thus
 ent quadratic at any feasible s_0 gives a lower bound
 y a feasible descent direction is relevant: an outward-
 zero, while an inward-pointing direction contributes
 ther by minimizing the tangent quadratic over the
 ear term. The stated lower bound remains valid if the
 ite endpoint, because relaxing the displacement can

is no larger than $D(\lambda)$. A feasible action gives a
 valuation. The scalar interval $[-d, d]$ covers every
 only a box whose lower bound exceeds a feasible
 iving lower bound preserve a global enclosure. Root
 inally $|H_p(a) - H_{\hat{p}}(a)| \leq 2 \sum_i \delta_i$ on the action box.
 um lower can each worsen by at most this amount,

Q.E.D.

interval automatic differentiation propagates both
 n layer is evaluated as $(e^{2z} - 1)/(e^{2z} + 1)$ with a
 ve $1 - \tanh^2 z$, all in multiprecision intervals. The
 mathematical formula, including the exact $\sqrt{d}$, not as
 is gives a componentwise interval for the derivative
 e stored floating-point gradient gives the δ_i . The two
 xteen-dimensional learned checkpoints and the same
 errors are below 4.584×10^{-17} and 7.877×10^{-17}
 t whole-state residual bounds.

atic terminal terms give finite positive expectations. The
 ore the terminal date and has at most quadratic growth.
 identity, 100 truncation followed by monotone Gaussian
 bound without assuming a smooth solution of the original

ic terminal component. Its Gaussian normalization con-

$$\frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3.$$

ar sum divided by $\sqrt{d}$ has mean $m = 2\sqrt{d}/(r_0 + 4)$ and
 al coordinates disappear, leaving

$$4/r_0) - .3 - \lambda \log \mathbb{E} \exp\{-(.2/\lambda) \cos(m + sZ_1)\}.$$

osition. An upper enclosure for the positive scalar ex-
 y, because $-\lambda \log$ is decreasing. That lower endpoint is
 l value. *Q.E.D.*

and $R = \lceil \sqrt{2(\eta + 50)} \rceil$. The omitted Gaussian contri-
 interval ranges of the positive integrand times cell lengths
 successive 1,024, 4,096, and 16,384-cell calculations add
 apply the decreasing logarithm with outward rounding.
 certainty in the relaxed value. Relaxing the action bound
 structural gap even when this quadrature width is small.

REPLICATION AND SCOPE LEDGER

independent of the inherited killed-Euler discretization.
 ous-generator residual, an explicit source-polynomial
 and exponentially bounded artificial-face continuation.
 tic wealth, interval Gaussian polynomial moments, an
 r, outward time integration, and an exact-exit probability
 continuous portfolio controls algebraically and evaluates

external protocol, all tuning trials, all 72 holdout pairs, pinned. The validation program checks coverage of the nes, pinned author revision, finite outputs, refinement access lower bounds, and the explicitly false flexible and delivered R5 textual and numerical material is d earlier source files remain byte-for-byte unchanged. factured-payoff test is used as new original-economy

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 .tex; its Appendix proves the new results. The accom-
 mber 22, 2026 R8 numerical-methods report at commit
 n the R9 constructive-certification branch. Earlier main
 anced by this revision.

s the original-payoff actor and moment evaluator, fitted-
 er, restricted market-deflator verifier, scalar global action
 absolute-value relaxation, and frozen external-comparison
 l from their retained JSON records. Every new trained
 ples for each of the 72 holdout pairs are retained. Source
 nds, scope checks, and failed-pilot status are in the R9

at preserved in full. References to a preceding revision, its
 s conclusions refer to that historical record. In particular,
 ot evidence for original-payoff precision, and its six-seed
 r current inference by the frozen R9 holdout. Neither
 Historical finite safeguards remain distinct from the new
 sive, temporal-self, and game derivations retain their own
 no new large-scale extension result is inferred from the

t. Its stopping and settlement contract is specified as
ults identify three distinct objects: an exact policy-
ntation, and an independently evaluated policy. None
ds.

e verification pair. A supersolution controls the value
solution controls the payoff of the particular policy
ether with explicitly signed residual and boundary
smooth test functions, not smoothness of the optimal
uity of the optimal value at an irregular controlled
inuous-generator implementation and an operator-
implementation. The latter separates time, action,
olve, and arithmetic errors; a refinement difference is

accuracy to an economic comparative static. Raising
e optimal discounted adjustment budget. For policies
ossible reversal in this ordering is bounded by the
change in price. Value enclosures also bound optimal
ound the welfare gain from access to the adjustment
quired numerical precision depend on the economic
alibrated stopping tolerance.

ementary components. We re-evaluate the retained
ction safeguard, and separate time, state, and action
verification pairs for their deployment in the original
but too conservative to resolve the stated economic
lified-payoff experiment trains a bounded actor on
and exit geometry and obtains an all-state continuous
vex control problem in eight and sixteen dimensions
pinned author implementation of SOC-MartNet. The
analytical greedy-action labels, paired independent
gn. Its outcome measures realized cost at that budget,

and their combination have important antecedents.
ous-time policy improvement. Kim, Cho, Kim, and
provement for entropy-regularized stochastic control
d residual errors. The updated interior analysis of
onary second-order equations with bounded-domain

TH COSTLY PREFERENCE ADJUSTMENT

s relative risk aversion and x is financial wealth. An
 measurable process $a = (c, \theta, p)$ taking values in

$$, .8] \times [-.2, .2] \times [-.5, .8].$$

$$(1)$$

$$dt + .2p_t x_t dZ_t^x, \quad d\langle Z^u, Z^x \rangle_t = -.25 dt. \quad (2)$$

omain is $D = (1.2, 2.8) \times (.5, 2)$, and the discount rate
 $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$. Its running payoff and

$$k > 0, \quad (3)$$

$$.1 \log x, \quad g(t, u, x) = G(u, x) - 8(1 - t). \quad (4)$$

$$) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \Big], \quad V_k(t, s) = \sup_{\pi} J_k^\pi(t, s). \quad (5)$$

opping rule, and settlement do not depend on k . Bounded
 efficiencies give a well-defined stopped state equation for
 offs on the stated domain give finite values. The cardinal
 are held fixed across comparisons. A transformation of
 e u is generally a different economy.

fied liquidation covenant, not an estimated preference
 subsequent wealth injection or continuation consumption.
 t preference diffusion cannot be confined to the stated

$= 0$ in the appropriate solution sense. All covariance is retained. A negative shock correlation by itself does not imply a negative response, and dividing a re-solved V_u by k does not change the optimal policy.

selection for a fixed jet: *Let v have the derivatives in v separately in the three control components. If $v_x > 0$, also $\theta^* = \text{proj}_{[-.2, .2]}(v_u/k)$. Write*

$$.25xv_{ux}, \quad h_2 = .04x^2v_{xx}.$$

0. If $h_2 \geq 0$, choose an endpoint maximizing $h_1p +$

marginal wealth value, zero curvature, convex portfolio. It removes action discretization from a continuous-time problem. It is not a closed-form optimizer of the killed-Euler, and it is not a global convergence theorem for shared information. The rule was present in the retained derivations; the branch is explicit.

PRESERVATION, AND FINITE CORRECTION

solves $R_\pi v = 0$ with the economic stopping contract. It maximizes the Hamiltonian of that fixed evaluation. It is exact, and exact policy iteration is $\pi_{j+1} = \mathcal{I}(\mathcal{E}\pi_j)$. If a policy is optimal, then $J^{\pi_{j+1}} \geq J^{\pi_j}$. Verification then gives $J^{\pi_{j+1}} \geq J^{\pi_j}$. More than this monotonicity. Under compactness of the jet class, closure under limit, comparison for the optimality principle. These are hypotheses on the exact operators. The stochastic gradient training. Appendix B gives the limiting parameter stationarity, policy convergence, and

implementations share the separation of evaluation and control, and are different.

for the declared finite economy

is a positive substochastic continuation matrix and $\delta = \delta_n$ is a positive scalar. Let $\hat{\pi}$ be a feasible policy and v_n a critic, with

$$\eta_n = \left\| \max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\hat{\pi}_n} v_{n+1} \right\|_{\infty}.$$

g: On this finite economy,

$$\leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

with the exact finite terminal data, $e_j = b = 0$. A state-
 $B_N = 0$ and

$$g_n = \max_a \mathcal{T}_n^a V_{h,n+1}^{\hat{\pi}} - V_{h,n}^{\hat{\pi}}.$$

the entire declared candidate economy. For the retained
 grid together with the frozen continuous neural proposal
 does not cover all continuous controls merely because
 network. Floating-point computation of the finite envelope
 arithmetic statement of the theorem.

guarded algorithm and its cost

evaluates the raw policy, computes all candidate gains, and
 unified threshold. It re-evaluates the corrected policy and
 meets the target. In the new re-audit the threshold is

$$\gamma = \frac{\varepsilon}{2 \sum_{n=0}^{N-1} \delta^n}. \quad (9)$$

THE CONTINUOUS ECONOMY

me must be stated before it can be certified. For the
 ble bilinear interpolation on the exact uniform state
 ck function is frozen in time, but is evaluated at the
 This is not a simulation that holds the initial action
 eewise-time, Lipschitz state-feedback rule for the
 t-exit contract.

$) = h$.

ion witnesses: U and L are bounded, continuous
 interior of each time slab. Their derivatives permit
 expectations are uniformly integrable. The policy π
 At interior slab interfaces either the witnesses are
 $, s)$ and $L(t^-, s) \leq L(t^+, s)$.

s verification pair: Under Assumption 1, suppose

$$\epsilon_U, \quad R_\pi L \geq -\epsilon_L,$$

$\geq g - b_U$ and $L \leq g + b_L$, where all four constants

$$-b_U + C_\rho(T - t)\epsilon_U, \quad (10)$$

$$-b_L - C_\rho(T - t)\epsilon_L, \quad (11)$$

$$-b_U + b_L + C_\rho(T - t)(\epsilon_U + \epsilon_L). \quad (12)$$

be continuous at every controlled boundary point.

networks, and need not have the same approximation
 mer witness, but verification requires an enclosure

sup_{s,a} $R_a G$ above by $-\alpha$, with $\alpha > 0$. For each frozen
by $-\beta$ on every state cell and time slab. Consequently,

$$1 - t), \quad 11\theta L = g - \beta C_\rho(1 - t) \quad (13)$$

$\alpha < 8$, $U \geq g$ on the stopped trace, while $L \leq g$. Their

$$+ (\beta - \alpha) C_\rho(1). \quad (14)$$

ms, powers, products, and all bilinear-action component
en binary64 constants interpreted as exact real numbers;
onal grid. The returned bound is rounded outward again
ifies the specified real-arithmetic feedback, not every
ochastic differential equation.

TABLE I
CONTINUOUS NDU WITNESS BOUNDS

α	β	Regret upper bound
864832	7.078878	14.091405
864832	1.211230	8.339561

d as continuously reevaluated bilinear feedback on each time slab.
n model and exact first exit using outward interval arithmetic. Both
xceed the .01 precision target.

are numerically wide. They replace an absent continuous
ficate, but do not establish regret below .01 or .05. In
sufficiently narrow welfare enclosure. The theorem and
n testable rather than hiding it in a missing discretization

the discretization accounting route

ous verification. Let $\mathcal{B}_n^*$ be the exact continuous-time
decision interval, allowing all admissible within-interval
group for the declared continuously implemented policy.

term is unknown, not zero. A coarse-fine solution
 computed tails. Nor does a regular-domain scheme
 r controlled exit boundary. The direct generator pair
 ing the continuous implemented policy instead, it does
 come vanish.

NS AT FINITE NUMERICAL ACCURACY

he notation. For each feasible policy write

$$\left[e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad B(\pi) = \mathbb{E} \int_t^\tau e^{-\rho(r-t)} \frac{\theta_r^2}{2} dr.$$

$$0 \leq B(\pi) \leq .02C_\rho(T-t).$$

approximate optimality: *If the feasible policies and
 V is convex and nonincreasing. For $k_1 < k_2$, optimal
 erally, if $V(k_i) - J_{k_i}^{\pi_i} \leq \varepsilon_i$, then*

$$B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \quad (16)$$

ξ_i , their possible reversal is bounded by the right side
 point with an attained optimizer, $V'(k) = -B(\pi_k)$.

further restrictions on the feasible policy set. The
 expenditure, not pointwise values of $\theta(t, u, x)$. The
 nic theorem for the original continuous contract; the
 ufficient accuracy to quantify its magnitude.

lgets, and access to adjustment: *Suppose $L(k) \leq$
 ined optimal adjustment budget at k belongs to*

$$\min \left\{ .02C_\rho(T-t), \frac{U(k_-) - L(k)}{k - k_-} \right\} \Bigg]. \quad (17)$$

he and the restricted $\theta \equiv 0$ values, the value of access
 $L_R]$, intersected with $[0, \infty)$.

ified running payoff. The external comparison estimates v ; it does not know the optimal value. These distinctions are subtle, not alternative names for the same error. We pulled the code from [source](#) commit 14ecbd48 in a clean Python 3.11 environment with NumPy 2.3.5, SciPy 1.17.0, and mpmath 1.3.0. Numerical integration retains the processor description, exact versions, source code, and rollout weights, all seed-level records, and the rollout results. It checks those committed records and fails on missing cells. The code runs on all seeds, not extra independent confirmatory seeds.

Policy correction through thresholded correction

We correct the policy without retraining or selecting a new policy using the thresholded-10 policy had a raw localized envelope of .0922420. The thresholded policy replaced 54.6602 percent of interior state-date decisions. The threshold in (9), at target .05, replaces only 4.9350 percent of the envelope. At target .01, the correction share is 4.9350 percent. Thus most of the earlier replacements were unnecessary given the low tolerance in this sample.

TABLE II
THRESHOLD CORRECTION OF RETAINED NDU POLICIES

Final bound	Override (%)	Audits	Seconds
0.042805	0.000	1	0.472
0.003278	8.796	2	0.849
0.006823	0.875	2	1.641
0.006761	4.935	2	1.630

ous-model certificates. Seconds include all listed audits and envelope construction. The table still enumerates the full 125-action grid plus the proposal. Historical results are not added across different environments.

of correction rather than concealing it. The corrected policy audit still enumerates 125 grid actions plus the neural network audit. The two audits use 2,362,200 value-action evaluations and 2,362,200 transitions each with four transition branches. The new policy is not

TABLE III
 APPROXIMATION DIAGNOSTICS

n_a	Value	Exit prob.	h^2/Δ
5	-1.451486	0.104315	0.1067
5	-1.549787	0.193358	0.2133
5	-1.608530	0.227492	0.4267
5	-1.549787	0.193358	0.2133
5	-1.444048	0.151095	0.0948
5	-1.391646	0.102389	0.0533
3	-1.606541	0.245260	0.2133
5	-1.549787	0.193358	0.2133
9	-1.536711	0.182784	0.2133

5). n_a is the number of nodes per action component. Four-sign
 changed; these differences do not bound continuous-model error.

ce of convergence. Its central value moves from
 dates, while the exit probability rises substantially.
 h^2/Δ increase, rather than shrink; the experiment ex-
 unjustified time order. Conversely, refining the state
 -1.54979 to -1.39165. Action refinement improves
 mation factors cannot be conflated. The operator-
 precision; agreement of two implementations of the
 its continuous-model bias.

lattice, and nine points in each action component,
 omy is re-solved. Table IV shows the same ordering
 with an active preference margin. This is a stronger
 point panel, but still not a verified continuous welfare
 rms in (15) remain unbounded by these calculations.
 r nor a continuous policy-convergence rate is inferred

tion transfers from an analytic example to an actually
dimensional action box, correlated shocks, state domain,
the running payoff to
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$$= \ell_2(s, a) - \sup_{b \in A} R_b g(t, s). \quad (18)$$

the chosen action, but depends on state and time. Thus
r relative to the original-payoff economy; it is not an
in the modified problem $\sup_a \tilde{R}_a g = 0$ and the explicit
, .8). Itô verification proves that the modified value is g

a constant bounded sigmoid consumption and portfolio
t output. The actor maximizes the Hamiltonian of the
labels. Six declared seeds each receive 4,000 Adam steps
tes per step. This is a controlled verification experiment,
dimensional scaling claim. The witness is known rather
nieved errors are learned.
gap is bounded over the entire state domain by outward
ain is bounded using

$$.8 - c)/x \leq (.8 - c)\{c^{-2.8} - .05\},$$

difference, and the adjustment gain equals $(\theta + .02(u -$
tition in u encloses the latter. The full regret is bounded
p enclosure. The time discretization of a rollout and a
enter this certificate, because its proof is for the original

problem target, with continuous regret bounds between
eparately reports componentwise control errors rather than
norm accuracy. This is the desired transfer test with a
control geometry. It does not identify the original NDU
ely by sharing the state equation.

$$2a_t dt + \sqrt{2}(.4)dW_t, \quad a_t \in [-1, 1]^d, \quad (19)$$

$$d^{-1} \sum_{i=1}^d (a_i^2 + .1 \sin^2 a_i) + .3 \sin \left(\sum_i a_i \right), \quad (20)$$

$$d^{-1} \sum_i (x_i - .5)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right). \quad (21)$$

nts. For example, at an interior vector whose sum is negative direction along the all-ones vector for both genuinely nonconvex in the action. No closed-form applied to either method, and the quadratic exponential hamiltonian.

ified `SOCMartNet.train` routine in the authors' at commit `991ea8dd`. Its complete module hashes ation consists of model and network adapters and update at the time budget. The authors' solver is rator. The pinned implementation uses a Hessian-use constant diffusion in this comparison. This is a used here, not an assertion that the published SOC-control.

ector and critic weights within each seed. Both use ed actor outputs, a terminal-exact critic, 16 training O uses three evaluation updates per improvement successors, and Adam with learning rate `.003`. SOC-RMSprop updates, and the declared martingale and rate is `.003/ $\sqrt{d}$` , the test-network learning rate is `.01`, d in the protocol. The test network has 600 outputs; ed rather than excluded from the method description.

cost at 128 audit steps; negative favors NBO. Each dimension has six non-random-number paths per pair, and ten seconds of training per uncertainty, not optimality loss. Maximum paired path standard errors 1098, 0.009106 respectively.

TABLE VII
TRAINING AND SETUP RESOURCES

Train (s)	Setup + train (s)	Outer updates
10.003	10.692	1536.7
10.012	10.015	465.3
10.004	10.689	1477.3
10.012	10.014	458.0

al actor/critic parameter counts within each dimension: 3,224/2,881 at t also trains its 600-output test network. Paired policy audit means are ate definitions differ across algorithms and are not equal units of work.

n NBO-minus-SOC cost is -1.71785 , with a six-seed $[-1.71785, -1.71785]$. At dimension sixteen it is -2.24732 , with interval $[-2.24732, -2.24732]$. Evidence favors NBO under the specified adapter and rated benefit in a coupled nonconvex problem, in contrast comparison. It is not a theorem of computational dominance, partNet hardware or learning curves, or a statement that

ash diagnostics, and retained failures

paths with common Brownian increments for the two each training seed. The primary outcome uses 128 time adjacent increments, so its difference from the finer audit ble VI reports the maximum paired path standard error l. The interval is based on six independent training/audit her than a distribution-free coverage claim. Paths are not

logical axes. Its entries summarize the cited analyses not a ranking by paper title. In particular, an interior a, a stopped-control regret theorem with a contractual n envelope address different mathematical objects.

TABLE VIII
AND THE OBJECTS COMPARED

r guarantee	Relation to the present evidence
for entropy- tions; iteration, DE residual er-	Neighboring policy- evaluation/improvement analysis; not the external implementation tested here
or control with ormation contri- ngularity condi-	Clarifies what interior collocation can control without supplying the stopped NDU boundary data
usable operator terminal data	Representation reuse is a different per- formance target from one trained policy at a fixed budget
value and con- Hamiltonian in-	Pinned author routine tested on coupled constrained nonconvex controls in $d =$ 8, 16
improvement, or witness ver-	Continuous stopped-model theorem, explicit economic error calibration, fi- nite correction costs, and paired author- code experiment

m, Kim, and Cho (2026), Lee and Kim (2025), Cai, Fang, and
ant diffusion and a ten-second training budget. Its scope does not
ursive utilities, games, or higher dimensions.

stinction between state dimension and action dimen-
avoid a tensor state grid during the high-dimensional

$$= \frac{\rho}{a} c^a (bV)^{1-a/b} - \frac{\rho b}{a} V, \quad (22)$$

at when $a \neq 0$; see Epstein and Zin (1989), Duffie and
 $V + f(c, V) = 0$, not an additive-reward equation with
the transform $V = \exp(W)/b$ enforces the sign domain.
unds on consumption, and a monotone implicit step are
tz bound on the evaluation operator.

ffine evaluation. If the action operators $\mathcal{T}_n^a$ are monotone
s L_n on a verified continuation-value interval, then (8)
 m , and the terminal factor by $\prod_{m=n}^{N-1} L_m$. Appendix D
architecture alone does not verify the required compact

e stationary homothetic anchors $p = .3$ and $c/x = .0455$
implicit recursive reference, and the R5 trained recursive
.25 and .0502 against a .01 target were corrected using a
results establish finite-model performance of the corrected
tion cost or a scaled neural recursive-portfolio solution.
are preserved rather than replaced by an additive-control

sophisticated temporal selves

present-bias parameter β and ordinary discount δ , the
differ:

$$[u(c)\Delta + \beta\delta\mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]], \quad (23)$$

$$\Delta + \delta\mathbb{E}[V_{n+1}(F(s, c_n(s), \varepsilon))]. \quad (24)$$

With two consumption dates, terminal log wealth, zero
rst-date share is $1/(1 + 2\beta)$, hence .4166667 at $\beta = .7$.
etric- β evaluation gives .4566210, a different behavioral
ated residual check, and feasible-deviation calculation
rbrium verification concerns each acting self's feasible
fictitious commitment planner who can change all future

standard Gaussian probes, put $\bar{q}_K = K^{-1} \sum_k z'_k A z_k$.

$$= (c - \text{tr } A/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (25)$$

residuals retains a variance penalty that does not means give an unbiased cross-product estimator onnegative sample objective or a uniform residual and the retained Monte Carlo regression are given in

, an exact contraction can use m directional second es K probe operations and the loading operations. allocation coverage, optimization iterations, and the r total cost. Avoiding a dense Hessian is useful, but crossover or eliminate the cost of verification. The h off-diagonal drift, rewards, and shocks are useful quadratic value family and faster Riccati comparator l.

CONCLUSION

ximation, an improvement rule, and a verification pped preference–portfolio economy, this separation an audit, one-sided continuous value witnesses, and t comparisons at finite numerical accuracy. The new uthor-code comparison address different questions: ized performance on coupled nonconvex controls. riginal-payoff certificate or the substantial cost of a

e recursive utility, sophisticated temporal selves, and valuation and equilibrium conditions. The remaining ciently narrow original-model witnesses, or enclose

for localization or at the economic stopping time. Localization. The hypotheses on the witness and uniform integrability of the optimal value is invoked. The admissible action. The terminal inequality $g \leq U + b_U$ and

$$\int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T-t)\epsilon_U.$$

argument is unnecessary. For $w = L$, use the specified policy $g \leq L - b_L$, and the same integration identity to obtain the regret bound, and nonnegativity follows from the definition

on each slab and sum. Forward-time jumps contribute the witness identity. They are nonpositive under Assumption 1, backward-witness jumps have the opposite sign. This explains why we may not silently omit its interface errors. The actual policy is the same because Itô's formula is applied to the witnesses, not *Q.E.D.*

Spatial residual potentials

$R_a U \leq q_U$ and a nonnegative potential obeys

$$R_a U - \rho z_U \leq -q_U \quad \text{for every } a.$$

dependence of R_a on the witness. Its boundary inequality is clearly, if $R_\pi L \geq -q_L$ and $z_{L,t} + \mathcal{L}^\pi z_L - \rho z_L \leq -q_L$, then $R_a U \leq U + z_U$ and $L - z_L$. A potential solving an occupation time inequality by reachable discounted occupation; an approximate differential inequality is enclosed. This construction is a histogram into a theorem.

The explicit original-economy witnesses

$$\rho C_\rho(1-t) = 1 - e^{-\rho(1-t)},$$

$$(-\rho)[-aC_\rho(1-t)] = a$$

box; enlarging the action range can only make the dependency can likewise enlarge the bound, never of floating-point simulation conformance is needed: stored constants and exact interpolation weights. A d be required for another implementation.

of of Proposition 1

sum of

$$v_u\theta - \frac{k}{2}\theta^2, \quad h_1p + \frac{h_2}{2}p^2.$$

introduces no product between control components. $c^{-u} - v_x$ and strictly negative second derivative everywhere. Otherwise its unique unconstrained gives the constrained optimum. The adjustment objective $k > 0$. The portfolio objective is strictly concave ex when $h_2 > 0$; the stated vertex or endpoint rule is ability and additivity make these separate maximizers
Q.E.D.

l-geometry regret bound

or the original running payoff. The modified residual the feasible explicit optimizer. Applying (26) first to s $\tilde{V} = g$. For the learned policy,

$$\int_t^\tau e^{-\rho(r-t)} \{H_g^* - R_\pi g\}(r, S_r) dr.$$

$k = 2$ the adjustment term completes the square to differences are exactly those bounded in Section 6.3. s $C_\rho(1 - t)$ gives the recorded continuous regret rate interval consequences, not inferred by reversing
Q.E.D.

is $\|v_{j_m+1} - v^*\|_{\mathcal{L}(V)}$, and continuity of evaluation gives
of the improved limiting policy therefore reads

$$R_{\mathcal{I}(\bar{V})}\bar{V} = 0.$$

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at selector implies $\sup_a R_a \bar{V} = 0$. Boundary comparison
equality of limiting values, not equality of consecutive
closure, and strong evaluation continuity are hypotheses,
chitecture or Adam trajectory. *Q.E.D.*

Proof of Theorem 1

$\|\mathcal{T}_n^a v - \mathcal{T}_n^a w\|_\infty \leq \delta \|v - w\|_\infty$. The maximized operator
the candidate set includes the enacted action,

$$\max_a \|\mathcal{T}_n^a v_{n+1}\|_\infty \leq e_n + \eta_n.$$

with optimal backward induction yields

$$\leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} (e_j + \eta_j).$$

tion gives the same inequality with only e_j in the sum.

$V_{h,n}^* - V_{h,n}^\pi$. Exact evaluation implies $g_n \geq 0$. For every
nce inequality gives

$$\leq g_n + \delta \max_a P_n^a d_{n+1}.$$

positivity give $d_n \leq B_n$. The reference optimal value is
cy evaluation, transition matrices, and feasible gains are
Q.E.D.

correction and arithmetic scope

, Theorem 1 gives a regret bound at most $\gamma \sum_n \delta^n$. Thus
uniform condition holds. Correcting a subset of decisions
ition, since the re-evaluated continuation changes. The

the exact surviving-continuation map has Lipschitz constant L and the exact continuation is received at the admissible within-step controls preserves this bound. Then

$$E_n^\pi \leq \kappa_n^\pi + \delta E_{n+1}^\pi.$$

one lift of the finite gap prove (15). If $\mathcal{B}^{(0)}, \dots, \mathcal{B}^{(m)}$ are the exact semigroup to the lifted numerical operator, accuracy on a specified continuation w by $\sum_i \|\mathcal{B}^{(i)} - \mathcal{B}\|$ is bounded; changing the tested continuation between steps changes the change invalidates the telescoping argument. *Q.E.D.*

FOR ECONOMIC COMPARISONS

and nonincreasing in k . The supremum is therefore bounded by $.02C_\rho(T-t)$ because $|\theta| \leq .2$. If π attains

$$B(\pi) = V(k) - (k' - k)B(\pi).$$

a differentiability point every attained optimizer has

Proof of Theorem 3

of the rival candidate policy give

$$\geq A(\pi_2) - k_1 B(\pi_2) - \varepsilon_1,$$

$$\geq A(\pi_1) - k_2 B(\pi_1) - \varepsilon_2.$$

$\{B(\pi_2) - B(\pi_1)\} \leq \varepsilon_1 + \varepsilon_2$. The exact monotonicity of the two budget measurement discrepancies yields the independence of k is essential: otherwise a counterexample is not available.

$k - k_-$
 appropriate outer value enclosure gives (17), and intersecting
 valid. Inclusion of restricted policies in the flexible set
 traction ¹²⁴ gives the other endpoints. No optimizer
 cy, or pointwise action ordering is required. *Q.E.D.*

ASSIVE AND EQUILIBRIUM GAIN BOUNDS

equation values is invariant under all action operators, each
 has Lipschitz constant at most $L_n \geq 0$. The maximized
 g the two approximate-equation comparisons in the finite

$$e_n + \eta_n + L_n \|V_{n+1}^* - v_{n+1}\|_\infty,$$

$$e_n + L_n \|V_{n+1}^\pi - v_{n+1}\|_\infty.$$

error formula stated in the text. These hypotheses must be
 zation. In particular, the sign condition $bV > 0$ does not
 he driver close to zero and does not supply a uniform L_n .
 d regard player i 's unilateral deviations over all future
 own payoff and transition law. The preceding certificate
 g this bound for every player gives an approximate Nash
 in the norm. Checking only a single-date action deviation
 s not replace the all-future-date best-response calculation.
 e acting self's criterion uses the β -weighted continuation
 valuation uses δ . A unilateral deviation certificate must
 ture selves held fixed. Substituting a geometrically β -
 es that equilibrium problem, so its small residual would
 d supplement supplies the full two-date algebra and the
Q.E.D.

ATION AND LIMITATIONS OF LOCAL TRAINING

$(0, I)$, diagonalize A orthogonally. Rotational invariance
 mply $\mathbb{E} z' A z = \text{tr } A$ and $\text{Var}(z' A z) = 2\|A\|_F^2$. The mean

in shared network weights.
 has the same logical role. A small integrated strong
 -1 does not select the viscosity solution $1 - |x|$ of
 Missing viscosity inequalities cannot be replaced by
 boundary specifications, and regression checks for
 in the preservation supplement.
 proximation theorem requires bounded iterates, ap-
 dynamics, martingale-noise control, and step-size
 variant-set statement for the limiting slow dynamics.
 Additional argument ruling out nonoptimal attractors.
 asserting that every observed gradient trajectory has

y and its feasible deviation gains. We study both the
 a positive finite-horizon operator formulation. The latter
 e action maximization explicit and provides a reference
 ation can be assessed.

First, we give a quantitative connection between policy
 feasible improvement, and loss of economic value. The
 opped controlled diffusion and requires uniform, rather
 s. Its finite-horizon counterpart is directly computable
 a. Second, we establish a comparative-static result for
 en the adjustment cost enters as a price of a discounted
 price reduces the optimal budget. This conclusion holds
 e differentiability of the value function or a pointwise
 nt. Third, we implement the separated method, including
 as HJB benchmark and trained nonlinear actor and critic
 rtfolio problem. Independent policy evaluation separates
 full policy vector.

s the interior preference and wealth dynamics of the
 at the boundary by an explicit early-liquidation contract.
 diffusion, so a bounded drift cannot keep it in a closed
 , the original consumption floor is incompatible with a
 ed to the lower endpoint. Liquidation is an economically
 flection or a capital injection; it is specified as such,
 sensitivity. No transition replenishes wealth and then

oles. A continuous-time Merton benchmark tests the HJB
 raph. A two-state NDU problem tests a trained nonlinear
 ndent exhaustive stochastic Bellman solver. A re-solved
 omic comparative static. Recursive utility, sophisticated
 y game exercise additional operator interfaces. A coupled
 ecks dynamic cross-state dependence at dimensions 4, 8,
 general economic scope without conflating an algebraic
 d a trained solution of a continuous nonlinear economy.

elation to the literature

dents already combine policy improvement and neural
 no (2025) study physics-informed neural policy iteration

portfolio choice (Merton, 1969), recursive preferences (Epstein and Zin, 1992), and the motivating endogenous-preference methods remain important comparators in computational finance (see, e.g., Schied and Schott, 2017). In our two-state experiment, exhaustive search is computationally intractable. That result is reported, not converted into a numerical representation can nevertheless be evaluated as a numerical approximation, and an independently measured value loss can be used to assess the quality of the approximation.

Our main contribution (Han, Jentzen, and E, 2018) is about the equations that a control analyst can write a closed-form control. Semilinearity of the BSDE and its derivative. Eliminating a control analytically can be done, but, conversely, an admissible BSDE representation need not exist. This paper studies the differentiation costs conditionally, including the cost of a trace approximation.

PROBLEM AND ITS BOUNDARY

Let $D \subset \mathbb{R}^d$ be a compact feasible domain, let $A(s)$ be a compact feasible set of admissible feedback. Up to $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$,

$$dV(r) = \mu(r, S_r, \pi_r) dr + \Sigma(r, S_r, \pi_r) dW_r. \quad (1)$$

Assuming the stopped process and integrability of the stated payoff. For a given discount rate ρ , define

$$J^\pi = \mathbb{E} \left[\int_t^\tau \mu(r, S_r, \pi_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V^* = \sup_{\pi} J^\pi. \quad (2)$$

As an output of a projection routine. At T , it equals the value of the payoff.

$$R_a(v) = v_t + \mathcal{L}^a v + \ell^a - \rho v.$$

Fixed-policy evaluation has $R_\pi(V^\pi) = 0$, while optimal policy has the same declared parabolic-boundary data. A reflected BSDE with an appropriate derivative boundary condition; the condition with a value penalty.

missible, it solves

$$Y_r^\pi) dr - Z_r^\pi dW_r, \quad Y_\tau^\pi = g(\tau, S_\tau). \quad (3)$$

and integrability of this policy-wise equation are prerequisites. A Markov state must contain any variable on which the quadratic variation is known; the executed models here use recursive extensions. The recursive extension and its sign domain

NEURAL BELLMAN OPERATORS

v_ξ is trained for a fixed enacted policy by its evaluation loss. The actor parameters ψ are updated against a

$$\mathbb{E}_\pi(v_\xi)^2 + \lambda_B \mathbb{E}_{\nu_B} |v_\xi - g|^2, \quad (4)$$

$$\mathcal{H}^{\pi_\psi}[\text{stopgrad}(v_\xi, \nabla v_\xi, D^2 v_\xi)]. \quad (5)$$

parameterization or by explicit constrained maximization. A feasibility approximation and must be accompanied by a feasibility parameter gradient reaches ξ . In a game no player- i gradient

surrogate. It is not generally the lifetime-objective measure. Such an identification requires the appropriate differentiability conditions. Likewise, minimizing a parameter blocks solves a different optimization problem-loss and viscosity counterexamples and states what the

Finite operator realization

be a finite state set and A_h a finite feasible action set. a_n and surviving transitions in a substochastic kernel P_n^a .

$$\delta P_n^a w, \quad T_n w = \max_{a \in A_h} T_n^a w. \quad (6)$$

ator limit: *Let the policy space Π be compact in the
 tinuous into a precompact subset of $C^{1,2}$, endowed
 r time derivative, gradient, and Hessian. Assume a
 continuous at accumulation points, and satisfies the
 Assumption 1. If $\pi_{j+1} = I(\pi_j)$, every accumulation
 on limiting value solves the HJB boundary problem.*

values, not equality of successive limiting policies.
 nder the exact selector is a real hypothesis and is not

value certificate: *Under Assumption 1, let v be a test
 uniformly on the stopped domain,*

$$\}) \leq \eta, \quad |v - g| \leq b \quad \text{on the parabolic boundary.}$$

$$(h) = h,$$

$$s) \leq 2b + C_\rho(T - t)(2\epsilon + \eta). \quad (7)$$

actor stationary point is a maximizer, is required for

the terminal date: exit can occur earlier. Theorem 1
 o a value loss. A sampled mean-square loss does not
 residual Lipschitz constant and a covering radius can
 ch constants must be established, not inferred from a

olicy certificate: *For the finite operators (6), let $\hat{v}_N$
 ted policy π , define*

$$\eta_n = \|T_n \hat{v}_{n+1} - T_n^\pi \hat{v}_{n+1}\|_\infty.$$

$^2 + .2a$ on $[-2, 2]$ has a stable local ascent attractor near 1.024120. Their payoff gap is .3998746. Both stationary critic, diminishing two-timescale steps, and isolated actor-improvement fixed point.

turns the *action* maximization. If H is differentiable and set, then

$$) \leq \max_{a \in A} \langle \nabla H(a_0), a - a_0 \rangle. \tag{9}$$

ing-hyperplane gap and handles boundary optima. Point-ly in shared neural parameters. Our finite implementation e action directly. The separate asymptotic stochastic-iant-set conclusion and additional optimality conditions,

REFERENCES AND PORTFOLIO CHOICE

cient of relative risk aversion and X is financial wealth. nce adjustment θ , and the risky portfolio share π . Before

$$\tag{10}$$

$$dt + .2\pi X dZ^X, \quad d\langle Z^u, Z^X \rangle = -.25 dt. \tag{11}$$

the control box is $[.05, .8] \times [-.2, .2] \times [-.5, .8]$. The y is cardinally fixed as

$$^2, \quad G(u, X) = -.02(u - 2)^2 + .1 \log X.$$

ation are held fixed throughout the cost panel. An affine controlled preference state generally changes the economic

$$9V_u + [(.02 + .06\pi)X - c]V_X - \rho V \\
\left. \frac{1}{2}(.2\pi X)^2 V_{XX} - .25(.05)(.2)\pi X V_{uX} \right\}, \quad (13)$$

ree second-order contributions are retained.

d economic interpretation

indicated derivatives have the required signs,

$$\pi = -\frac{.06V_X}{.2^2 X V_{XX}} - \frac{.05(-.25)V_{uX}}{.2X V_{XX}}. \quad (14)$$

onstrained maximization. Consumption and adjust-
 portfolio objective is concave only when $V_{XX} \leq 0$;
 al. Appendix B gives the full selection rule, including
 ealth value.

gainst preference shocks correlated with returns. Its
 ck correlation nor an assumed positive V_{uX} alone
 atic. Likewise, dividing V_u by k does not determine
 n k . The following result gives a global economic

djustment: *Suppose the feasible policy set, stopping*
ed on k . Write

$$g(\tau, S_\tau) \Big] , \quad B(\pi) = \mathbb{E} \int_t^\tau e^{-\rho(\tau-r)} \frac{\theta_r^2}{2} dr.$$

convex and nonincreasing. If $k_1 < k_2$ and optimal
 a differentiability point with an attained optimizer,
 value weakly exceeds the value under the feasible

as a label.
 $= 1/(mD)$ for $J^\pi = -A^\pi/x$, where $D = .04 + .02 +$
has $m^* = .04125$, $\pi^* = .75$. Across seeds 0, 1, and 2
 8.25×10^{-8} and π within 8.29×10^{-10} . The maximum
 $[-.5, 2]$ is 4.70×10^{-9} . The scaled HJB residual is at most
es that actor backpropagation leaves the critic gradient
al benchmark with three trained parameters, not evidence
ation theorem. Its value is that the continuous generator,
actor graph can all be checked independently.

U transition and neural architecture

our equiprobable branches use independent signs $z_1, z_2 \in$

(15)

$$c]\Delta + .2\pi X\sqrt{\Delta}[-.25z_1 + \sqrt{1-.25^2}z_2]. \quad (16)$$

at its first intersection and terminates; otherwise con-
ulation. The running payoff is integrated with the exact
al step while holding the current flow fixed. This defines
exact Brownian first-passage integration. Interior drift
oundary corrections.

ave two hidden layers of width 48 with tanh activations.
five values for each of the three controls. The critic uses
e four boundary faces; it is represented as $G + (1-t)N_\xi$
ct prescribed settlement at boundary nodes. The finite
-switched representation to be a globally C^2 function.
te applied to it is Theorem 2, not an unverified use of

interior nodes of a 25×31 lattice, 500 actor Adam steps
by critic L-BFGS refinement. The learning rate is .004.
rozen. This experiment is a lattice-trained finite-operator
periment. Seeds 0, 1, and 2 are reported without selecting

reference-adjustment costs

25×31 state lattice, and the 125-action stochastic
d, along with the restricted $\theta = 0$ economy. Table II
ills by approximately 86 percent between $k = .5$ and

TABLE II
ION UNDER THE LIQUIDATION CONTRACT

$k = .5$	$k = 2$	$k = 8$
-1.351522	-1.372708	-1.404420
.016520	.011921	.002309
.064480	.043293	.011582
.2	.2	.1
.248935	.249938	.249920
.896927	.764993	.475637

ounting. Exit probabilities refer to this numerical liquidation
decision dates and interior lattice states, not over a population.

25, 31, 5)	−1.451486
49, 61, 5)	−1.344458
25, 31, 7)	−1.359628

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 control. The changes are material. Refining time alone while need not improve a diffusion approximation: the inter-scale as h^2/Δ . Action refinement and boundary-crossing report a verified finite-model neural approximation and not a numerically certified continuous-diffusion value. Theorem 3 does apply to the continuous contract under its

6 or 10, with the twelve-date baseline discretization held 909 or -1.379622 . These are different contracts rather to the same contract. The complete exit and adjustment frequent boundary refinements can be evaluated without zero-error result.

UTILITY AND EQUILIBRIUM EXTENSIONS

1. *Recursive utility*

$\gamma > 0$. In the normalization used here, the Epstein–Zin

$$= \frac{\rho}{a} c^a (bV)^{1-a/b} - \frac{\rho b}{a} V, \tag{17}$$

0. The architecture $V = \exp(W)/b$ enforces the sign-itive margin for bV , along with consumption bounds, assertion without executing the transform is not a domain

is $V_t + \mathcal{L}^{c,\pi} V + f(c, V) = 0$. Formal stationary homoth-1.5, .08, .02, .2) are $\pi = .3$ and $m = .0455$. An algebraic- t -horizon solution. We retain these anchors and add an-ve-utility calculation. The latter has horizon .5, ten dates, n fractions, and solves each fixed-action implicit value ts maximum implicit-equation residual is 2.46×10^{-13} ; d terminal sign transform differs from the terminal target

β . For two consumption dates followed by terminal
ary continuation has coefficient 2 on log wealth at
consumption share is $1/(1 + 2\beta)$. At $\beta = .7$ this is
gives .4566210. The corrected executed evaluation of
feasible one-shot search finds no positive gain. Strict
dition establish global optimality for this particular
would provide only a lower bound on exploitability.
to detect an error that vanishes at $\beta = 1$.

c capacity competition

es both the evaluation operator and an independent
e Cournot benchmark has Nash output 1/3 and joint-
on test confirms that the rival gradient is absent from

game. Two firms have capacities in $\{0, .5, 1\}$, output
d investment $i \in \{0, 1\}$ that adds .5 capacity when
.5} with transition matrix having diagonal .8 and
 $-q_i - q_j) - .2i$, discount is .96, and the horizon is
ation selects a pure Nash equilibrium in every state;
in three state-date problems. The maximum feasible
ro in this finite game. The full profile and selection
ilibria. This is not a neural dynamic-game training
the latter would require a market-size normalization
the operator formulation and its economic questions

ICES AND COUPLED DYNAMICS

s z_k , write $q_k = z_k' A z_k$ and $\bar{q}_K = K^{-1} \sum q_k$. The

$$= (a - \text{tr } A/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (20)$$

ance penalty independent of K . With $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ and
 K , not 8.5 for every K . The replication runs 30,000

sions are respectively 6.60×10^{-14} , 1.55×10^{-14} , and linear-quadratic dynamic regression, not the earlier statics faster, and the known quadratic value family makes the ned sparse grid superiority or general high-dimensional

8. CONCLUSION

the improvement, and independent economic evaluation as distinction yields explicit value-loss bounds and prevents γ point from being mistaken for a solution certificate. 1, an explicit liquidation contract makes the stochastic agree; the re-solved adjustment-cost panel implements discounted preference-adjustment budget. Actual neural moments, temporal-self recursion, trace estimands, and cutable.

ve utility and strategic interaction without assigning them ther accuracy must be earned through the declared error e finite stochastic economy, while continuous-diffusion cal models, derivations, figures, and prior reports remain ap distinguishing corrected equations, supplementary evidence. This structure permits the next review to assess nomy rather than reconstruct their intended relationship.

AND THE SCOPE OF THE VALUE CERTIFICATES

Proof of Proposition 1

selector inequality is $\mathcal{H}^{I(\pi_j)}[V_j] \geq \mathcal{H}^{\pi_j}[V_j]$. Comparison ty imply $V_{j+1} \geq V_j$. Precompactness of the evaluation e at each state-time point the monotone sequence has a

er convergent subsequence in the declared strong norm. by monotonicity and pointwise uniqueness of the limit. merges uniformly to $\bar{V}$; in fact it converges in the declared quence separated from its unique possible strong cluster Derivatives of any strong cluster point are the derivatives

on a compact, selector-closed policy class. Neither optimizer automatically meets those assumptions. The d by Theorem 2. For recursive drivers the comparison driver and value domain; the proof does not obtain

Proof of Theorem 1

st function under any feasible policy a and stop at τ .

$$v)(\tau, S_\tau) + \int_t^\tau e^{-\rho(r-t)} R_a(v)(r, S_r) dr \Big]. \quad (21)$$

the integrability hypothesis justifies removal of the $R_\pi(v) \geq -\epsilon$ and the boundary difference is at least every feasible alternative, $R_a(v) \leq R_\pi(v) + \eta \leq \epsilon + \eta$. Thus $J^a \leq v + b + C_\rho(T - t)(\epsilon + \eta)$. Taking a ng proves (7). Attainment of that supremum is not

of the true value or differentiability of the policy with legitimate test function, a legitimate stopped process, g policy can travel. A residual measured under the alternative policy. This is why an off-policy uniform asure bound, is essential for the displayed worst-case

ations. The theorem bounds the stopped payoff as s a classical extension to every boundary face. The th under $\pi = 0$. We do not assert that all boundary usion, or that the continuous NDU value belongs to boundary starting state is settlement by definition; generalized viscosity boundary interpretation. The eses would still have to be verified before numerical not used to bypass this requirement.

imply that T_n^π and T_n are monotone and δ -Lipschitz in d_n^π . The evaluation defect gives $d_n^\pi \leq e_n + \delta d_{n+1}^\pi$, and hence

$$\delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} e_j.$$

with respect to T_n is at most $e_n + \eta_n$ in absolute value, $(T_n \hat{v}_{n+1} - T_n^\pi \hat{v}_{n+1})$. The same backward estimate

$$\leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} (e_j + \eta_j).$$

It. For exact re-evaluation, both e and b vanish; the code gain and accumulates it with the factor δ .

onal continuation coefficient: its payoff, including the a_n . The substochastic surviving row sums remain at most their immediate settlement operator in both the optimizer before applies to the finite model even though its boundary acted.

al-economy error. Suppose, for the same policies and establishes $|J^\pi - J_h^\pi| \leq \zeta$ uniformly and $|V^* - V_h^*| \leq \zeta_*$. This decomposition identifies the missing bridge without coarse-fine value difference is neither ζ nor ζ_* without a

. Proof of Theorem 3

$A(\pi) - kB(\pi)$ is affine and nonincreasing in k , since functions is convex; the pointwise supremum of nonin- t an initial state where optimizers exist, their optimality

$$B(\pi_1) \geq A(\pi_2) - k_1 B(\pi_2),$$

$$B(\pi_2) \geq A(\pi_1) - k_2 B(\pi_1).$$

separates across c, θ, π . Let their bounds be (c_-, c_+) ,
 minimizes $c^{1-u}/(1-u) - V_X c$, whose second derivative
 is the projection of $V_X^{-1/u}$ onto $[c_-, c_+]$. If $V_X \leq 0$,
 um is c_+ . Adjustment maximizes $V_u \theta - k\theta^2/2$ and

$_{2}\pi^2/2$, where

$$-.25)XV_{uX}, \quad a_2 = .2^2X^2V_{XX}.$$

a_2 . If $a_2 > 0$, compare the two endpoints. If $a_2 = 0$,
 of a_1 , with any feasible action optimal when $a_1 = 0$.
 t create an inaccessible local search problem in this
 ad enumerates its declared action grid; it never labels
 action optimum.

maximand, concavity gives $H(a) \leq H(a_0) +$
 proves (9). If an interior maximizer and an m -
 e familiar gradient-square bound $H^* - H(a_0) \leq$
 he square. The supporting-hyperplane formulation is
 argument transfers concavity from actions to shared

cy and preference reachability

enote the covariance matrix of the two-dimensional
 atisfies

$$S) = C\Delta, \quad C_{uX} = \sigma_u \sigma_X \pi X \rho_{uX}.$$

uX give the correct constant and linear generators
 Δ) from the drift outer product. At $(u, X) = (2, 1)$
 s generators are

$$5225, -.6651, -.553625).$$

25, .0125. The covariance error is below 10^{-12} and
 nes at the predicted rate. These tests include, rather
 nent interior state, the generator contribution of u^2 is

y cash flows and recorded exits

fraction is the minimum positive fraction at which $s + \lambda d$ at a crossing, the continuation value is $g(t + \lambda \Delta, s + \lambda d)$, $-\exp(-\rho \lambda \Delta)]/\rho$, and there is no continuation policy $X = .51$, chooses $c = .8$, $\pi = 0$, and verifies that every ictly below one and then terminates. The program never er consumption date under the pre-exit contract. event from a post-correction state. The adjustment budget r a surviving branch interpolated to a boundary node, the f the finite kernel. This interpolation effect is a numerical settlement fee is a model primitive; comparing fees 6, 8, on of contracts.

REGREGATION, TEMPORAL SELVES, AND STRATEGIC INTERACTION

rsive aggregator and its domain

> 0 , expand the aggregator as in (17). Its consumption

$$)=\rho c^{a-1}(bV)^{1-a/b}>0.$$

$$=\frac{\rho}{a}(b-a)c^a(bV)^{-a/b}-\frac{\rho b}{a}.$$

a bounded away from zero and bV bounded away from is is a local Lipschitz conclusion on the specified range, or all values. The transform $V = e^W/b$ enforces $bV > 0$, terminal compatibility, comparison, or transversality. selves (3); in the Markov setting its equation is $v_t + \mathcal{L}^\pi v + v$ with constant L , standard comparison of a perturbed ves a finite-horizon error envelope of the form

$$^{-t})b+\int_t^Te^{L(r-t)}\epsilon\,dr$$

olved numerically before maximizing over the action
the wealth lattice, continuation uses the specified
undary data at that date. It is not a solution on an
l node the recorded value is $-.29658656$ and the
bound $.01$. This active bound is reported rather than

residual tests are actual evaluated quantities. Neural
ent solves scalar implicit reference equations rather
general recursive portfolio model remains part of the
derivations; its stationary ratios $.3$ and $.0455$ remain
orizon verification is supplied.

nnuation and acting-self utility

ry log-wealth utility in the two-consumption-date
. Starting from $A_N = 1, B_N = 0$, the sophisticated

$$\delta A_{n+1}, \quad B_n = \log \alpha_n + \delta[A_{n+1} \log(1 - \alpha_n) + B_{n+1}].$$

ses $\beta\delta$. With $\delta = 1$ and two decisions, $A_1 = 2$ and
valuation coefficient instead produces the different
two formulas coincide, so this special case cannot

t the same frozen continuation policy whose equilib-
ation with a different grid's optimized value changes
cated-self computation, all feasible actions can be
y evaluated future policy. For continuous actions a
lower bound unless concavity, a global oracle, or an
search gap.

es and equilibrium selection

f with its own objective and transition law. A Nash

$$\pi_{-i}) - J_i(\pi_i, \pi_{-i}),$$

limit analysis, the demand and capacity scaling, investment must be fixed before taking a limit. Symmetric equilibrium nor an equilibrium-selection theorem. Those extensions rather than filled with the historical illustrative high-dimensional game or a solved competitive limit is e.

ESTIMANDS AND COMPUTATIONAL COST

metric A , diagonalization and the independent squared- and $\text{Var}(z'Az) = 2\|A\|_F^2$. Independence across K probes mean. Expanding the squared residual then proves (20). Contribution to variance vanishes, so the formula must be . correction

$$a\bar{q}_K + \frac{1}{4K(K-1)} \sum_{i \neq j} q_i q_j. \tag{22}$$

$\mathbb{E}U_K = (a - \text{tr } A/2)^2$. The statistic is unbiased but can have a different variance and optimization behavior. , not a blanket replacement justified by unbiasedness mean and standard error separately. When A depends on variance term itself varies with those parameters; a squared ge the expected minimizer.

directional second derivative and its parameter gradient for ns, exact contraction can be organized as m directional ching costs. A K -probe contraction costs roughly KC_{jet} onal $O(d)$ solver. Total work additionally multiplies by and any restarts. Peak memory, warm-up, and accelerator a hardware comparison. The current CPU timers do not remains null.

ression. For the finite-horizon quadratic-cost economy, due $x'Px + c$. Given a frozen next evaluator, the actor ge-plus-continuation cost. Gaussian shocks contribute

problem whose true value is zero and whose critic is v . A joint surrogate $v^2 - \lambda v$ has minimizer $v = \lambda/2$. The example isolates the logical issue: Optimizing the value parameter need not preserve the evaluation changes the bias but does not prove exact evaluation. A value parameter, so it does not create this particular

not a stability theorem. The derivative of $R^2/2$ is λ , and step sizes govern a finite optimizer. Value inside this derivative. A zero target does not make the

al does not select a viscosity solution

consider $|v'| = 1$. The nonnegative distance function on $|x| - 1$ has the same boundary values and satisfies viscosity supersolution test at zero: the constant test $|\phi'| - 1 = -1 < 0$. $-\varepsilon^2 - \sqrt{1 + \varepsilon^2}$ have exactly the boundary data and residual $\int_{-1}^1 (|v'_\varepsilon| - 1)^2 dx$ tends to zero by dominated convergence. Thus boundary agreement, not establish viscosity selection. Monotone, stable, comparison can provide a different route (Barles and Perthame) but interpolation alone does not verify all of those

can approximate a sufficiently smooth function and relevant approximation hypotheses. This does not boundary regularity, close a finite policy class under a stochastic optimization finds the approximant. The retained approximation is only in that conditional sense.

ements and suboptimal attractors

1, a conventional ODE analysis requires bounded martingale-difference noise with bounded conditional

parameterizations and data flow

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 The learned log-amplitude and its actor has two learned
 variation through the value with respect to wealth, including
 rs are separate. Its independently evaluated policy has
 ical controls are used only after training to measure error.
 ic nonlinear networks represent a finite Bellman operator.
 o 125 logits through two 48-unit tanh layers. Its critic
 es to boundary faces, through two 48-unit tanh layers to
 text-date critics are frozen, the actor target uses only their
 nt critic is fitted to the enacted actor's policy-evaluation
 izer and evaluator are not imported as training labels.
 ample certificates because all interior finite states are
 uator changes the computational path, not the economic
 nPy and PyTorch implementations of the transition and
 on deterministic data. Cross-grid reference calculations
 e not been relabeled a held-out neural accuracy result.

Interpreting metrics

al reference, a critic error against the evaluated policy, a
 ilateral gain are different estimands. Policy component
 n, adjustment, and portfolio share. Boundary-node actor
 nt; full-grid maxima and interior summaries must be
 A finite exhaustive best-response maximum is a finite-
 ver only some continuous deviations is merely a lower

s are separate fields. A program that completes can miss
 solution certificate. A null numerical error has an explicit
 ous-diffusion error or a recursive reference that did not
 is used only for a quantity actually calculated, such as
 a tested monotonicity violation.

its weights, metrics, and raw arrays are retained in the
 ee-seed sources and weights are separate. The attempted
 e. Wall-clock times are metadata and are excluded from

10	10	0.0496	18.19	
0	3	0.0064	34.64	
1	2	0.0471	16.86	
2	146	2	0.0461	20.03
0	2	0.0050	18.52	
2	2	0.0339	17.95	
0	2	0.0065	13.66	
1	2	0.0440	15.70	

the number of time steps. Pass counts use the historical finite-candidate
 timing measure; this is not the new full safeguard re-audit timing.

TABLE II
 QUADRATIC-CONTROL COMPARISON

Sampling upper	Train (s)	Audit (s)	MiB
0.0018	14.93	3.45	377.5
0.0007	12.59	6.61	376.9
0.0055	14.29	3.84	378.6
0.0032	11.67	5.69	377.7
0.0145	16.29	3.72	381.3
0.0117	13.34	5.64	382.1

benchmark with an analytic greedy control and an in-house PINN-PI
 reflects reference or simulation uncertainty, not performance exceeding
 in each row pair. This result is retained despite the favorable R8
 convex-control comparison.

Δ	V_0	B_0	Exit
33	-1.624355	0.012266	0.1543
67	-1.510132	0.012248	0.1558
33	-1.391646	0.011969	0.1024
67	-1.353683	0.012234	0.0987

al panel. The table supplies numerical differences, not validated
s or a separated convergence order.

TABLE V

CURSIVE-UTILITY POLICIES

ound	Final log loss	m_0	p_0
0.0000	0.0000	2.506	0.955
0.0000	0.0000	2.442	1.093

og-value target. The final candidate-set bounds use repeated
h failures are not relabeled as neural successes.

TABLE VI

D R5 NEURAL GAME

ver 2, $t = 0$	All subgames	Seconds
0.00673	0.00673	13.87
0.00305	0.00368	14.04

exact continuous best-response auditing for the declared finite
s-state equilibrium scalability result.

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